Stability of meshfree MUSCL schemes on point clouds: what is proved and what is only measured

Working note for P. Jung (companion to the mathematical addendum)

Draft, 29 October 2026 (revised after review)

Combines `stability_note.tex` and `high_order_stability_note.tex` (28–29 October 2026)

Abstract

We collect what is known about the stability of the meshfree MUSCL schemes of [9, 10] and separate it strictly into results with a proof (Part I) and results that are only supported by computations (Part II).

Proved. (1) Through positivity, on arbitrary clouds: a maximum principle for scalar laws under stage-wise MOOD with the strict discrete maximum principle, for every design order; ℓ^1 bounds for the Euler equations up to the conservation defect; a local entropy inequality for the first-order scheme. (2) For linear advection, the first-order aligned operator generates a Markov chain on the particles, and the scheme is contractive in every ℓ^p norm weighted with its stationary distribution π, on arbitrary clouds on which the chain has a positive stationary distribution. The same structure shows that the first-order scheme converges to a solution with a *wrong* transport speed on irregular clouds, at a rate set by the size of a corrector that we only measure (for random jitter the measured sizes fit $a\,h + b\,h^{1/2}$ with a small b). (3) For the high-order schemes ($K \geq 2$, linear advection, no MOOD): von Neumann stability of order two on lattices; an exact reduction of stability on periodic clouds to a uniform bound on $m \times m$ Bloch semigroups; stability on smoothly deformed lattices with constant stencil topology (unconditionally for $K = 2$ in one dimension); and, in one dimension, stability on deformed lattices with localised insertion/merging defects up to a residual, given local Lyapunov certificates for every local spacing; such certificates are computed by semidefinite programming and checked a posteriori, which covers the undeformed lattice with defects but not yet a whole range of spacings. A residual-free statement for lattices with defects whose separation grows like $|\log h|$ holds under a hypothesis of exponential decay of the residuals, which no finite computation can verify and which the computed residuals rule out for some defect types. (4) Two obstructions explain why the energy method in a volume norm cannot work on jittered clouds, and a local-symbol expansion explains why even orders are fragile.

Measured only. Spectra, Kreiss constants and Bloch symbols on jittered clouds show no visible growth in the settings of the paper (jitter $0.2h$), with Kreiss constants 1.002–1.33, except an extremely weak Bloch instability ($\alpha h \leq 10^{-9}$) of one periodic pattern at $K = 2$; weak instabilities ($\alpha h \approx 2 \cdot 10^{-6}$) for $K = 2$ and failures for $K = 4$ when the stencil topology varies (jitter $0.4h$); order three is stable in every setting tested. On lattices, orders three to five are stable except order four with the geometric orientation in two dimensions. The first-order scheme advects 0.1–0.45% too slowly on perturbed clouds, and a moment correction of the dissipation removes this. Stability of the high-order schemes on random clouds remains unproved.

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1 Overview

1.1 The question and how this note is organised

The question is whether the schemes of [10], analysed in the addendum [9] (cited as [A] with its numbering), are stable, and how stability relates to the positivity that [A] proves. The answer depends on the notion of stability (nonlinear bounds or Lax–Richtmyer stability of the linearised scheme), on the order (first order or $K \geq 2$) and on the cloud (arbitrary, lattice, periodic, smoothly deformed, with defects, randomly jittered).

- **Part I (Sections 3 to 9)** contains only statements with proofs. Where a proof rests on a hypothesis that is checked numerically and not proved (a corrector bound, a frozen symbol for $K \geq 3$, a semidefinite certificate), the hypothesis is stated in the theorem and flagged in Table 1.
- **Part II (Sections 10 to 13)** contains the numerical evidence, written as numbered *Observations*. Nothing in Part II is used in a proof in Part I, except as the explicitly named hypotheses.
- **Part III (Sections 14 and 15)** draws consequences for the scheme and the paper and lists the open questions.

All numbers come from an independent NumPy/SciPy implementation of the linear scheme described in Appendix A; it tests the scheme as defined in [A], not HyperCloudCore.jl.

1.2 Bottom line

(a) *Positivity gives nonlinear bounds on every cloud.* Everything that [A] proves by writing an update as a convex combination of admissible states also bounds a convex functional: a maximum principle (Theorem 3.1), an entropy inequality (Proposition 3.5) and, with conservation, ℓ^1 bounds for Euler (Proposition 3.3). These hold for the high-order schemes under MOOD, because MOOD falls back to the positive first-order scheme wherever a bound would fail.

(b) *For linear problems, positivity is a Markov chain.* The first-order aligned operator is a Markov generator, and its stationary distribution π gives the norm in which the scheme contracts (Theorem 4.1). No antisymmetry and no regular cloud are needed.

(c) *Stability is not convergence on irregular clouds.* The same chain has a drift: the first-order scheme transports with a wrong velocity, by a defect that does not vanish as $h \to 0$ (Theorem 4.4, measured in Observation 11.2). A moment correction of the dissipation fixes it (Proposition 4.5).

(d) *High order: proofs exist for structured clouds only.* Lattices (Theorem 5.2), periodic clouds (Theorem 6.1), smooth deformations of lattices (Theorem 7.3) and, in 1-D, deformations with localised defects (Theorem 8.7). On randomly jittered clouds the energy method in any volume norm provably fails (Propositions 9.1 and 9.2), and stability is only measured (Section 12).

(e) *Odd orders are safer, and stencil topology matters more than jitter.* All instabilities found are at even order (explained by Proposition 9.3) and occur when the neighbour count varies from particle to particle.

2 Setting and notions of stability

2.1 The scheme

We use the notation of [A]: interior particles I, neighbourhoods $\nu(i)$, offsets $x_{ij} = x_j - x_i$, derivative weights $c_{ij}^{(d)}$ of degree q_i with zero row sums [9, Lem. 3.3], interface states U_{ij}, the Rusanov flux [9, (4.3)] with orientation $\sigma_{ij,d}$ and the scheme [9, (4.4)], $\mathrm{d}U_i/\mathrm{d}t = \mathcal{L}_i(U) = -2\sum_j c_{ij} \cdot (F_{ij}^* - F(U_i))$. We abbreviate $\sum_{j,d} = \sum_{j\in\nu(i)} \sum_{d=1}^{D}$. Particle volumes are $V_i > 0$ with $\sum_i V_i = |\Omega|$, and $\|U\|_V^2 = \sum_i V_i |U_i|^2$ is the discrete L^2 norm. Jittered clouds are lattices with spacing h whose points are displaced uniformly in $[-\varepsilon h, \varepsilon h]$ per coordinate ("jitter εh"); the support radius is κh.

The first-order operator. With piecewise constant states $U_{ij} = U_i$, $U_{ji} = U_j$ and zero row sums,

$$\mathcal{L}_i^1(U) = \sum_{j,d} \Big[s_{ij,d}\, \sigma_{ij,d} c_{ij}^{(d)}\, (U_j - U_i) - c_{ij}^{(d)} \big(F_d(U_j) - F_d(U_i)\big) \Big], \tag{2.1}$$

and on aligned edges $\sigma_{ij,d} c_{ij}^{(d)} = |c_{ij}^{(d)}|$. By [9, Prop. 5.6], this is the operator that a particle at order one sees under EPD1 and EPD2.

Linear advection. For $F_d(U) = a_d U$ with a constant $a \in \mathbb{R}^D$ and the Rusanov speed $s_{ij,d} = |a_d|$, the scheme is a linear system $\mathrm{d}U/\mathrm{d}t = L_h U$ with an $N \times N$ matrix L_h. The first-order aligned operator is

$$(QU)_i = \sum_{j\in\nu(i)} q_{ij}\, (U_j - U_i), \qquad q_{ij} = \sum_{d=1}^{D} \big(|a_d|\, |c_{ij}^{(d)}| - a_d\, c_{ij}^{(d)} \big) \geq 0. \tag{2.2}$$

For the high-order schemes, L_h denotes the operator with all particles at the design order $K \geq 2$, the divergence degree $q = q_{\mathrm{div}}(K)$, unregularised fits, $s_d = |a_d|$, no limiter and no MOOD. It splits into a central and a dissipative part,

$$(L_h U)_i = \underbrace{-\sum_{j,d} a_d\, c_{ij}^{(d)} \big(U_{ij} + U_{ji} - 2U_i\big)}_{(C_h U)_i} + \underbrace{\sum_{j,d} |a_d|\, \sigma_{ij,d} c_{ij}^{(d)} \big(U_{ji} - U_{ij}\big)}_{(S_h U)_i}, \tag{2.3}$$

with the interface states $U_{ij} = \mathcal{R}_i(\frac{1}{2} x_{ij})$ of the reconstruction fits of degree $K-1$ [9, (4.1)]. $L_h \mathbf{1} = 0$ by the zero row sums [9, Lem. 3.3(ii)] and the reproduction of constants. All operators are homogeneous of degree -1 under dilation: if the cloud is scaled by θ, L_h becomes L_h/θ [9, Lem. 3.3(iii)].

2.2 Notions of stability

Lax–Richtmyer stability. A family L_h on clouds X_h is *stable* if

$$\|e^{tL_h}\|_{H_h} \leq C\, e^{\omega t} \quad \text{for all } t \geq 0 \text{ and all } h, \tag{2.4}$$

with C, ω independent of h and norms $\|\cdot\|_{H_h}$ uniformly equivalent to $\|\cdot\|_V$ [7]; the fully discrete version replaces e^{tL_h} by powers of the one-step operator. Together with consistency this gives convergence. For a translation-invariant operator on a periodic lattice, L_h is circulant with symbol $\lambda(\theta)$, $\theta \in [-\pi, \pi]^D$, and stability in $\|\cdot\|_V$ with $C = 1$, $\omega = 0$ is the von Neumann condition $\mathrm{Re}\, \lambda(\theta) \leq 0$.

Table 1: Part I: proved results. "Rests on" names every ingredient that is *not* proved here. All results assume the idealisations of [9, Rem. 1.1] (unregularised fits, time step as assumed); the linear results assume linear advection, periodic boundaries, no limiter and no MOOD.

Statement	Notion	Clouds	Status / rests on
Nonlinear, from positivity (Section 3)			
Theorem 3.1: maximum principle, stage-wise MOOD, scalar laws, any order	ℓ^∞	arbitrary	**proved** for the strict DMP ($\delta = 0$, no overrides); does not cover the relaxed DMP with u2 overrides used in the runs
Proposition 3.3: ℓ^1 bounds for Euler	ℓ^1 + invariant domain	arbitrary	**proved**; exact for antisymmetric weights, otherwise up to the conservation defect (a posteriori)
Proposition 3.5 and Corollary 3.7: entropy inequality, first order	entropy	arbitrary	**proved**; needs an upper bound of the Riemann fan; global form up to pair defects
First order, linear advection (Section 4)			
Theorem 4.1: Markov structure	$\ell^p(\pi)$ contraction	arbitrary (positive stationary π)	**proved**; L^2 constant $\sqrt{\rho_h}$, whose uniformity in h is only measured (Observation 11.1)
Proposition 4.3: truncation error	consistency	arbitrary	**proved**; $O(1)$ on irregular stencils
Theorem 4.4: drift, convergence to the wrong speed	convergence	arbitrary periodic	**proved under a numerically checked hypothesis**: rate $h + \eta_h$ with the corrector size $\eta_h = \max \lvert\delta\rvert$, measured $\eta_h \approx a\,h + b\,h^{1/2}$ with small b (Observation 11.2)
Proposition 4.5: moment-corrected dissipation	positivity, consistency	arbitrary	**proved** (existence of the correction needs x_i inside the convex hull of its neighbours)
High order, linear advection (Sections 5 to 8)			
Lemma 5.1: central part skew-symmetric	structure	lattices	**proved**
Theorem 5.2: order two	von Neumann	lattices	**proved (computer-assisted, exact)** in 1-D for $\kappa < 10$ (exact rational arithmetic); **proved** for nearest neighbours in any dimension
Theorem 6.1, Proposition 6.3, and Corollary 6.4: Bloch reduction	Lax–Richtmyer	periodic (supercell repeated)	**proved**; turns stability into a finite-dimensional check, done numerically for given patterns (Observation 12.3)
Theorem 7.3: smoothly deformed lattices	Lax–Richtmyer	$x = \varphi(s)$, constant topology	**proved** given frozen von Neumann stability (D4): proved for $K = 2$ in 1-D (Corollary 7.4), **numerical only** for $K = 3, 4$ (1-D) and $K = 2, 3$ (2-D); assumes the torus form of the Lax–Nirenberg inequality, whose adaptation from $\ell^2(h\mathbb{Z}^D)$ is not written out (Theorem 7.2)
Theorem 8.7 and Corollary 8.8: deformed lattices with localised defects	Lax–Richtmyer up to a residual	1-D, defects separated by σh	**proved under a numerically checked hypothesis** for $\varphi = \mathrm{id}$: local certificates from an SDP, verified a posteriori in floating point at $g = 1$ (Observation 13.2). For $\varphi \neq \mathrm{id}$ the theorem needs certificates for every $g \in [g_-, g_+]$; they are computed only at $g = 0.85, 1, 1.15$, without a bound in between, so the deformed case is a statement given such certificates. Strict dissipativity (D5) checked numerically for $\kappa = 2.5$. Residual-free (Corollary 8.8(i)) only for $\varphi = \mathrm{id}$ and only under an exponential decay of the residuals that no computation can verify and that the computed residuals rule out for $K = 2$ (merge, insert) and $K = 4$ aligned (shift, insert)
Structural and negative results (Section 9)			
Proposition 9.1: energy method in a volume norm fails	numerical abscissa	arbitrary	**proved** as an inequality in the defect d; the rate $\omega_V \geq c\varepsilon^2/h$ for jitter εh needs $\lVert d\rVert_V \gtrsim \varepsilon/h$, which is **numerical only** (Table 9)

Table 2: Part II: numerical results only (no proof). "Settings of the paper": jitter $0.2h$ with $\kappa = 2.5$ or 5.5, as in [10]. Kreiss constants are maxima over sampled times (and Bloch symbols over sampled θ), hence lower bounds of the suprema. Clouds of different sizes are drawn with the same seed (`make_cloud` in `linop.py`), so their first jitter values coincide and the clouds are not independent across N; this matters only for the extreme-value statements of Observation 11.1.

Question	Clouds	Finding
Orders three to five, von Neumann (Observation 10.1)	lattices	stable with the aligned orientation; with the geometric orientation unstable only for $K = 4$ in 2-D at $\kappa = 3.5$ (and marginally 5.5). RK2 is unstable for $K = 4, 5$
Spread ρ_h of π (Observation 11.1)	perturbed	1.21–1.34 in 1-D, 1.6–1.9 in 2-D (growing slowly with n) in the settings of the paper; grows to 16 when some particles have only three neighbours
Drift of the first-order scheme (Observation 11.2)	perturbed	corrector $\max \lvert\delta\rvert/h$ grows from 0.19 to 0.64 for $N = 400$ to 102400 (a factor 3.4 over a 256-fold refinement), roughly as $a + b\sqrt{N}$ with $a \approx 0.16$, $b \approx 1.4 \cdot 10^{-3}$; wave speed 0.1–0.45% too slow at every resolution; the moment correction reduces it to $3 \cdot 10^{-8}$
Spectra, Kreiss constants, $K = 2$–5 (Observations 12.1 and 12.2)	jittered	jitter $0.2h$: $\alpha = 0$ to round-off, $\mathcal{K}_V = 1.002$–1.33, flat in h. Jitter $0.4h$: weak $1/h$ instability for $K = 2$ ($\alpha h \approx 2 \cdot 10^{-6}$), blow-up for $K = 4$ (geometric) on near-interpolatory stencils; $K = 3$ stable
Bloch symbols of jittered supercells (Observation 12.3)	periodic	jitter $0.2h$: stable refinement families for $K = 3, 4$, constants ≤ 1.17; $K = 2$ weakly unstable on one of two patterns ($\alpha h \leq 9 \cdot 10^{-10}$); jitter $0.4h$: genuine weak instability for $K = 2$, failures for $K = 4$
Local symbols (Observation 12.4)	jittered	local diffusion coefficient of $K = 2$ fluctuates with std 0.10–0.18; zero for $K = 3$, as predicted
Weight π, diagonal Lyapunov weights (Observations 12.6 and 12.7)	jittered	π removes the conservation-defect obstruction; the SDP finds no diagonal Lyapunov weight for $K = 3$
Deformed lattices (Observation 13.1)	$x = \varphi(s)$	$\omega_V = O(1)$ and decreasing, $\mathcal{K}_V - 1 < 10^{-4}$ (1-D) and $\leq 5 \cdot 10^{-4}$ (2-D); varying topology gives $\omega_V \propto 1/h$
Certificates, defect arrays, superposition, random defects (Observations 13.2 to 13.5)	lattices with defects	residual consistent with exponential decay for $K = 3$ (all types) and $K = 4$ geometric (down to the solver accuracy); not for $K = 2$ at insert and (beyond $\varrho = 12$) merge defects, whose periodic arrays are weakly unstable, nor at shift defects with $g \neq 1$; not for $K = 4$ aligned at shift and insert defects (weakly unstable arrays; at insert defects the computed residual stalls near 10^{-8}); $K = 3$ stable at every defect density; $K = 4$ fails at density 0.1; 2-D: sum of squares found, certificates not computed

Energy method, spectral quantities. For a positive weight $H = (H_i)$ let $\langle U, W\rangle_H = \sum_i H_i \bar{U}_i W_i$, $L^{*_H} = H^{-1}L^{\mathsf{T}}H$ the adjoint, and

$$\omega_H(L) = \max_{U\neq 0} \frac{\mathrm{Re}\langle U, LU\rangle_H}{\|U\|_H^2} \tag{2.5}$$

the *numerical abscissa*, the largest eigenvalue of the H-symmetric part $\frac{1}{2}(L + L^{*_H})$. The energy method is the inequality $\frac{\mathrm{d}}{\mathrm{d}t}\|U\|_H^2 \le 2\omega_H\|U\|_H^2$, which gives (2.4) with $C = 1$, $\omega = \omega_H$; it proves stability if and only if $\omega_{H_h}(L_h)$ is bounded independently of h. The *spectral abscissa* $\alpha(L) = \max \mathrm{Re}\,\mathrm{spec}\, L$ satisfies $\alpha \le \omega_H$ for every H and $\|e^{tL}\|_H \ge e^{t\alpha}$. The *Kreiss constant* $\mathcal{K}_H(L) = \sup_{t\ge 0}\|e^{tL}\|_H$ (in this note the semigroup bound, or maximal transient growth; in the literature the name often denotes the resolvent quantity $\sup_{\mathrm{Re}\, z>0} \mathrm{Re}\, z\, \|(z-L)^{-1}\|$, which bounds it up to a factor eN) is finite if and only if $\alpha \le 0$ and the imaginary eigenvalues are semisimple; (2.4) with $\omega = 0$ is $\sup_h \mathcal{K}_V(L_h) < \infty$ [24]. These are the quantities measured in Part II.

Remark 2.1 (Stability is a property of a family of clouds). If $X_h = hX_1$ for a fixed finite cloud X_1, then $L_h = L_1/h$ and $\sup_{h,t}\|e^{tL_h}\| = \mathcal{K}(L_1)$: the family is stable if and only if $\alpha(L_1) \le 0$ with semisimple imaginary eigenvalues. A measurement of α and $\mathcal{K}$ on given clouds therefore decides stability for those clouds and their dilations; the open question is always the uniformity over a family (finer and finer clouds that are not dilations of each other). A positive αh that does not decrease with h is an instability with growth rate proportional to $1/h$.

Nonlinear problems. Here we use bounds that the scheme preserves: invariant domains, maximum principles, ℓ^1 bounds and entropy inequalities.

Part I
Proved results

3 Nonlinear stability from positivity (any order, arbitrary clouds)

3.1 A maximum principle for stage-wise MOOD

The proof of [9, Thm. 6.3] uses of PAD only that a flagged particle is lowered until the first-order operator applies, and of the first-order operator only that it maps admissible states to admissible states. For a scalar conservation law the first-order operator satisfies the stronger local-extremum-diminishing property of [9, Cor. 5.3], and the DMP [9, (6.4)] is the matching detection criterion. This gives a maximum principle for the high-order scheme.

Theorem 3.1 (Maximum principle of stage-wise MOOD)**.** *Consider a scalar conservation law, $M = 1$, with smooth fluxes f_d, and let the explicit Runge–Kutta method have $\mathcal{C} > 0$, with MOOD applied by [9, Alg. 1]. Assume:*

(M1) the interface rule is EPD1 or EPD2, or $K = 1$ with any rule;

(M2) the aligned Rusanov flux is used, in either divergence mode, with signs taken from the weights in use [9, Rem. 5.5], and the derivative fits of all interior particles are unregularized on Π_{q_i}-unisolvent neighbourhoods;

(M3) $s_{ij,d} \ge \max\{|f_d'(w)| : w$ between $Y_{k-1,i}$ and $Y_{k-1,j}\}$ at every checkpoint (true for the Rusanov speed $s_{ij,d} = \max\{|f_d'(Y_{k-1,i})|, |f_d'(Y_{k-1,j})|\}$ if every f_d is convex or concave, for instance for linear advection and Burgers' equation);

(M4) the detection consists of the DMP (3.1) below, with $\delta = 0$ and without overrides (and, optionally, PAD);

(M5) the boundary operator maps interior states in $[m, M]$ to ghost states in $[m, M]$, where $m = \min_l U_l^0$ and $M = \max_l U_l^0$ over all particles (for instance periodic boundaries, or Dirichlet data in $[m, M]$);

(M6) at every checkpoint k, every particle i with $k_i = 1$ at the end of the checkpoint satisfies $2\tau \sum_{j,d} s_{ij,d} |c_{ij}^{(d)}| \le 1$ with $\tau = \Delta t / \mathcal{C}$.

Here the DMP with $\delta = 0$ flags $W_{k-1,i}$ if and only if

$$W_{k-1,i} \notin \Big[\min_{l \in \bar{\nu}(i)} Y_{k-1,l},\ \max_{l \in \bar{\nu}(i)} Y_{k-1,l} \Big]. \tag{3.1}$$

Then every checkpoint terminates with $W_{k-1,i}$ in the interval (3.1) for all $i \in I$, and $m \le Y_{k,l} \le M$ and $m \le U_l^n \le M$ for all stages, all steps and all particles, for every distribution of the orders that MOOD produces. In particular $\|U^n\|_\infty \le \|U^0\|_\infty$.

Proof. We follow the induction of [9, Thm. 6.3] and show that $Y_k \in [m, M]$ at all particles and that $W_{k-1,i}$ lies in the interval (3.1). For $k = 1$, $Y_1 = U^n \in [m, M]$ by the induction over the steps (and by (M5) for the ghosts). Let $k \ge 2$, and consider $i \in I$ and the last pass of checkpoint k in which $i \in T$. By exact locality, $W_{k-1,i} = Y_{k-1,i} + \tau \mathcal{L}_i(Y_{k-1})$ with the final orders. If $W_{k-1,i}$ violated (3.1), it would have been flagged in that pass; with $k_i > 1$, i would have been lowered and checked again, contradicting the choice of the pass. Hence $k_i = 1$. By (M1) and [9, Prop. 5.6], $\mathcal{L}_i$ is the first-order operator (2.1), all edges of i are aligned (since $k_i = 1 < K$ or $K = 1$), and by (M2) the weights in use have zero row sums and $\sigma_{ij,d} c_{ij}^{(d)} = |c_{ij}^{(d)}|$. The scalar part of [9, Cor. 5.3], whose hypotheses are (M3) and (M6), gives

$$W_{k-1,i} = Y_{k-1,i} + \tau \sum_{j,d} \big(s_{ij,d} |c_{ij}^{(d)}| - c_{ij}^{(d)} a_{ij,d} \big) (Y_{k-1,j} - Y_{k-1,i})$$

with non-negative coefficients whose sum is at most one, so $W_{k-1,i}$ lies in (3.1), a contradiction. Thus all $W_{k-1,i}$ satisfy (3.1), and since $Y_{k-1} \in [m, M]$ at all particles (interior by induction, ghosts by (M5)), $W_{k-1,i} \in [m, M]$. The substeps $W_1, \dots, W_{k-2}$ are unchanged since their checkpoints. By [9, Lem. 6.1], $Y_{k,i} = v_k U_i^n + \sum_{j<k} \alpha_{kj} W_{j,i}$ is a convex combination of values in $[m, M]$. Ghost values of Y_k lie in $[m, M]$ by (M5). □

Remark 3.2 (Scope). (i) The theorem holds for every design order K, both divergence modes and every cloud on which the fits are unregularized. It is a genuinely nonlinear statement: it bounds the high-order scheme because MOOD makes it nonlinear, and it says nothing about the scheme without MOOD.

(ii) With the relaxed DMP [9, (6.4)], $\delta > 0$, a substep may exceed the local bounds by δ, and particles whose neighbourhood range is below δ^3 are not checked at all. Without the plateau exemption the argument gives $m - n\,s\,\delta \le U^n \le M + n\,s\,\delta$ after n steps of an s-stage method, a bound that grows linearly; with the exemption we have no bound. The u2 overrides for smooth extrema [10] accept values outside the local bounds by design, and the theorem does not cover them. The runs of [10] use both relaxations, so Theorem 3.1 covers a stricter variant of what is run.

(iii) For systems there is no maximum principle; [9, Thm. 6.3] is the corresponding statement for the invariant domain.

(iv) *Time step.* (M6) is twice as strict as the positivity condition [9, (5.1)]. With the time step of the implementation, $\Delta t = \mathrm{CFL} \cdot \min_i \big(D \sum_d \lambda_d \sum_j |\hat{c}_{ij}^{(d)}| \big)^{-1}$ [9, Rem. 6.5(iii)], it holds if $\mathrm{CFL} \le D\,\mathcal{C}/2$ (RK3: CFL ≤ 0.5 in 1-D and ≤ 1 in 2-D; SSPRK(5,4): CFL $\le 0.754\,D$), up to the difference between the weights with and without constant term. Runs with CFL above this bound are not covered even in the strict variant.

3.2 ℓ^1 bounds for the Euler equations

Positivity of density and pressure turns conservation into bounds.

Proposition 3.3 (ℓ^1 bounds from positivity and conservation)**.** *Let the hypotheses of [9, Thm. 6.3] hold for the Euler equations [9, (2.2)] on a periodic cloud (no ghost particles), let $V_i > 0$, and let $\mathcal{M}^n = \sum_i V_i U_i^n = (\mathcal{M}_\rho^n, \mathcal{M}_m^n, \mathcal{M}_E^n)$. Then for all n*

$$\sum_i V_i|\rho_i^n| = \mathcal{M}_\rho^n, \qquad \sum_i V_i|E_i^n| = \mathcal{M}_E^n, \qquad \sum_i V_i|m_i^n| \le \sqrt{2\,\mathcal{M}_\rho^n\,\mathcal{M}_E^n}, \qquad \sum_i V_i \frac{p_i^n}{\gamma-1} \le \mathcal{M}_E^n. \tag{3.2}$$

(i) If the generalized face vectors $S_{ij} = 2V_i c_{ij}$ are antisymmetric, $S_{ij} = -S_{ji}$, and the pairs with $x_{ij,d} = 0$ satisfy the condition after [9, Prop. 4.4], then $\mathcal{M}^n = \mathcal{M}^0$, and (3.2) are uniform bounds in terms of the initial data.

(ii) In general $\mathcal{M}^n = \mathcal{M}^0 + \Delta^n$, where Δ^n is the accumulated second sum of the balance [9, (4.6)], evaluated at the stages with the weights $\Delta t\, b_k$. The bounds (3.2) then hold with $\mathcal{M}_\rho^0 + |\Delta_\rho^n|$ and $\mathcal{M}_E^0 + |\Delta_E^n|$.

Proof. By [9, Thm. 6.3], $\rho_i^n > 0$ and $p_i^n > 0$, hence $E_i^n = p_i^n/(\gamma-1) + |m_i^n|^2/(2\rho_i^n) > 0$, which gives the first two identities and the last inequality, and $|m_i^n| \le \sqrt{2\rho_i^n E_i^n}$. The Cauchy–Schwarz inequality gives $\sum_i V_i\sqrt{2\rho_i E_i} \le \sqrt{2\sum_i V_i\rho_i \sum_i V_i E_i}$. For (i) and (ii), $\mathcal{M}^{n+1} - \mathcal{M}^n = \Delta t \sum_k b_k \sum_i V_i \mathcal{L}_i(Y_k)$ by [9, (6.1)], and $\sum_i V_i\mathcal{L}_i(Y)$ is the right-hand side of [9, (4.6)], whose first sum is empty without ghost particles and whose second sum vanishes under (i) by the discussion after [9, Prop. 4.4]. □

Remark 3.4. (i) Proposition 3.3 is the precise sense in which positivity implies stability for the Euler equations: without positivity, conservation of $\sum_i V_i U_i$ bounds nothing, since positive and negative values may cancel. It gives no ℓ^∞ bound, and density may concentrate; the same holds for positive finite volume schemes.

(ii) In (ii), Δ^n is an a posteriori quantity. An a priori bound would need bounds of the fluxes, i.e. ℓ^∞ bounds, which we do not have. The conservation defects measured in [10] (relative mass defects of order 10^{-4}–10^{-3} in the frozen mode, and up to a few percent in the coupled mode with MOOD) quantify how far the bounds are from the exact ones.

3.3 An entropy inequality for the first-order scheme

For a convex entropy η with entropy fluxes q_d, $\eta'(U)F_d'(U) = q_d'(U)$, the positivity proof can be redone with Riemann averages in place of the states $U_j - \sigma F_d(U_j)/s$ of [9, Prop. 5.2], as for first-order schemes on graphs [5]. For $U_L, U_R \in \mathcal{G}$ and a direction d, let $\lambda_d^{\max}(U_L, U_R)$ be the largest absolute wave speed of the one-dimensional Riemann problem $\partial_t V + \partial_\xi F_d(V) = 0$ with the data U_L for $\xi < 0$ and U_R for $\xi > 0$.

Proposition 3.5 (Local entropy inequality)**.** *Let $\mathcal{G}$ be convex and invariant under the Riemann problems of all directions, let $i \in I$ use piecewise constant states and the aligned orientation on all its edges, with zero-sum weights, and let $U_l \in \mathcal{G}$ for $l \in \bar\nu(i)$. Let*

$$s_{ij,d} \ge \max\{\lambda_d^{\max}(U_i, U_j), \lambda_d^{\max}(U_j, U_i)\} \quad \text{for all } j, d, \qquad 2\tau \sum_{j,d} s_{ij,d}|c_{ij}^{(d)}| \le 1. \tag{3.3}$$

Then $U_i^+ = U_i + \tau\mathcal{L}_i^1(U) \in \mathcal{G}$ and, for every entropy pair (η, q),

$$\eta(U_i^+) \le \eta(U_i) + \tau \sum_{j,d} \Big[s_{ij,d}|c_{ij}^{(d)}|\big(\eta(U_j) - \eta(U_i)\big) - c_{ij}^{(d)}\big(q_d(U_j) - q_d(U_i)\big)\Big]. \tag{3.4}$$

Proof. For $c_{ij}^{(d)} \neq 0$ let $\sigma = \operatorname{sign} c_{ij}^{(d)}$, $s = s_{ij,d}$ and $\bar{U}_{ij,d} = \frac{1}{2}(U_i + U_j) - \sigma\big(F_d(U_j) - F_d(U_i)\big)/(2s)$. If $\sigma = 1$, let V be the solution of the Riemann problem with $U_L = U_i$, $U_R = U_j$; if $\sigma = -1$, with $U_L = U_j$, $U_R = U_i$. Its waves stay in $|\xi| \le st$, so integrating the conservation law over $(-\frac{1}{2}, \frac{1}{2}) \times (0, t^*)$ with $t^* = 1/(2s)$ gives $\int_{-1/2}^{1/2} V(\xi, t^*)\,\mathrm{d}\xi = \bar{U}_{ij,d}$ [8], hence $\bar{U}_{ij,d} \in \mathcal{G}$ by invariance and convexity. Integrating the entropy inequality $\partial_t \eta(V) + \partial_\xi q_d(V) \le 0$ over the same rectangle and applying Jensen's inequality gives

$$\eta(\bar{U}_{ij,d}) \le \int_{-1/2}^{1/2} \eta(V(\xi, t^*))\,\mathrm{d}\xi \le \tfrac{1}{2}\big(\eta(U_i) + \eta(U_j)\big) - \frac{\sigma}{2s}\big(q_d(U_j) - q_d(U_i)\big)$$

in both cases. Next, $2s|c_{ij}^{(d)}|(\bar{U}_{ij,d} - U_i) = s|c_{ij}^{(d)}|(U_j - U_i) - c_{ij}^{(d)}(F_d(U_j) - F_d(U_i))$, because $\sigma|c_{ij}^{(d)}| = c_{ij}^{(d)}$. By (2.1) and the aligned orientation,

$$U_i^+ = \Big(1 - 2\tau \sum_{j,d} s_{ij,d}|c_{ij}^{(d)}|\Big)U_i + \sum_{j,d} 2\tau s_{ij,d}|c_{ij}^{(d)}|\,\bar{U}_{ij,d},$$

a convex combination by (3.3). Hence $U_i^+ \in \mathcal{G}$, and Jensen's inequality with the bound for $\eta(\bar{U}_{ij,d})$ gives (3.4). □

Remark 3.6. (i) Proposition 3.5 needs an upper bound of the maximum wave speed of the Riemann problem, which is a stronger requirement than the hypothesis $s \ge \lambda_d(U_j)$ of [9, Prop. 5.2]. For scalar laws with convex or concave f_d, the Rusanov speed, the larger of $|f_d'(U_i)|$ and $|f_d'(U_j)|$, is such a bound. For the Euler equations the Rusanov speed $\max\{\lambda_d(U_i), \lambda_d(U_j)\}$ is not guaranteed to bound the Riemann fan in general; guaranteed and inexpensive upper bounds are given in [6]. The time step condition carries a factor two compared with [9, (5.1)].

(ii) The right-hand side of (3.4) is the first-order operator applied to the pair (η, q) instead of (U, F). This is a non-conservative cell entropy inequality: summing it with volumes does not telescope.

Corollary 3.7 (Global entropy balance)**.** *Let all interior particles satisfy the hypotheses of Proposition 3.5 on a periodic cloud. Then*

$$\sum_i V_i \eta(U_i^+) \le \sum_i V_i \eta(U_i) - \tau \sum_{\{i,j\}} \sum_{d=1}^{D} \Big[d_{ij,d}\big(q_d(U_i) + q_d(U_j)\big) - s_{ij,d}\, e_{ij,d}\big(\eta(U_j) - \eta(U_i)\big)\Big] \quad (3.5)$$

with the pair defects $d_{ij,d}$, $e_{ij,d}$ of [9, Prop. 4.4]. If the volume-weighted weights are antisymmetric, the total entropy $\sum_i V_i \eta(U_i)$ does not increase, also under every SSP Runge–Kutta method with $\Delta t/\mathcal{C}$ in place of τ. For a linear symmetric hyperbolic system, $F_d(U) = A_d U$ with $A_d = A_d^{\mathsf{T}}$ and $s_{ij,d} \ge \rho(A_d)$, the choice $\eta = \frac{1}{2}|U|^2$ makes the first-order aligned scheme contractive in $\|\cdot\|_V$ on every cloud with antisymmetric volume-weighted weights.

Proof. Multiply (3.4) by V_i and sum. The sum of the right-hand sides is $\tau \sum_i V_i \mathcal{L}_i^1$ evaluated for (η, q), which by [9, Prop. 4.4], applied with F replaced by q and U by η (the proof uses only the form (2.1)), equals the pair sum in (3.5). Antisymmetry gives $d_{ij,d} = e_{ij,d} = 0$. For a Runge–Kutta method with $\mathcal{C} > 0$, every stage is a convex combination of U^n and forward Euler substeps [9, Lem. 6.1], and convexity of η carries the bound over. For linear symmetric systems with $s \ge \rho(A_d)$ the Riemann fan lies in $|\xi| \le \rho(A_d)\,t$, and $(\frac{1}{2}|U|^2, \frac{1}{2}U \cdot A_d U)$ is an entropy pair. □

Without antisymmetry, (3.5) bounds the entropy production by the pair defects, which are $O(h^{D-1})$ per pair on irregular clouds while $V_i = O(h^D)$; summed over the $O(h^{-D})$ pairs this does not give an h-independent bound. For linear advection, Section 4 replaces the volumes by a better weight and removes this obstacle.

4 The first-order scheme on arbitrary clouds (linear advection)

4.1 The first-order operator as a Markov generator

Theorem 4.1 (Markov structure)**.** *Let $a \in \mathbb{R}^D$, let the derivative weights of all particles have zero row sums, and let Q be the first-order aligned operator (2.2) on a periodic cloud.*

(i) $q_{ij} \geq 0$ for $j \neq i$ and $Q\mathbf{1} = 0$: Q is the generator of a continuous-time Markov chain on the particles, e^{tQ} is a stochastic matrix for $t \geq 0$, and $\min_l U_l(0) \leq U_i(t) \leq \max_l U_l(0)$.

(ii) If the directed graph with the edges $i \to j$ for $q_{ij} > 0$ is strongly connected, there is a unique $\pi \in \mathbb{R}^N$ with $\pi_i > 0$, $\sum_i \pi_i = 1$ and $\pi^{\mathsf{T}} Q = 0$, and $\sum_i \pi_i U_i(t)$ is conserved.

(iii) Let $\pi > 0$ be any stationary distribution, $\pi^{\mathsf{T}} Q = 0$ (by (ii) it exists under strong connectivity; on a lattice with a parallel to an axis the chain is not irreducible, but the uniform π is stationary). For $1 \leq p < \infty$ and $\|U\|_{\pi,p}^p = \sum_i \pi_i |U_i|^p$, $\|e^{tQ}U\|_{\pi,p} \leq \|U\|_{\pi,p}$, and

$$\frac{\mathrm{d}}{\mathrm{d}t}\|U\|_{\pi,2}^2 = -\sum_i \sum_{j \in \nu(i)} \pi_i \, q_{ij} \, (U_j - U_i)^2 \leq 0. \tag{4.1}$$

(iv) The forward Euler step $U \mapsto (\mathrm{Id} + \tau Q)U$ with $\tau \max_i \sum_j q_{ij} \leq 1$, and every Runge–Kutta method with $\mathcal{C} > 0$ for which $\tau = \Delta t/\mathcal{C}$ satisfies this condition, are contractive in $\|\cdot\|_{\pi,p}$ for $1 \leq p \leq \infty$.

(v) With $\rho_h = \max_i(\pi_i/V_i)/\min_i(\pi_i/V_i)$, $\|e^{tQ}\|_V \leq \sqrt{\rho_h}$, and the same bound holds for the steps in (iv). Hence the first-order scheme is stable in the sense of (2.4), with $C = \sqrt{\rho}$ and $\omega = 0$, on every family of clouds with $\rho_h \leq \rho$.

Proof. (i) $|a_d||c| - a_d c \geq 0$ for all real c, and $Q\mathbf{1} = 0$ by construction. A matrix with nonnegative off-diagonal entries and zero row sums generates a semigroup of stochastic matrices: $e^{tQ} = e^{-t\mu}e^{t(Q+\mu\mathrm{Id})}$ with $\mu = \max_i \sum_j q_{ij}$ and $Q + \mu\mathrm{Id} \geq 0$ entrywise, and $e^{tQ}\mathbf{1} = \mathbf{1}$. (ii) For an irreducible generator on a finite state space the stationary distribution exists, is unique and positive [20, Ch. 3]; $\frac{\mathrm{d}}{\mathrm{d}t}\pi^{\mathsf{T}}U = \pi^{\mathsf{T}}QU = 0$. (iii) Let $P = e^{tQ}$, which is stochastic with $\pi^{\mathsf{T}}P = \pi^{\mathsf{T}}$. Jensen's inequality for the convex function $|\cdot|^p$ and the probability vector $(P_{ij})_j$ gives $|(PU)_i|^p \leq \sum_j P_{ij}|U_j|^p$; multiplying by π_i and summing gives the claim. For (4.1), expand the right-hand side, $\sum_{i,j} \pi_i q_{ij}(U_j^2 - 2U_iU_j + U_i^2)$, and use $\sum_i \pi_i q_{ij} = \pi_j \sum_k q_{jk}$, which is $\pi^{\mathsf{T}}Q = 0$ written out; the result is $-2\sum_i \pi_i U_i (QU)_i = -\frac{\mathrm{d}}{\mathrm{d}t}\|U\|_{\pi,2}^2$. (iv) Under the time step condition, $\mathrm{Id} + \tau Q$ is stochastic with $\pi^{\mathsf{T}}(\mathrm{Id} + \tau Q) = \pi^{\mathsf{T}}$, and the argument of (iii) applies; for $p = \infty$ stochasticity suffices. Runge–Kutta methods with $\mathcal{C} > 0$ are convex combinations of such steps [9, Lem. 6.1]. (v) $\min_l(\pi_l/V_l)\|U\|_V^2 \leq \|U\|_{\pi,2}^2 \leq \max_l(\pi_l/V_l)\|U\|_V^2$. □

Remark 4.2. (i) Theorem 4.1 is the linear form of the positivity argument: the hypothesis (b) of [9, Prop. 5.2], $\sigma c \geq 0$, is exactly what makes $q_{ij} \geq 0$. With the geometric orientation on perturbed clouds about 3% of the weights have the wrong sign [9, Rem. 5.4], Q is not a generator, and the argument fails. The weights may have any degree, and no antisymmetry is needed; the price is the weight π, which is not the volume.

(ii) On a periodic lattice, Q is circulant, its column sums equal its row sums, and π is uniform: $\rho_h = 1$ with $V_i = h^D$. On perturbed clouds π must be computed. The theory for chains in random environments suggests that ρ_h stays bounded when the cloud is uniformly well conditioned and grows slowly (like an extreme value of local quantities) when nearly degenerate stencils occur with positive probability; this is *not proved* here, and ρ_h is only measured (Observation 11.1).

(iii) The norms $\|\cdot\|_{\pi,p}$ depend on a. For variable coefficients, nonlinear problems and systems there is no common stationary weight; this is where the antisymmetric weights of Corollary 3.7 would be needed.

4.2 Consistency and convergence of the first-order scheme

Stability does not give convergence without consistency, and the first-order scheme is not consistent in the pointwise sense on irregular clouds.

Proposition 4.3 (Truncation error of the first-order operator)**.** *Let [9, Ass. 2.1] hold, let the derivative weights reproduce linear functions [9, Lem. 3.3(i)], and let u be smooth. With the first moments of the dissipation weights,*

$$\beta_i^d = \sum_{j\in\nu(i)} |c_{ij}^{(d)}|\, x_{ij} \in \mathbb{R}^D, \tag{4.2}$$

the first-order operator satisfies

$$(Qu)_i + a\cdot\nabla u(x_i) = \sum_{d=1}^{D} |a_d|\,\beta_i^d\cdot\nabla u(x_i) + r_i, \qquad |r_i| \le C\,h, \tag{4.3}$$

with u evaluated at the particles. On centrally symmetric stencils $\beta_i^d = 0$. On irregular stencils $\beta_i^d = O(1)$, so the truncation error is $O(1)$.

Proof. With $u(x_j) - u(x_i) = x_{ij}\cdot\nabla u(x_i) + \rho_{ij}$, $|\rho_{ij}| \le \frac{1}{2}\|\nabla^2 u\|_\infty |x_{ij}|^2$, and $\sum_j c_{ij}^{(d)} x_{ij} = e_d$, $(Qu)_i = \sum_d |a_d|\beta_i^d\cdot\nabla u(x_i) - a\cdot\nabla u(x_i) + \sum_{j,d}(|a_d||c_{ij}^{(d)}| - a_d c_{ij}^{(d)})\rho_{ij}$, and the last sum is $O(h^{-1})\,O(h^2)$ by [9, Lem. 3.5]. On a centrally symmetric stencil $|c_{ij'}| = |c_{ij}|$ by [9, Lem. 3.7(i)] and $x_{ij'} = -x_{ij}$. □

The β_i^d fluctuate from particle to particle on perturbed lattices. Fluctuations average out, but their π-mean does not, and it changes the transport speed.

Theorem 4.4 (Drift of the first-order scheme)**.** *Let [9, Ass. 2.1] hold on periodic clouds, let Q be irreducible with stationary distribution π, and let $\bar\beta^d = \sum_i \pi_i \beta_i^d$. Let δ^d be the solutions of the corrector equations $Q\delta^d = \beta^d - \bar\beta^d$ (componentwise, $\delta_i^d \in \mathbb{R}^D$, normalized by $\sum_i \pi_i\delta_i^d = 0$; they exist because $\pi^{\mathsf T}(\beta^d - \bar\beta^d) = 0$), and let $\eta_h = \max_{i,d}|\delta_i^d|$ be the* corrector size*. Let u be a smooth periodic solution of*

$$\partial_t u + a_{\mathrm{eff}}\cdot\nabla u = 0, \qquad a_{\mathrm{eff}} = a - \sum_{d=1}^{D} |a_d|\,\bar\beta^d, \tag{4.4}$$

and U the solution of $\mathrm{d}U/\mathrm{d}t = QU$. Then

$$\|U(t) - u(t)\|_{\pi,2} \le \|U(0) - u(0)\|_{\pi,2} + C(1+t)\,(h + \eta_h), \tag{4.5}$$

with C depending on the constants of [9, Ass. 2.1], κ, $|a|$ and u. In particular, if $\eta_h \to 0$, the scheme converges at the rate $h + \eta_h$ to the solution of (4.4), and it converges to the solution of the advection equation with velocity a only if $\sum_d |a_d|\bar\beta^d \to 0$.

Proof. Let $e = U - u$. Since $\partial_t u = -a_{\mathrm{eff}}\cdot\nabla u$, Proposition 4.3 gives $\mathrm{d}e/\mathrm{d}t = Qe + \tau$ with $\tau_i = (Qu)_i + a_{\mathrm{eff}}\cdot\nabla u(x_i) = \sum_d |a_d|(\beta_i^d - \bar\beta^d)\cdot\nabla u(x_i) + r_i$. Let $\psi_i = \sum_d |a_d|\,\delta_i^d\cdot\nabla u(x_i)$. Then

$$(Q\psi)_i = \sum_d |a_d|\Big[(Q\delta^d)_i\cdot\nabla u(x_i) + \sum_j q_{ij}\,\delta_j^d\cdot\big(\nabla u(x_j) - \nabla u(x_i)\big)\Big],$$

and the second term is bounded by $\sum_d |a_d|\,\eta_h\,\|\nabla^2 u\|_\infty \sum_j q_{ij}|x_{ij}| \le C\eta_h$, because $\sum_j q_{ij}|x_{ij}| \le 2|a|\kappa h\sum_{j,d}|c_{ij}^{(d)}| \le C$ by [9, Lem. 3.5]. Hence $\tau = Q\psi + r'$ with $|r_i'| \le C(h+\eta_h)$, and $\tilde e = e + \psi$ satisfies $\mathrm{d}\tilde e/\mathrm{d}t = Q\tilde e + r' + \partial_t\psi$ with $|\partial_t\psi_i| \le \eta_h\,|a|\,\|\nabla\partial_t u\|_\infty$. By Duhamel's formula and Theorem 4.1(iii), $\|\tilde e(t)\|_{\pi,2} \le \|\tilde e(0)\|_{\pi,2} + Ct(h+\eta_h)$, and $\|\psi\|_{\pi,2} \le C\eta_h$ gives (4.5). □

On a lattice $\bar\beta^d = 0$. Whether $\bar\beta^d \neq 0$ on irregular clouds, and how large it is, is a property of the cloud; it is measured in Observation 11.2 (0.1–0.45% of the speed, not decreasing with h), together with the corrector size. Heuristically, δ is a discrete integral of $\beta - \bar\beta$ along the flow; the part of β that is linear in the jitter is a discrete difference and telescopes, the quadratic part does not, so for independent jitter one expects $\eta_h \sim h^{1/2}$ up to logarithms. The measurements are consistent with this but do not show it: they fit $\eta_h \approx a\,h + b\,h^{1/2}$ with a small b, so that the $h^{1/2}$ term dominates only for $N \gtrsim 10^4$ (Observation 11.2). If $\eta_h \sim h^{1/2}$, the theorem gives convergence to the wrong speed at the rate $h^{1/2}$.

Proposition 4.5 (Moment-corrected dissipation)**.** *The defect is in the dissipation, not in the derivative weights. Replace the dissipation coefficients $s_{ij,d}|c_{ij}^{(d)}|$ by $\gamma_{ij}^d = s_{i,d}|c_{ij}^{(d)}| + \epsilon_{ij}^d$ with $\epsilon_{ij}^d \geq 0$ and*

$$\sum_{j\in\nu(i)} \epsilon_{ij}^d\, x_{ij} = -s_{i,d}\,\beta_i^d,$$

where all neighbours use the same dissipation speed $s_{i,d}$ (for linear advection $s_{i,d} = |a_d|$; in general $s_{i,d} = \max_j s_{ij,d}$, which only adds dissipation). Such ϵ_{ij}^d exist whenever $-\beta_i^d$ lies in the cone spanned by the offsets, for instance if x_i lies in the interior of the convex hull of its neighbours. For (iii) the corrections must be $O(1/h)$ uniformly in i, which needs a uniform interior-cone condition (x_i uniformly away from the boundary of the convex hull of its neighbours); existence alone does not give it. Then:

(i) the positivity statement of [9, Prop. 5.2] holds with the time step condition $\tau \sum_{j,d} \gamma_{ij}^d \leq 1$;

(ii) the scalar maximum principle, Theorem 3.1 and Theorem 4.1 hold, with the time step conditions taken for the corrected coefficients: (M6) becomes $2\tau \sum_{j,d} \gamma_{ij}^d \leq 1$, and in Theorem 4.1(iv) $\tau \max_i \sum_j q_{ij} \leq 1$ for the entries $q_{ij} = \sum_d (\gamma_{ij}^d - a_d c_{ij}^{(d)})$ (the uncorrected conditions do not suffice, since $\gamma_{ij}^d \geq s_{i,d}|c_{ij}^{(d)}|$);

(iii) the first moment vanishes, and (4.3) holds with $\beta = 0$: the scheme is consistent of order one.

Proof. (i) $U_i^+ = (1 - \tau\sum\gamma)U_i + \tau\sum_{j,d}\gamma_{ij}^d\big(U_j - c_{ij}^{(d)} F_d(U_j)/\gamma_{ij}^d\big)$ and $\gamma_{ij}^d \geq s_{ij,d}|c_{ij}^{(d)}|$, so these states are admissible by [9, Lem. 5.1] (for $c_{ij}^{(d)} = 0$ they are U_j). (ii) All coefficients stay non-negative, which is all that the proofs use. (iii) The proof of Proposition 4.3 with γ in place of $s|c|$ produces the first moment $\sum_j \gamma_{ij}^d x_{ij} = s_{i,d}\beta_i^d - s_{i,d}\beta_i^d = 0$. □

The correction is not implemented in HyperCloudCore.jl; its numerical effect is in Observation 11.3. It costs one small nonnegative solve per first-order particle, but it makes the dissipation coefficients less symmetric and therefore enlarges the pair defects $e_{ij,d}$ of [9, Prop. 4.4], i.e. the conservation error, exactly where the first-order operator acts under MOOD (near discontinuities). An implementation should report the mass defect with and without it; it is safest as an option for pure first-order runs.

5 High order on lattices

On a periodic Cartesian lattice with spacing h, with all particles at the same order K, the same divergence degree q, unregularized fits and no limiter, L_h is translation invariant. With the splitting (2.3), call an orientation *odd* if $\sigma_{ij',d} = -\sigma_{ij,d}$ whenever $c_{ij}^{(d)} \neq 0$, for the involution $j \mapsto j'$ of [9, Def. 3.6]. The geometric orientation is odd on lattices (pairs with $x_{ij,d} = 0$ have $c_{ij}^{(d)} = 0$, by the argument of [9, Lem. 3.7(i)] for the reflection in the d-th coordinate, under which $c^{(d)}$ is odd), and so is the aligned one, by [9, Lem. 3.7(i)].

Lemma 5.1 (Skew-symmetric central part)**.** *For any K and q and any odd orientation, $C_h^{\mathsf{T}} = -C_h$ and $S_h^{\mathsf{T}} = S_h$. Hence the symbol of C_h is imaginary, that of S_h real, $\operatorname{Re}\lambda(\theta) = \hat{S}(\theta)$, and the semi-discrete scheme is stable, $\|e^{tL_h}\|_V \le 1$, if and only if $S_h \le 0$.*

Proof. Index the particles by $\mathbb{Z}^D$ (modulo the period), $x_i = hi$, and let $(RU)_i = U_{-i}$ be the point reflection. All fits are the same linear maps of the data at the offsets, and by central symmetry they commute with $y \mapsto -y$, as in the proof of [9, Prop. 4.3]; hence the interface states of the reflected data are $U_{ij}[RU] = U_{(-i)(-j)}[U]$. The weights depend only on the offset and are odd, $c^{(d)}_{(-i)(-j)} = -c^{(d)}_{ij}$, by [9, Lem. 3.7(i)]. Therefore

$$(C_hRU)_i = -\sum_{j,d} a_d c^{(d)}_{ij}\big(U_{(-i)(-j)} + U_{(-j)(-i)} - 2U_{-i}\big) = \sum_{j,d} a_d c^{(d)}_{(-i)(-j)}\big(\cdots\big) = -(C_hU)_{-i},$$

i.e. $C_hR = -RC_h$. In terms of the stencil, $(C_hU)_i = \sum_l \beta_l U_{i+l}$, this reads $\beta_{-l} = -\beta_l$. The transpose of a real convolution has the stencil β_{-l}, so $C_h^{\mathsf{T}} = -C_h$. For S_h, the product $\sigma_{ij,d}c^{(d)}_{ij}$ of two odd functions of the offset is even, and the same computation gives $S_hR = RS_h$, an even stencil and $S_h^{\mathsf{T}} = S_h$. A real circulant matrix is skew-symmetric if and only if its symbol is imaginary, and symmetric if and only if it is real; circulant matrices are normal, so $\|e^{tL_h}\|_V = \max_\theta e^{t\operatorname{Re}\lambda(\theta)}$. □

For $K = 1$ the dissipation is $S_hU_i = \sum_{j,d} s_d|c^{(d)}_{ij}|(U_j - U_i)$ with symmetric weights, a negative semidefinite graph Laplacian. For $K \ge 2$ the jumps $U_{ji} - U_{ij}$ are high-order differences, and the sign of S_h is not obvious; with the geometric orientation and $q \ge 3$, some σc are negative [9, Rem. 5.4(i)], and S_h may be indefinite.

Theorem 5.2 (Order two on lattices)**.** *Let $K = 2$ (so $q = 1$, and the geometric and the aligned orientation coincide on lattices), $s_d = |a_d|$, and let the weight function w be positive.*

(i) In one dimension, for every support radius κh with $1 \le \kappa < 10$, $\operatorname{Re}\lambda(\theta) \le 0$ for all θ.

(ii) In D dimensions with $1 \le \kappa < \sqrt{2}$ (neighbours $\pm he_d$ only), $\operatorname{Re}\lambda(\theta) = -\frac{1}{2h}\sum_d |a_d|\,(1 - \cos\theta_d)^2$.

In both cases $\|e^{tL_h}\|_V \le 1$. The proof of (i) is computer-assisted in exact rational arithmetic; the proof of (ii) is by hand.

Proof. On a lattice the moment matrices of the degree-one fits are diagonal by the reflection symmetries, so $c^{(d)}_{ij} = w_{ij}x_{ij,d}/M_d$ with $M_d = \sum_j w_{ij}x^2_{ij,d}$, and the reconstruction slope is $g_{i,d} = \sum_j c^{(d)}_{ij}(U_j - U_i)$; then $U_{ij} = U_i + \frac{1}{2}x_{ij}\cdot g_i$ and $\operatorname{sign} c^{(d)}_{ij} = \operatorname{sign} x_{ij,d}$. For $U_l = e^{i\theta\cdot x_l/h}$ and $h = 1$ one finds $g_i = \mathrm{i}\,S(\theta)U_i$ with $S_d = \sum_j c^{(d)}_{ij}\sin\varphi_j$, $\varphi_j = \theta\cdot x_{ij}$, and, by pairing j with j',

$$\operatorname{Re}\lambda(\theta) = \sum_d |a_d| \sum_j |c^{(d)}_{ij}|\Big[\cos\varphi_j - 1 + \tfrac{1}{2}\,(x_{ij}\cdot S)\sin\varphi_j\Big]. \tag{5.1}$$

(ii) Only $j = \pm e_d$ contribute to direction d, with $|c^{(d)}| = \frac{1}{2}$ and $S_d = \sin\theta_d$, so the bracket is $\cos\theta_d - 1 + \frac{1}{2}\sin^2\theta_d = -\frac{1}{2}(1 - \cos\theta_d)^2$. (i) Let the offsets be $\pm n$, $n = 1, \dots, m$, $m < 10$, and $\mu_n = 2w_n n/M \ge 0$, so that $\sum_n n\mu_n = 1$. Then (5.1) becomes, with $t = \theta/2$,

$$-\frac{\operatorname{Re}\lambda}{|a|} = \Big(\sum_n n\mu_n\Big)\Big(\sum_n \mu_n(1 - \cos 2nt)\Big) - \tfrac{1}{2}\Big(\sum_n n\mu_n \sin 2nt\Big)\Big(\sum_n \mu_n \sin 2nt\Big) = \sum_{n,m} \mu_n\mu_m\, E_{nm}(t),$$

$E_{nm}(t) = n\sin^2 mt + m\sin^2 nt - (n+m)\sin nt\cos nt\,\sin mt\cos mt$.

With $c = \cos t$, $\sin nt = \sin t\, U_{n-1}(c)$ and $\cos nt = T_n(c)$ (Chebyshev polynomials), $E_{nm}(t) = \sin^2 t\, P_{nm}(c)$ with the integer polynomial $P_{nm} = nU^2_{m-1} + mU^2_{n-1} - (n+m)U_{n-1}T_nU_{m-1}T_m$. For $1 \le m \le n \le 9$, $P_{nm} \ge 0$ on $[-1, 1]$, which we verified in exact rational arithmetic (`certify_K2_1d.py`: square-free decomposition and Sturm counting of the roots in $(-1, 1)$ of odd multiplicity). For instance, $E_{12}(t) = 2\sin^2 t\,(6c^2 + 1)(1 - c^2) \ge 0$. Hence all $E_{nm} \ge 0$, and $\operatorname{Re}\lambda \le 0$ for $\mu \ge 0$. □

Remark 5.3. $P_{10,1}$ takes negative values, so the entrywise argument stops at $\kappa = 10$; this does not imply instability, since only the quadratic form on $\mu \geq 0$ matters (a random search, Observation 10.1, found no violation). For $\kappa \geq \sqrt{2}$ in two dimensions, (5.1) couples the directions through $x_{ij} \cdot S$, and we have no proof. Orders $K \geq 3$ on lattices are only treated numerically (Observation 10.1).

6 High order on periodic clouds: Bloch reduction

Let $Y = \{y_1, \dots, y_m\} \subset [0,1)^D$ be a *supercell* with m particles in arbitrary position, and consider the infinite periodic cloud $Y + \mathbb{Z}^D$ with nominal spacing $h_1 = m^{-1/D}$ (any fixed value will do). For $\ell \in \mathbb{N}$ let

$$X_\ell = \tfrac{1}{\ell}\,(Y + \mathbb{Z}^D) \;\text{ on the torus } \mathbb{T}^D = \mathbb{R}^D/\mathbb{Z}^D, \qquad h = h_1/\ell, \tag{6.1}$$

which has $m\ell^D$ particles, indexed by (p,k) with $p \in \{1,\dots,m\}$ and $k \in \mathbb{Z}^D_\ell$. We assume that the support radius is smaller than the period of the infinite cloud, $\kappa h_1 < 1$, and that the stencils of the infinite cloud are unisolvent as required by [A]. By translation invariance of the infinite cloud and homogeneity, the operator of the scheme on $Y + \mathbb{Z}^D$ has the block structure $L^{(1)}_{(p,k),(q,k')} = A(k'-k)_{pq}$ with $m \times m$ blocks $A(k)$, nonzero only for $|k|_\infty \leq k_0$ (two stencil layers, $k_0 \leq \lceil 2\kappa h_1 \rceil$), and

$$(L_\ell)_{(p,k),(q,k')} = \ell\, A(k' - k \bmod \ell)_{pq} \qquad \text{for } \ell > 2k_0. \tag{6.2}$$

The *Bloch symbol* of the cloud is the trigonometric polynomial

$$\hat{A}(\theta) = \sum_{k \in \mathbb{Z}^D} A(k)\, e^{\mathrm{i}\theta\cdot k} \in \mathbb{C}^{m\times m}, \qquad \theta \in \mathbb{T}^D_{2\pi} = [-\pi,\pi]^D. \tag{6.3}$$

Let $V_Y = (V_p)$ be positive volumes of the supercell and $V_{(p,k)} = V_p/\ell^D$ the periodic volumes of X_ℓ; $\|\cdot\|_{V_Y}$ is the weighted norm on $\mathbb{C}^m$.

Theorem 6.1 (Bloch reduction)**.** *Let $\Theta_\ell = \frac{2\pi}{\ell}\{0,\dots,\ell-1\}^D \subset \mathbb{T}^D_{2\pi}$. For $\ell > 2k_0$ and $t \geq 0$,*

$$\|e^{tL_\ell}\|_V = \max_{\theta\in\Theta_\ell} \big\|e^{t\ell\hat{A}(\theta)}\big\|_{V_Y}, \tag{6.4}$$

and therefore

$$\sup_{\ell>2k_0}\ \sup_{t\geq 0} \|e^{tL_\ell}\|_V = \sup_{\theta\in\mathbb{T}^D_{2\pi}}\ \sup_{s\geq 0} \big\|e^{s\hat{A}(\theta)}\big\|_{V_Y} =: \mathcal{K}_Y \in [1,\infty]. \tag{6.5}$$

The refinement family (L_ℓ) is stable in the sense of (2.4), for some C and ω, if and only if $\mathcal{K}_Y < \infty$; in that case (2.4) holds with $C = \mathcal{K}_Y$ and $\omega = 0$.

Proof. The discrete Fourier transform in the cell index, $\hat{U}_p(\theta) = \ell^{-D/2}\sum_{k\in\mathbb{Z}^D_\ell} U_{(p,k)} e^{-\mathrm{i}\theta\cdot k}$, $\theta \in \Theta_\ell$, is unitary from $\ell^2(\mathbb{Z}^D_\ell)$ onto $\ell^2(\Theta_\ell)$ for each p, and since the volumes do not depend on k, $\|U\|_V^2 = \ell^{-D}\sum_{\theta\in\Theta_\ell}\|\hat{U}(\theta)\|^2_{V_Y}$. By (6.2), $\widehat{L_\ell U}(\theta) = \ell\hat{A}(\theta)\hat{U}(\theta)$ (for $\ell > 2k_0$ the sum over $k' - k \bmod \ell$ coincides with the sum over $\mathbb{Z}^D$), hence L_ℓ is unitarily equivalent to $\ell\bigoplus_{\theta\in\Theta_\ell}\hat{A}(\theta)$ in the weighted norms, which gives (6.4).

For (6.5), let $f(\theta) = \sup_{s\geq 0}\|e^{s\hat{A}(\theta)}\|_{V_Y}$. By (6.4) with $s = t\ell$, the left-hand side equals $\sup_\ell \max_{\theta\in\Theta_\ell} f(\theta) = \sup_{\theta\in\Theta} f(\theta)$ with $\Theta = \bigcup_{\ell>2k_0}\Theta_\ell$, which is dense in $\mathbb{T}^D_{2\pi}$. The function f is a supremum of continuous functions of θ ($\hat{A}$ is a polynomial in $e^{\pm\mathrm{i}\theta_d}$), hence lower semicontinuous, and for a lower semicontinuous function the supremum over a dense set equals the supremum over the whole space: if $c < f(\theta_0)$, the open set $\{f > c\}$ contains θ_0 and therefore a point of Θ.

If $\mathcal{K}_Y < \infty$, (2.4) holds with $C = \mathcal{K}_Y$, $\omega = 0$. Conversely, let $\|e^{tL_\ell}\|_V \leq Ce^{\omega t}$ for all $\ell > 2k_0$ and t. If $\theta \in \Theta_{\ell_1}$, then $\theta \in \Theta_{j\ell_1}$ for all $j \in \mathbb{N}$, and (6.4) gives $\|e^{s\hat{A}(\theta)}\| \leq Ce^{\omega s/(j\ell_1)}$ for all s and j, hence $f(\theta) \leq C$. So $f \leq C$ on Θ, and by the density argument on $\mathbb{T}^D_{2\pi}$. □

Remark 6.2. (i) For $m = 1$, $\hat{A}(\theta)$ is the von Neumann symbol $h_1\lambda(\theta)$ of Section 5, and $\mathcal{K}_Y < \infty$ if and only if $\operatorname{Re}\lambda \le 0$ (a normal operator). For $m > 1$ the symbols are not normal, and $\operatorname{Re}\operatorname{spec}\hat{A}(\theta) \le 0$ is necessary but not sufficient for a bound uniform in θ; the quantity that matters is the Kreiss constant of the $m \times m$ family.

(ii) The theorem is a statement about *refinement by repetition*: the irregular pattern inside the cell is fixed and reproduced at every scale, as in a mesh generated by mapping a reference pattern to the cells of a coarse grid. A cloud with fresh random jitter at every resolution is not of this form; but every finite periodic cloud is the $\ell = 1$ member of such a family, and $\hat{A}(\theta)$ for $\theta \neq 0$ describes all its refinements by repetition.

(iii) The weighted norm can be taken as any periodic volume norm, since all are uniformly equivalent on $\mathbb{C}^m$. Stability does not depend on this choice, but the value of $\mathcal{K}_Y$ does.

(iv) Bloch (Floquet) reduction is classical for periodic media [16]; the only point specific to the scheme is the block-circulant structure (6.2), which follows from translation invariance of the infinite cloud and the exact dilation homogeneity of the fits. It holds for every orientation, every divergence mode, both EPD rules with all particles at order K, and also for the first-order operator.

Proposition 6.3 (Sufficient condition)**.** *Suppose that for every $\theta \in \mathbb{T}^D_{2\pi} \setminus \{0\}$ all eigenvalues of $\hat{A}(\theta)$ have negative real part, and that 0 is a simple eigenvalue of $\hat{A}(0)$ (its eigenvector is $\mathbf{1}$) while the other eigenvalues of $\hat{A}(0)$ have negative real part. Then $\mathcal{K}_Y < \infty$.*

Proof. Let $\mathcal{M}_\delta = \{M \in \mathbb{C}^{m\times m} : \alpha(M) \le -\delta\}$. On a compact subset of $\mathcal{M}_\delta$ the resolvent $(z - M)^{-1}$ is continuous on the compact set of (M, z) with z on the boundary of a fixed rectangle containing all spectra in $\operatorname{Re} z \le -\delta/2$, so by the Dunford integral $\|e^{sM}\| \le C_1 e^{-\delta s/2}$ uniformly; hence $\sup_{s\ge 0}\|e^{sM}\| = \max_{s\in[0,S]}\|e^{sM}\|$ for a fixed S, which is a continuous function of M. By continuity of the eigenvalues there is $\delta_0 > 0$ such that for $|\theta| \le \delta_0$ the spectrum of $\hat{A}(\theta)$ consists of one simple eigenvalue $\lambda_0(\theta)$ near 0 and the rest in $\operatorname{Re} z \le -2\delta$ for some $\delta > 0$; λ_0 and its Riesz projector $P(\theta)$ are analytic in θ [12, Ch. II], and $\operatorname{Re}\lambda_0(\theta) \le 0$ by hypothesis ($= 0$ at $\theta = 0$, < 0 otherwise). Then $e^{s\hat{A}(\theta)} = e^{s\lambda_0(\theta)}P(\theta) + e^{s\hat{A}(\theta)}(I - P(\theta))$; the first term is bounded by $\|P(\theta)\| \le C_2$, and the second is $e^{s\hat{A}(\theta)}(I - P(\theta)) = \frac{1}{2\pi\mathrm{i}}\oint_\Gamma e^{sz}(z - \hat{A}(\theta))^{-1}\mathrm{d}z$ over a fixed contour Γ in $\operatorname{Re} z \le -\delta$ enclosing the rest of the spectrum for all $|\theta| \le \delta_0$; the resolvent is continuous on the compact set of (θ, z), so this term is bounded by $C_3 e^{-\delta s}$. For $|\theta| \ge \delta_0$ the spectral abscissa of $\hat{A}(\theta)$ is a continuous negative function on a compact set, so $\hat{A}(\theta) \in \mathcal{M}_{\delta'}$ for some $\delta' > 0$, and the first part applies again. □

Corollary 6.4 (Instability)**.** *If $\operatorname{Re}\lambda > 0$ for an eigenvalue λ of $\hat{A}(\theta_*)$ at some θ_*, then there are $\gamma > 0$ and ℓ_0 such that $\|e^{tL_\ell}\|_V \ge e^{\gamma t\ell} = e^{\gamma h_1 t/h}$ for all $\ell \ge \ell_0$ and $t \ge 0$: the refinement family is unstable with growth rate proportional to $1/h$.*

Proof. By continuity, $\alpha(\hat{A}(\theta)) \ge \gamma > 0$ on a neighbourhood of θ_*, which contains a point of Θ_ℓ for all large ℓ; apply (6.4) and $\|e^{sM}\| \ge e^{s\alpha(M)}$. □

Theorem 6.1 reduces stability on a periodic cloud to a finite-dimensional, θ-dependent check. For a given pattern this check is carried out numerically (sampled θ, floating point) in Observation 12.3; it is not a proof for any specific pattern.

7 High order on smoothly deformed lattices

The second family on which stability can be proved consists of clouds $X_h = \varphi(S_h)$, $S_h = h(\mathbb{Z}^D + \frac{1}{2}) \cap [0,1)^D$, for a fixed diffeomorphism φ of the torus, $\varphi(s) = s + \psi(s)$ with a 1-periodic displacement ψ. Such clouds are irregular (no two stencils are congruent) but *coherent*: the

stencil of particle i depends smoothly on its position. The idea is classical [14, 17]: a scheme with smoothly varying coefficients is stable if its frozen-coefficient symbols are, and the tool is the sharp Gårding inequality for difference operators.

Assumption 7.1. Let $\varphi \in C^3(\mathbb{T}^D;\mathbb{R}^D)$ be a diffeomorphism with $\det D\varphi \geq c_0 > 0$, and $A(s) = D\varphi(s)$.

(D1) *Constant stencil topology.* There is a finite symmetric set $\mathcal{N} \subset \mathbb{Z}^D \setminus \{0\}$ and $\delta_T > 0$ with $|A(s)l| \leq \kappa - \delta_T$ for $l \in \mathcal{N}$ and $|A(s)l| \geq \kappa + \delta_T$ for $l \notin \mathcal{N}$, for all s.

(D2) *Uniform unisolvency.* For every s the stencil $\{A(s)l : l \in \mathcal{N}\}$ is Π_{K-1}- and Π_q-unisolvent.

(D3) *Constant orientation signs.* For every $l \in \mathcal{N}$ and d, either the orientation $\sigma_{i,i+l,d}$ is the same for all particles i and all h, or the frozen dissipation coefficient of the pair vanishes identically, $\gamma_l^{(d)}(s) = 0$ for all s (as for pairs with $x_{ij,d} \equiv 0$ under separable deformations). For the geometric orientation the first alternative holds if $(A(s)l)_d \neq 0$ for all s.

(D4) *Frozen stability.* For every s the scheme on the affine lattice $A(s)\mathbb{Z}^D$ (with the neighbour set $\mathcal{N}$, weights $w(|A(s)l|/\kappa)$ and the orientation signs of (D3)) has a von Neumann symbol $\mu_s(\theta)$ with $\operatorname{Re}\mu_s(\theta) \leq 0$ for all $\theta \in \mathbb{T}^D_{2\pi}$.

(D1)–(D3) are geometric conditions on φ. (D4) is a condition on the scheme: it is *proved* for $K = 2$ in one dimension (Corollary 7.4) and only *checked numerically* otherwise (Observation 13.1).

By (D1) and $x_{i,i+l} = \varphi(s_i + hl) - \varphi(s_i) = h\int_0^1 A(s_i + \tau hl)l\,\mathrm{d}\tau$, there is h_0 such that for $h \leq h_0$ every particle has the neighbours $\{i + l : l \in \mathcal{N}\}$. The frozen operator of (D4) is the operator of the scheme on a centrally symmetric lattice, so Lemma 5.1 applies to it: its central part is skew-symmetric and its dissipative part symmetric, $\operatorname{Re}\mu_s(\theta)$ is the symbol of the dissipative part, and $\operatorname{Re}\mu_s(\theta) = \sum_l \beta_l^{\mathrm{d}}(s)\cos(\theta\cdot l)$ with real, even coefficients $\beta^{\mathrm{d}}_{-l} = \beta^{\mathrm{d}}_l$, while the central coefficients are odd, $\beta^{\mathrm{c}}_{-l} = -\beta^{\mathrm{c}}_l$. In one dimension $A(s) = \varphi'(s)$ is a scalar and the affine lattice is a rescaled lattice; (D4) is then the lattice condition of Section 5 for the weights $w(\varphi'(s)|l|/\kappa)$.

We need the sharp Gårding inequality for difference operators with smooth coefficients.

Theorem 7.2 (Lax–Nirenberg [17], see also [25])**.** *Let $P_h = \sum_{l\in F} b_l(s)\,T_h^l$, $F \subset \mathbb{Z}^D$ finite, $(T_h^l u)(s) = u(s + hl)$, act on grid functions on S_h, with coefficients $b_l \in C^2(\mathbb{T}^D)$ (complex, or matrix valued) and symbol $p(s,\theta) = \sum_l b_l(s)e^{\mathrm{i}\theta\cdot l}$. If $\operatorname{Re}p(s,\theta) \geq 0$ for all s,θ (in the matrix case: the Hermitian part), then*

$$\operatorname{Re}\langle u, P_h u\rangle_{\ell^2} \;\geq\; -K_{\mathrm{LN}}\,h\,\|u\|^2_{\ell^2} \qquad \text{for all } u, \tag{7.1}$$

with K_{LN} depending only on F and the C^2 norms of the b_l.

The theorem is stated in [17, 25] for $\ell^2(h\mathbb{Z}^D)$; the proofs (Friedrichs symmetrization with a smooth, compactly supported kernel) are local and carry over to periodic coefficients and periodic grid functions on the torus with the same constant. We use it in this form without writing out the adaptation. (In one dimension, Remark 8.3 gives an elementary replacement under an extra hypothesis.)

Theorem 7.3 (Stability on smoothly deformed lattices)**.** *Under Assumption 7.1, with the volumes $V_i = \det A(s_i)\,h^D$, there are $h_0 > 0$ and $\omega \geq 0$, independent of h, such that*

$$\|e^{tL_h}\|_V \leq e^{\omega t} \qquad \text{for all } t \geq 0,\ h \leq h_0. \tag{7.2}$$

Proof. Step 1: structure of the coefficients. For $h \leq h_0$ the stencils are translates of $\mathcal{N}$ in the index lattice, so $L_h = h^{-1}\sum_{l\in\mathcal{N}_2} B_l(s_i;h)\,T_h^l$ with $\mathcal{N}_2 = (\mathcal{N}\cup\{0\}) + (\mathcal{N}\cup\{0\})$ (the reconstruction of a neighbour j involves the neighbours of j). The coefficient $hL_{i,i+l}$ is a fixed smooth function Φ_l of the scaled offsets $x_{i,i+l'}/h$ and $x_{i+l',i+l'+l''}/h$, $l', l'' \in \mathcal{N}$ (the fits are rational functions of the

offsets and of the weights $w(|x|/\kappa h)$, smooth wherever the scaled moment matrices are invertible, which by (D2) and compactness holds on a uniform neighbourhood of the frozen configurations; the orientation signs are constant by (D3)). The scaled offsets are $\Lambda_{l'}(s_i;h) = \int_0^1 A(s_i+\tau h l')l'\,\mathrm{d}\tau$, which are C^2 functions of s_i with C^2 norms bounded uniformly in h (because $\varphi \in C^3$) and satisfy $\Lambda_{l'}(s;h) = A(s)l' + O(h)$. Hence $B_l(\cdot;h) \in C^2(\mathbb{T}^D)$ uniformly in h, and

$$B_l(s;h) = \beta_l(s) + h\,B'_l(s;h), \qquad \sup_h \|B'_l(\cdot;h)\|_{C^0} < \infty, \tag{7.3}$$

where $\beta_l(s) = \Phi_l(A(s)l', A(s)l'')$ are the coefficients of the frozen operator of (D4); $\beta_l \in C^2$. The pairs of (D3) with identically vanishing frozen dissipation contribute coefficients that are $O(h)$ by the same expansion and are included in hB'_l. The splitting (2.3) induces $B_l = B_l^{\mathrm{c}} + B_l^{\mathrm{d}}$ and $\beta_l = \beta_l^{\mathrm{c}} + \beta_l^{\mathrm{d}}$ with the parities stated above.

Step 2: the central part. Write $L_h = h^{-1}(P_h^{\mathrm{c}} + P_h^{\mathrm{d}})$ accordingly. The V-adjoint of $P_h^{\mathrm{c}} = \sum_l B_l^{\mathrm{c}}(s_i;h)T^l$ is $\sum_l \frac{V_{i+l}}{V_i}\overline{B_{-l}^{\mathrm{c}}(s_{i+l};h)}\,T^l$, so the coefficients of $P_h^{\mathrm{c}} + (P_h^{\mathrm{c}})^{*_V}$ are

$$B_l^{\mathrm{c}}(s_i;h) + \frac{\det A(s_i+hl)}{\det A(s_i)}\,B_{-l}^{\mathrm{c}}(s_i+hl;h) = \beta_l^{\mathrm{c}}(s_i) + \beta_{-l}^{\mathrm{c}}(s_i) + O(h) = O(h),$$

using (7.3), the Lipschitz continuity of β_{-l}^{c} and of $\det A$, and $\beta_{-l}^{\mathrm{c}} = -\beta_l^{\mathrm{c}}$. A difference operator with finitely many $O(h)$ coefficients has V-norm $O(h)$, hence $\mathrm{Re}\langle U, P_h^{\mathrm{c}}U\rangle_V \le C_1 h\|U\|_V^2$.

Step 3: the dissipative part. By (7.3), $P_h^{\mathrm{d}} = P_0^{\mathrm{d}} + hP_1^{\mathrm{d}}$ with $P_0^{\mathrm{d}} = \sum_l \beta_l^{\mathrm{d}}(s_i)T^l$ and $\|P_1^{\mathrm{d}}\|_V \le C_2$. Further, $\langle U, P_0^{\mathrm{d}}U\rangle_V = \langle u, \tilde{P}u\rangle_{\ell^2}$ with $u = V^{1/2}U$ and $\tilde{P} = V^{1/2}P_0^{\mathrm{d}}V^{-1/2} = \sum_l \big(\det A(s_i)/\det A(s_i+hl)\big)^{1/2}\beta_l^{\mathrm{d}}(s_i)T^l = P_0^{\mathrm{d}} + O(h)$ in operator norm. The operator $-P_0^{\mathrm{d}}$ has C^2 coefficients and the real symbol $-\sum_l \beta_l^{\mathrm{d}}(s)\cos(\theta\cdot l) = -\mathrm{Re}\,\mu_s(\theta) \ge 0$ by (D4). Theorem 7.2 gives $\mathrm{Re}\langle u, P_0^{\mathrm{d}}u\rangle_{\ell^2} \le K_{\mathrm{LN}}h\|u\|_{\ell^2}^2$, and therefore $\mathrm{Re}\langle U, P_h^{\mathrm{d}}U\rangle_V \le (K_{\mathrm{LN}} + C_2 + C_3)h\|U\|_V^2$.

Step 4. Together, $\mathrm{Re}\langle U, L_hU\rangle_V \le \omega\|U\|_V^2$ with $\omega = C_1 + K_{\mathrm{LN}} + C_2 + C_3$, so $\frac{\mathrm{d}}{\mathrm{d}t}\|U\|_V^2 \le 2\omega\|U\|_V^2$ and (7.2) follows. □

Corollary 7.4 (Order two in one dimension)**.** *Let $D = 1$, $K = 2$, w positive, $\kappa \ge 1$, and let $\varphi \in C^3(\mathbb{T})$ be a diffeomorphism such that the number of neighbours on each side, $\lfloor \kappa/\varphi'(s)\rfloor$, is the same for all s and lies in $\{1,\dots,9\}$ (and $\kappa/\varphi'(s)$ stays away from integers). Then (7.2) holds.*

Proof. (D1) is the hypothesis, (D2) holds for any two distinct points, (D3) holds for the geometric orientation since $x_{i,i+l}$ has the sign of l, and also for the aligned one: in one dimension a degree-one fit with constant term has $c_{i,i+l} = w_{i,i+l}(x_{i,i+l} - \bar{x}_i)/\sum_j w_{ij}(x_{ij} - \bar{x}_i)^2$ with the weighted mean offset $\bar{x}_i = O(h^2)$ on a C^2 deformation, so $\operatorname{sign} c_{i,i+l} = \operatorname{sign} l$ for $h \le h_0$. (D4) is Theorem 5.2(i), whose proof is valid for every set of positive weights $\mu_n \ge 0$ and offsets $\pm n$, $n \le m \le 9$, hence for the frozen lattice $\varphi'(s)\mathbb{Z}$, which has $m = \lfloor \kappa/\varphi'(s)\rfloor \le 9$ neighbours per side by hypothesis (for $\varphi' < 1$ this can exceed $\lfloor\kappa\rfloor$), with weights $w(\varphi'(s)n/\kappa)$: the symbol of the rescaled lattice is the symbol of the unit lattice divided by $\varphi'(s)$. □

Remark 7.5. (i) *What the theorem does and does not say.* It gives (2.4) with $C = 1$ and an ω that depends on the C^3 norm of φ; it does not give $\omega = 0$, and indeed $\omega_V(L_h)$ is positive (Observation 13.1). The mechanism is exactly the one that fails on jittered clouds: there, the coefficients are not Lipschitz on the scale h, the commutator terms of Step 2 are $O(1)$ instead of $O(h)$, and Proposition 9.1 shows that no volume weight can repair this.

(ii) *Frozen instabilities.* The exceptions found on lattices ($K = 4$, geometric orientation, $\kappa = 3.5$ in 2-D, Observation 10.1) violate (D4), and the theorem does not apply; Corollary 6.4 shows that any periodic refinement of such a lattice is unstable.

(iii) *Fully discrete.* The theorem is semi-discrete. Semi-boundedness of $L_h - \omega I$ transfers to explicit Runge–Kutta methods of order three and four under a CFL condition $\Delta t\|L_h\| \le c$ by the energy arguments of Levy and Tadmor [18, 22]; we do not pursue this.

(iv) *Varying topology.* If φ' crosses a value where the neighbour count changes, the coefficients jump and (D1) fails; then ω_V is measured to grow like $1/h$ (Observation 13.1).

8 High order on deformed lattices with localised defects (1-D)

The clouds produced by particle management (insertion into holes, merging of close pairs) are not jittered lattices: away from the places where a particle was inserted or removed they are smooth deformations of a lattice, and the irregularity is concentrated in *defects* of bounded size. This section proves stability for such clouds in one dimension, given local certificates. The mechanism: the dissipation form of the lattice is written as a sum of *local* nonnegative squares (Fejér–Riesz), each defect is handled by a finite semidefinite programme that produces a local, non-diagonal Lyapunov weight, and the local certificates of different defects superpose because the squares are local. The obstruction of Proposition 9.1 is circumvented, not contradicted: the weight is a volume weight only away from the defects.

What is proved and what is computed. Lemmas 8.1, 8.2 and 8.4 and Theorem 8.7 and Corollary 8.8 are proved. The *inputs* of Theorem 8.7, the certificates of Definition 8.5, are computed by a semidefinite programme in floating point and verified a posteriori by floating-point eigenvalues (Observation 13.2). Six of them (merge, insert and shift at $K = 3$, $\varrho = 12$; shift and insert at $K = 2$ and merge at $K = 4$, $\varrho = 16$) were also checked in exact rational arithmetic (`exact_cert.py`, `data/exact_cert_results.txt`): the 50-digit $\lambda_{\max}$ agrees with the float residual to 5–10 digits and $tI - E|_{Z'} \succ 0$, $W_Z - c_W g I \succ 0$ hold exactly. Two gaps remain for a fully rigorous statement: with a floating-point Fejér–Riesz factor b, E has entries of 10^{-15} outside $Z' \times Z'$, which have to be bounded or removed with an exact factor, and the remaining certificates have not been checked exactly. Hypothesis (D5) below is likewise checked numerically for $\kappa = 2.5$.

Throughout this section $D = 1$ unless said otherwise and the operator is written in the units $h = 1$ (Remark 2.1): $L = hL_h$, and a weight W stands for $W_h = hW$. The frozen lattice of spacing g is $g(\mathbb{Z} + \frac{1}{2})$ with the neighbourhood radius κ; its operator is $L^{(g)} = g^{-1}L[\kappa/g]$ by dilation, where $L[\kappa']$ is the operator of the unit lattice with radius κ', and its row is $\ell_g(k) = L^{(g)}_{i,i+k}$, $|k| \le d$. We assume (D1)–(D4) of Assumption 7.1 for $g \in [g_-, g_+]$ (constant topology $\mathcal{N} = \{\pm 1, \dots, \pm d_{\mathcal{N}}\}$, unisolvency, constant signs, $\operatorname{Re}\mu_g \le 0$) and in addition

(D5) *Strict dissipativity off* $\theta = 0$. With $m \ge 1$ the order of vanishing of $\operatorname{Re}\mu_g$ at $\theta = 0$ ($\operatorname{Re}\mu_g(\theta) = O(\theta^{2m})$, m given by Proposition 9.3, the same for all g), there is $c_0 > 0$ with $-\operatorname{Re}\mu_g(\theta) \ge c_0(1 - \cos\theta)^m$ for all θ and all $g \in [g_-, g_+]$.

By Proposition 9.3 applied to the lattice, $\operatorname{Re}\mu_g(\theta) = O(\theta^{2m})$ with $2m = K + 2$ for even K and $2m = K + 1$ for odd K, so (D5) asks that the leading dissipation coefficient does not vanish and that there is no other zero of $\operatorname{Re}\mu_g$ on $[-\pi, \pi]$.

8.1 The lattice dissipation form as a sum of local squares

Lemma 8.1 (Local squares)**.** *Under (D4)–(D5) there is, for every $g \in [g_-, g_+]$, a real polynomial $b_g(z) = \sum_{k=0}^{d} b_k(g) z^k$ with*

$$-g\big(\ell_g(k) + \ell_g(-k)\big) = \sum_{k'} b_{k'}(g)\, b_{k'+k}(g) \quad (|k| \le d), \qquad \textit{i.e.} \qquad -2g \operatorname{Re}\mu_g(\theta) = |b_g(e^{\mathrm{i}\theta})|^2, \tag{8.1}$$

and $g \mapsto b_g$ is Lipschitz on $[g_-, g_+]$. Consequently, for the periodic lattice of spacing g with the volume weight $V = gI$, every grid function u satisfies

$$-u^{\mathsf{T}}\big(VL^{(g)} + (L^{(g)})^{\mathsf{T}}V\big)u = \sum_j B_j^{(g)}(u), \qquad B_j^{(g)}(u) = \Big(\sum_{k=0}^{d} b_k(g)\, u_{j+k}\Big)^2 \ge 0. \tag{8.2}$$

Proof. $f_g(\theta) = -2g\,\mathrm{Re}\,\mu_g(\theta) = -g\sum_k(\ell_g(k)+\ell_g(-k))e^{\mathrm{i}k\theta}$ is a real, even trigonometric polynomial of degree d, nonnegative by (D4). By (D5), $f_g(\theta) = (1-\cos\theta)^m q_g(\theta)$ with a trigonometric polynomial $q_g \geq c_0 > 0$ of degree $d-m$ (division of the cosine polynomial f_g, as a polynomial in $x = \cos\theta$, by $(1-x)^m$). The Fejér–Riesz theorem [4, Sec. 1.12] gives $q_g = |r_g(e^{\mathrm{i}\theta})|^2$ with r_g a real polynomial of degree $d-m$ without zeros in the closed unit disc (the outer factor), and $1-\cos\theta = \frac{1}{2}|1-e^{\mathrm{i}\theta}|^2$, so $b_g(z) = 2^{-m/2}(1-z)^m r_g(z)$. The outer factor of a strictly positive trigonometric polynomial is a real-analytic function of its coefficients ($r_g = \exp(\frac{1}{2}P_+\log q_g)$ with the Riesz projection P_+), and the coefficients of q_g are Lipschitz in g because ℓ_g is (the fits are smooth functions of the scaled offsets gk and of the weights $w(gk/\kappa)$, cf. Step 1 of the proof of Theorem 7.3), whence the Lipschitz continuity of b_g. Expanding $|b_g|^2$ gives the first identity in (8.1), and summing $B_j^{(g)}$ over j collects exactly the coefficients of the periodic Toeplitz form $-(VL^{(g)} + (L^{(g)})^{\mathsf{T}}V)$, whose $(i, i+k)$ entry is $-g(\ell_g(k)+\ell_g(-k))$. □

Lemma 8.2 (Deformed lattices, local form)**.** *Let $\varphi \in C^2(\mathbb{T})$ satisfy (D1)–(D5) with $g(s) = \varphi'(s) \in [g_-, g_+]$, let $X_h = \varphi(S_h)$ and $V_i = g(s_i)h$. Then there are C and h_0 such that for $h \leq h_0$ and all u,*

$$u^{\mathsf{T}}\big(VL_h + L_h^{\mathsf{T}}V\big)u \;\leq\; -\sum_j B_j^{(g_j)}(u) + C\,h\,|u|^2, \qquad g_j = g(s_j), \tag{8.3}$$

and more precisely the matrix $R = VL_h + L_h^{\mathsf{T}}V + \sum_j B_j^{(g_j)}$ has bandwidth $2d$ and entries $|R_{ij}| \leq Ch$.

Proof. By Step 1 of the proof of Theorem 7.3, $hL_{i,i+k} = \Phi_k$(scaled offsets of i and its neighbours) with Φ_k smooth and the scaled offsets equal to $g_i k' + O(h)$, so $hL_{i,i+k} = \ell_{g_i}(k) + O(h)$ with $O(h)$ uniform (here $\varphi \in C^2$ suffices since only the first-order expansion of the offsets is used). Hence $(VL_h + L_h^{\mathsf{T}}V)_{i,i+k} = g_i\ell_{g_i}(k) + g_{i+k}\ell_{g_{i+k}}(-k) + O(h) = g_i(\ell_{g_i}(k)+\ell_{g_i}(-k)) + O(h)$ by the Lipschitz continuity of $g \mapsto g\ell_g$. On the other hand $(\sum_j B_j^{(g_j)})_{i,i+k} = \sum_{k'} b_{k'}(g_{i-k'})b_{k'+k}(g_{i-k'}) = \sum_{k'} b_{k'}(g_i)b_{k'+k}(g_i) + O(h)$ by Lemma 8.1, which by (8.1) equals $-g_i(\ell_{g_i}(k)+\ell_{g_i}(-k)) + O(h)$. The sum R therefore has $O(h)$ entries and bandwidth $2d$, and $|u^{\mathsf{T}}Ru| \leq (4d+1)\max|R_{ij}|\,|u|^2$. □

Remark 8.3. Since $\sum_j B_j \geq 0$, Lemma 8.2 gives $\mathrm{Re}\langle U, L_hU\rangle_V \leq \frac{C}{2g_-}\|U\|_V^2$: an elementary proof of Theorem 7.3 in one dimension, with $\varphi \in C^2$ instead of C^3 and without the sharp Gårding inequality, at the price of hypothesis (D5). What it adds is the *local* form of the dissipation, which is what the defects need.

8.2 Local certificates

A *defect type* τ is a finite modification of the lattice inside a window: a set $Y_\tau \subset \mathbb{R}$ which differs from $\mathbb{Z}+\frac{1}{2}$ in finitely many points and coincides with it outside the window $(z-\rho_0, z+\rho_0)$ around the defect centre z ($z = 0$ for merge and shift, $z = 1$ for insert). The three types used below are

$$\text{merge: } Y = (\mathbb{Z}+\tfrac{1}{2})\backslash\{\tfrac{1}{2}\}, \qquad \text{insert: } Y = (\mathbb{Z}+\tfrac{1}{2})\cup\{1\}, \qquad \text{shift: } Y = (\mathbb{Z}+\tfrac{1}{2})\backslash\{\tfrac{1}{2}\}\cup\{\tfrac{1}{2}+\delta\}\ (\delta = 0.3),$$

with $\rho_0 = 1$; merging two particles into one at their midpoint is, up to a shift of the lattice, the merge type. The *defect cloud* of type τ at spacing g is gY_τ (radius κ); its operator $L_{\tau,g}$ coincides, in every row at distance $> (\rho_0 + 2\kappa)g$ from the centre $z = 0$ (merge, shift) or $z = 1$ (insert), with the lattice operator $L^{(g)}$, because a row depends on the cloud within distance 2κ [9, (4.4)], i.e. within $2\kappa/g$ spacings. *Units.* The support radius is κ in physical units (as in `ho_defect_cert.py`); to keep the zone constants in units of the spacing, κ in the zone radii, separations and distances of this section stands for κ/g_-, the radius in spacings ($d = 4$ for $\kappa = 2.5$, $g \in [0.85, 1.15]$; no relation between d and κ is needed). For a *zone radius* $\varrho \geq \rho_0$ let

$$\begin{aligned} Z &= \{y \in gY_\tau : |y-z| \leq \varrho g\}, \qquad Z' = \{y \in gY_\tau : |y-z| \leq (\varrho + d + 4\kappa)g\}, \\ \mathcal{J}_{\mathrm{far}} &= \{j \in g(\mathbb{Z}+\tfrac{1}{2}) :\ \{j, j+g, \dots, j+dg\} \cap Z = \emptyset\}. \end{aligned} \tag{8.4}$$

For a symmetric matrix W with $W_{yy'} = g\delta_{yy'}$ whenever $y \notin Z$ or $y' \notin Z$ ("$W = gI$ outside Z") define

$$E_\tau(W) = WL_{\tau,g} + L_{\tau,g}^{\mathsf{T}} W + \sum_{j \in \mathcal{J}_{\mathrm{far}}} B_j^{(g)}, \tag{8.5}$$

where $B_j^{(g)}$ denotes the matrix of the square in (8.2), supported on $\{j, \dots, j+dg\} \subset gY_\tau$ (all these sites are lattice sites outside Z, hence particles).

Lemma 8.4 (Support)**.** *$E_\tau(W)_{yy'} = 0$ unless $y, y' \in Z'$, and $E_\tau(W)$ is affine in the block W_Z.*

Proof. Let $y \notin Z'$, so $|y - z| > (\varrho + d + 4\kappa)g$. The entry (y, y') of $WL + L^{\mathsf{T}}W$ vanishes unless $|y - y'| \le 2\kappa g$; for such y', both y and y' are outside Z (so $W = gI$ there) and at distance $> (\rho_0 + 2\kappa)g$ from z, so both rows are lattice rows and the entry equals $g(\ell_g(k) + \ell_g(-k))$, $y' = y + kg$. The squares containing y have $j \in \{y - dg, \dots, y\}$, so their sites $j, j+g, \dots, j+dg$ lie within dg of y, hence at distance $> (\varrho + 4\kappa)g \ge \varrho g$ from z and outside Z: these squares are all far, and their contribution to the entry (y, y') is the full lattice sum $\sum_{k'} b_{k'} b_{k'+k} = -g(\ell_g(k) + \ell_g(-k))$ by (8.1). The two cancel. Affinity in W_Z is clear. □

Definition 8.5 (Local certificate)**.** A *certificate* of type τ, spacing g, zone radius ϱ and residual $\mathrm{res} \ge 0$ is a symmetric block $W_Z \in \mathbb{R}^{Z \times Z}$ with

$$c_W\, gI \preceq W_Z \preceq C_W\, gI, \qquad E_\tau(W)\big|_{Z' \times Z'} \preceq \mathrm{res}\, I, \tag{8.6}$$

for the extension W of W_Z by gI. The smallest residual $\mathrm{res}_\tau(g, \varrho)$ for given (c_W, C_W) is the optimal value of a semidefinite programme in the $\frac{1}{2}|Z|(|Z|+1)$ entries of W_Z.

Proposition 8.6 (The residual cannot be zero)**.** *If $L_{\tau,g}$ has no bounded left null vector that is constant outside Z, then $\mathrm{res}_\tau(g, \varrho) > 0$.*

Proof. $\mathbf{1}^{\mathsf{T}} E_\tau(W) \mathbf{1} = 2\,\mathbf{1}^{\mathsf{T}} WL\mathbf{1} + \sum_{\mathrm{far}} B_j(\mathbf{1}) = 0$ because $L\mathbf{1} = 0$ and $b_g(1) = 0$ ($m \ge 1$). (On the infinite cloud this is read as follows: the left side is the finite sum $\sum_{i,j \in Z'} E_{ij}$, since $E_\tau(W)$ is supported on $Z' \times Z'$. Testing with a finitely supported u equal to 1 on a neighbourhood of Z' wider than the stencil gives $u^{\mathsf{T}} E_\tau(W) u = \sum_{i,j \in Z'} E_{ij}$; on the other side Lu and $B_j(u)$ vanish wherever u is locally constant, and near the edge of the support of u the terms add up to the lattice identity (8.2), so the sum is 0.) Hence $E \preceq 0$ forces $E\mathbf{1} = 0$, i.e. $L^{\mathsf{T}} W \mathbf{1} = 0$, i.e. $W\mathbf{1}$ is a left null vector (Proposition 9.1 in local form), and it equals g outside Z. The optimum of the SDP is attained, since the feasible set $c_W gI \preceq W_Z \preceq C_W gI$ is compact, so $\mathrm{res}_\tau = 0$ would produce such a vector. □

Away from the defect, $\pi^{\mathsf{T}} L = 0$ is a linear recurrence with constant coefficients, so $\pi - \bar{\pi}$ decays exponentially but generically does not vanish; this is confirmed numerically (Observation 13.2). The best one can expect is therefore $\mathrm{res}_\tau(g, \varrho) \to 0$ exponentially in ϱ.

8.3 Superposition

Theorem 8.7 (Lattices with defects)**.** *Let $D = 1$, let $\varphi \in C^2(\mathbb{T})$ satisfy (D1)–(D5) with $g = \varphi' \in [g_-, g_+]$, and let $\mathcal{T}$ be a finite set of defect types with certificates $(W_{\tau,g}, \mathrm{res}_\tau(g))$ of zone radius ϱ for all $\tau \in \mathcal{T}$, $g \in [g_-, g_+]$, with common bounds (c_W, C_W) and $\mathrm{res}^* = \sup_{\tau,g} \mathrm{res}_\tau(g)$. Let S_h be obtained from the lattice $h(\mathbb{Z} + \frac{1}{2})$ by placing defects of types $\tau_A \in \mathcal{T}$ at centres z_A, $A \in \mathcal{A}_h$, with*

$$|z_A - z_B| \;\ge\; \sigma h, \qquad \sigma = 2\varrho + 2d + 12\kappa + 2 \qquad (A \ne B), \tag{8.7}$$

and let $X_h = \varphi(S_h)$. Then there are h_0 and C, depending on φ, κ, K, ϱ and $\mathcal{T}$ but not on h and not on the number or the positions of the defects, such that for $h \le h_0$ the weight W_h equal to

$g(s_i)h$ outside the zones and to $h\,W_{\tau_A,g(z_A)}$ on the zone of A satisfies

$$\min(c_W,1)\,g_-h \le W_h \le \max(C_W,1)\,g_+h, \qquad \mathrm{Re}\langle U, L_hU\rangle_{W_h} \le \Big(C+\frac{\mathrm{res}^*}{2\min(c_W,1)g_-}\,\frac{1}{h}\Big)\|U\|^2_{W_h}, \tag{8.8}$$

and consequently

$$\|e^{tL_h}\|_V \le \frac{g_+}{g_-}\sqrt{\frac{\max(C_W,1)}{\min(c_W,1)}}\,\exp\Big(\Big(C+\frac{\mathrm{res}^*}{2\min(c_W,1)g_-}\,\frac{1}{h}\Big)t\Big) \qquad (t\ge 0,\ h\le h_0). \tag{8.9}$$

Proof. Write everything in units $h=1$ and let $u\in\mathbb{R}^{X_h}$. Let Z_A, Z'_A be the zones of (8.4) around z_A (in the reference coordinate s) and $\mathcal{J}_{\mathrm{far}}$ the set of lattice sites j such that $\{j,\dots,j+d\}$ meets no Z_A. We decompose

$$u^{\mathsf T}(WL+L^{\mathsf T}W)u = -\sum_{j\in\mathcal{J}_{\mathrm{far}}} B_j^{(g_j)}(u) + u^{\mathsf T}Ru, \qquad R := WL+L^{\mathsf T}W+\sum_{j\in\mathcal{J}_{\mathrm{far}}} B_j^{(g_j)},$$

and show that $R=\sum_A R_A + R_0$ with R_A supported on $Z'_A\times Z'_A$, $R_A = E_{\tau_A}(W_{\tau_A,g(z_A)})+O(h)$ entrywise, and R_0 banded with $O(h)$ entries.

Far pairs. Let (i,j) be a pair of particles with $|s_i-s_j|\le 2\kappa$ (otherwise all entries vanish) such that neither lies in any Z'_A. Then both rows are rows of the deformed lattice without defects (their 2κ-neighbourhoods contain no defect, since $|s_i-z_A|>\varrho+d+4\kappa\ge\rho_0+2\kappa$), both weights are volume weights, and every square through i or j is far. By the entrywise statement of Lemma 8.2, $R_{ij}=O(h)$.

Pairs near a defect. Let $i\in Z'_A$ and $|s_i-s_j|\le 2\kappa$. By (8.7), $|s_j-z_B|\ge\sigma-(\varrho+d+4\kappa)-2\kappa = \varrho+d+6\kappa+2>\varrho+d+4\kappa$ for $B\ne A$, so $j\notin Z'_B$: R_{ij} belongs to the block $Z''_A\times Z''_A$ with $Z''_A=\{|s-z_A|\le\varrho+d+6\kappa\}$, and these blocks are disjoint for different A, since $\sigma>2(\varrho+d+6\kappa)$. Within Z''_A the cloud X_h is, in the scaled coordinate $(x-\varphi(z_A))/h$, the defect cloud $g(z_A)Y_{\tau_A}$ up to $O(h)$ displacements (since $\varphi(s)-\varphi(z_A)=g(z_A)(s-z_A)+O(h^2\cdot\varrho'^2)$ for $|s-z_A|\le\varrho' h$; the $O(h)$ terms below are therefore $O(\varrho^2 h)$, with constants depending on ϱ), so the rows of L at particles of Z''_A are those of $L_{\tau_A,g(z_A)}$ up to $O(h)$; the weights are $W_{\tau_A,g(z_A)}$ on Z_A and $g(s_i)=g(z_A)+O(h)$ outside; and the far squares meeting Z''_A have coefficients $b(g_j)=b(g(z_A))+O(h)$ and are exactly the far squares of the single-defect cloud. Hence $R|_{Z''_A\times Z''_A}=E_{\tau_A}(W_{\tau_A,g(z_A)})+O(h)$, and by Lemma 8.4 the first term is supported on $Z'_A\times Z'_A$.

Conclusion. With $\mathrm{res}_A=\mathrm{res}_{\tau_A}(g(z_A))\le\mathrm{res}^*$ and the certificate inequality,

$$u^{\mathsf T}Ru \le \sum_A \mathrm{res}_A|u_{Z'_A}|^2 + C'h|u|^2 \le (\mathrm{res}^*+C'h)|u|^2 \le \frac{\mathrm{res}^*+C'h}{\min(c_W,1)g_-}\,u^{\mathsf T}Wu,$$

where C' collects the $O(h)$ entries over a band of width $4d+1$ (far pairs) and the disjoint blocks Z''_A (bounded size). Restoring the units h ($L_h=L/h$, $W_h=hW$) gives (8.8) with $C=C'/(2\min(c_W,1)g_-)$ and $\frac{\mathrm{d}}{\mathrm{d}t}\|U\|^2_{W_h}\le 2(C+\mathrm{res}^*/(2\min(c_W,1)g_-h))\|U\|^2_{W_h}$, and (8.9) follows from the norm equivalence in (8.8) together with $g_-h\le V_i\le g_+h$: $\|U\|^2_V\le g_+h|U|^2\le\frac{g_+}{\min(c_W,1)g_-}\|U\|^2_{W_h}$ and $\|U\|^2_{W_h}\le\frac{\max(C_W,1)g_+}{g_-}\|U\|^2_V$, so the constant is $\frac{g_+}{g_-}\sqrt{\max(C_W,1)/\min(c_W,1)}$. □

Corollary 8.8. *In the setting of Theorem 8.7:*

(i) Let $\varphi=\mathrm{id}$ (lattices with defects, $g=1$). If $\mathrm{res}_\tau(1,\varrho)\le C_0e^{-c\varrho}$ uniformly in τ, with bounds (c_W,C_W) independent of ϱ, then the family (L_h) is Lax–Richtmyer stable, $\|e^{tL_h}\|_V\le C_1e^t$, for every sequence of defect configurations whose zone radius satisfies $\varrho_h\ge c^{-1}\log\big(C_0/(2\min(c_W,1)g_-h)\big)$ (with $g_-=1$ here), i.e. whose defects are separated by at least $(2c^{-1}|\log h|+\sigma_0)h$ with a constant σ_0 that depends on C_0, c, c_W and κ: a defect density up to $c/(2|\log h|)$ per particle. For a deformed lattice the same choice of ϱ_h gives $\|e^{tL_h}\|_V\le C_1\exp\big(C'(1+\log^2(1/h))\,t\big)$, because the constant C of Theorem 8.7 grows like ϱ^2.

(ii) For a fixed zone radius ϱ, every configuration with separation σh satisfies (8.9): the growth rate exceeds the deformation rate C by at most $\mathrm{res}^*(\varrho)/(2\min(c_W,1)g_-h)$.

Proof. (i) Choose ϱ_h with $C_0e^{-c\varrho_h} \le 2\min(c_W,1)g_-h$ and apply Theorem 8.7. For $\varphi = \mathrm{id}$ the clouds inside the blocks Z''_A coincide with the single-defect clouds, so $R_A = E_{\tau_A}(W_{\tau_A})$ exactly, $R_0 = 0$ and $C = 0$, independently of ϱ. For a deformed lattice $C = O(\varrho_h^2)$ by the remark in the proof. (ii) is (8.9). □

The hypothesis of (i) is asymptotic in ϱ and cannot be verified by finitely many certificates, so (i) is a conditional statement. The computed residuals are consistent with it for $K = 3$ (all three types) and for $K = 4$ with the geometric orientation, where they reach the solver accuracy; they rule it out for $K = 2$ at insert and merge defects and for $K = 4$ with the aligned orientation at shift and insert defects, whose periodic arrays are weakly unstable (Observation 13.3); and they stall for $K = 2$ at shift defects with $g \neq 1$ (Observation 13.2). With the computed residual at $\varrho = 12$ for $K = 3$, $\mathrm{res}^* = 1.1 \cdot 10^{-6}$ (merge, $g = 0.85$), (ii) gives an excess growth rate below $1.3{\cdot}10^{-5}/h$ with $c_W = 0.05$, and below $1.1{\cdot}10^{-6}/h$ with the actual lower bound $\lambda_{\min}(W_Z) \ge 0.59\,g$ of the computed certificates. The certificates are computed at $g = 0.85, 1, 1.15$ only. A certificate at g_0 serves every g with $|g - g_0|$ small, with residual $\mathrm{res}_\tau(g_0) + C|g - g_0|$, because E_τ depends Lipschitz-continuously on g (Lemma 8.1), but the constant C is not computed. The hypothesis of Theorem 8.7 is therefore established, in floating point, only for $\varphi = \mathrm{id}$; for $\varphi \neq \mathrm{id}$ the theorem is a statement given certificates for all $g \in [g_-, g_+]$. Closing this gap needs a bound on C (explicit in principle, since E_τ is a rational function of g through the fits and the Fejér–Riesz factor) or certificates on a grid of g with a computed interpolation error. In the superposition test of Observation 13.4 the certificates are computed at the exact local spacings of the particular clouds; this is a numerical check for those clouds, not a verification of the hypothesis.

Remark 8.9 (Two dimensions). The only one-dimensional ingredient of Theorem 8.7 is Lemma 8.1. In D dimensions a nonnegative trigonometric polynomial need not be a square, but it may be a *sum of squares* $\sum_s |b_s(\theta)|^2$, equivalently $f(\theta) = \sum_{m,m'} G_{mm'} e^{\mathrm{i}(m-m')\cdot\theta}$ with a positive semidefinite Gram matrix G over the monomials $m \in [-r,r]^D$; strictly positive polynomials always are, by Dritschel's theorem [1, 2], but the lattice dissipation vanishes at $\theta = 0$. If such a G exists, the local squares are $B_j(u) = \sum_{m,m'} G_{mm'} u_{j+m} u_{j+m'} \ge 0$, supported on $j + [-r,r]^2$, and Lemma 8.4, Definition 8.5 and the superposition argument of Theorem 8.7 hold verbatim for lattices with defects ($\varphi = \mathrm{id}$) with the separation $\sigma = 2\varrho + 4r + 12\kappa + 2$ in the maximum norm. A Gram matrix has been found numerically for $K = 2, 3$ (Observation 13.6), but no 2-D certificate has been computed, so there is no 2-D result. The extension to deformed lattices in 2-D additionally requires a sum-of-squares decomposition of the frozen symbol of every affine lattice $A(s)\mathbb{Z}^2$ with Lipschitz dependence on A, which is not automatic because the decomposition is not strictly feasible.

9 Limits of the energy method, and the parity of the order

The Markov argument of Theorem 4.1 for $K = 1$ is an energy estimate in the norm of the stationary distribution π. The following results show that for $K \ge 2$ no volume norm can play this role on jittered clouds, identify two mechanisms, and explain from the local symbol why even orders are fragile. They are proved; their quantitative consequences are measured in Section 12.

9.1 The conservation-defect obstruction

Proposition 9.1 (Lower bound for the numerical abscissa). *Let $L\mathbf{1} = 0$, let $H > 0$ be a diagonal weight with $|\Omega|_H = \sum_i H_i$, and let $d = L^{*_H}\mathbf{1}$, i.e. $d_j = H_j^{-1}\sum_i H_i L_{ij}$ (the defect density of H:*

$\frac{\mathrm{d}}{\mathrm{d}t}\sum_i H_iU_i = \langle d, U\rangle_H$). *Then*

$$\omega_H(L) \;\geq\; \frac{\|d\|_H^2}{9\,|\Omega|_H\,\|L\|_H}. \tag{9.1}$$

Proof. Let $e = \mathbf{1}/\|\mathbf{1}\|_H$, $g = L^{*_H}e = d/\|\mathbf{1}\|_H$, and $v = g/\|g\|_H$ (if $g = 0$ there is nothing to prove). For $U = e + \eta v$ with $\eta > 0$, using $Le = 0$ and $\langle e, Lv\rangle_H = \langle g, v\rangle_H = \|g\|_H$,

$$\mathrm{Re}\langle U, LU\rangle_H = \eta\|g\|_H + \eta^2\,\mathrm{Re}\langle v, Lv\rangle_H \;\geq\; \eta\|g\|_H - \eta^2\|L\|_H, \qquad \|U\|_H^2 \leq (1+\eta)^2.$$

With $\eta = \|g\|_H/(2\|L\|_H)$, which is $\leq \frac{1}{2}$ because $\|g\|_H = \|L^{*_H}e\|_H \leq \|L^{*_H}\|_H = \|L\|_H$ (the H-adjoint has the same H-norm and $\|e\|_H = 1$), the first bound is $\|g\|_H^2/(4\|L\|_H)$ and the second $\leq 9/4$, and $\|g\|_H^2 = \|d\|_H^2/|\Omega|_H$. □

On a jittered cloud with $H = V$, the defect density is $d_j = V_j^{-1}\sum_i V_iL_{ij}$, i.e. the conservation defects of [9, Prop. 4.4] per unit volume; d_j is of size ε/h for jitter εh (zero on lattices, where L has zero column sums), and $\|L\|_V = O(1/h)$ by [9, Lem. 3.5]. Hence, *if* $\|d\|_V^2 \geq c\,\varepsilon^2|\Omega|_V/h^2$ (a lower bound that we do not prove but measure, Table 9), $\omega_V(L_h) \geq c\,\varepsilon^2/h$: *the energy method in the volume norm fails for every order*, including $K = 1$ with the volume weight, which is why Theorem 4.1 needs π. For $K = 1$, π is the unique weight with $d = 0$, and the Markov structure makes the symmetric part negative. For $K \geq 2$ the left null vector π of L_h still exists and removes this obstruction, but it is not known to be positive or to make the symmetric part negative (measured in Observation 12.6).

9.2 The fluctuation obstruction: order two with nearest neighbours

Proposition 9.2 (Exact energy identity)**.** *Let $D = 1$, $K = 2$ (so $q = 1$), $a > 0$, and let every particle have exactly one neighbour on each side (true if κh lies between the largest gap and the smallest distance of second neighbours). Order the particles, let $g_i = x_{i+1} - x_i$ be the gaps, $w_i^{\pm}$ the weights of the right and left neighbour of i, and $V_i = \frac{1}{2}(g_{i-1} + g_i)$. Then:*

(i) The derivative weights are $c_{i,i\pm1} = \pm 1/(g_{i-1} + g_i)$, the volume-weighted weights are anti-symmetric, and the scheme is the finite volume scheme on the dual cells $[m_{i-1,i}, m_{i,i+1}]$ with the upwind reconstructed flux,

$$V_i\,\frac{\mathrm{d}U_i}{\mathrm{d}t} = -a\big(U_{i,i+1} - U_{i-1,i}\big), \qquad U_{i,i+1} = U_i + \tfrac{1}{2}g_i\,\gamma_i, \quad \gamma_i = \frac{w_i^+ g_i^2\sigma_i + w_i^- g_{i-1}^2\sigma_{i-1}}{w_i^+ g_i^2 + w_i^- g_{i-1}^2}, \tag{9.2}$$

where $\sigma_i = (U_{i+1} - U_i)/g_i$ is the slope on gap i and γ_i the weighted least-squares slope. In particular $L_h^{*_V}\mathbf{1} = 0$: the volume norm has no conservation defect.

(ii) With $m_i = \dfrac{w_i^- g_{i-1}^2 g_i^2}{w_i^+ g_i^2 + w_i^- g_{i-1}^2} > 0$,

$$\frac{\mathrm{d}}{\mathrm{d}t}\|U\|_V^2 = -a\,\mathcal{Q}(\sigma), \qquad \mathcal{Q}(\sigma) = \sum_i m_i\,\sigma_i(\sigma_i - \sigma_{i-1}) = \frac{1}{2}\sum_i m_i(\sigma_i - \sigma_{i-1})^2 + \frac{1}{2}\sum_i (m_i - m_{i+1})\,\sigma_i^2. \tag{9.3}$$

(iii) On a uniform grid $m_i \equiv m$ and $\mathcal{Q} = \frac{m}{2}\sum_i(\sigma_i - \sigma_{i-1})^2 \geq 0$: the scheme is contractive in $\|\cdot\|_V$, with fourth-order dissipation (Fromm's scheme, Theorem 5.2(ii)). On a non-uniform grid the quadratic form $\mathcal{Q}$ on $\mathbb{R}^N$ is indefinite unless m is constant. The slopes of periodic grid functions form the hyperplane $\sum_i g_i\sigma_i = 0$, on which $\mathcal{Q}$ may still be positive definite on coarse grids; for a 2-periodic pattern (m_i alternating between $m_{\mathrm{e}} \neq m_{\mathrm{o}}$), $\mathcal{Q}$ is indefinite on this hyperplane for all sufficiently large N.

Proof. (i) A degree-one fit with constant term through two points is interpolatory, which gives the weights, and $V_i c_{i,i+1} = \frac{1}{2} = -V_{i+1} c_{i+1,i}$. For $j = i+1$ the geometric orientation is $\sigma = +1$ and the Rusanov flux [9, (4.3)] with $s = a$ reduces to aU_{ij}, for $j = i-1$ to aU_{ji}; inserting into [9, (4.4)] gives (9.2). The reconstruction fit through U_i with the data $U_{i\pm1} - U_i = \pm g\sigma$ gives γ_i. Summing (9.2) over i gives $\frac{\mathrm{d}}{\mathrm{d}t}\sum_i V_i U_i = 0$. (ii) Multiply (9.2) by $2U_i$ and sum over the period; summation by parts gives $\frac{\mathrm{d}}{\mathrm{d}t}\|U\|_V^2 = -2a\sum_i U_i(U_{i,i+1} - U_{i-1,i}) = -2a\sum_i U_{i,i+1}(U_i - U_{i+1}) = 2a\sum_i U_{i,i+1}\, g_i\sigma_i$. With $U_{i,i+1} = U_i + \frac{1}{2}g_i\gamma_i$ and $2\sum_i U_i g_i\sigma_i = 2\sum_i U_i(U_{i+1} - U_i) = -\sum_i (U_{i+1} - U_i)^2 = -\sum_i g_i^2\sigma_i^2$,

$$\frac{\mathrm{d}}{\mathrm{d}t}\|U\|_V^2 = -a\sum_i g_i^2\sigma_i^2 + a\sum_i g_i^2\gamma_i\sigma_i = -a\sum_i \frac{w_i^- g_{i-1}^2 g_i^2}{w_i^+ g_i^2 + w_i^- g_{i-1}^2}\big(\sigma_i^2 - \sigma_{i-1}\sigma_i\big),$$

since $g_i^2\sigma_i^2 - g_i^2\gamma_i\sigma_i = g_i^2\sigma_i\big[(w^+ g_i^2 + w^- g_{i-1}^2)\sigma_i - w^+ g_i^2\sigma_i - w^- g_{i-1}^2\sigma_{i-1}\big]/(w^+ g_i^2 + w^- g_{i-1}^2)$. The second form of $\mathcal{Q}$ follows from $\sigma_i(\sigma_i - \sigma_{i-1}) = \frac{1}{2}(\sigma_i - \sigma_{i-1})^2 + \frac{1}{2}(\sigma_i^2 - \sigma_{i-1}^2)$ and a shift of the index in the telescoping sum. (iii) For constant m the second sum vanishes. In general, the symmetric matrix T of $\mathcal{Q}$ satisfies $(T\mathbf{1})_i = \frac{1}{2}(m_i - m_{i+1})$ and $\langle \mathbf{1}, T\mathbf{1}\rangle = 0$; if $T\mathbf{1} \neq 0$, then $\mathcal{Q}(\mathbf{1} - tT\mathbf{1}) = -2t\|T\mathbf{1}\|^2 + O(t^2) < 0$ for small $t > 0$. This test vector is not an admissible slope. For a 2-periodic pattern take Bloch vectors $\sigma_{2k} = xe^{\mathrm{i}\theta k}$, $\sigma_{2k+1} = ye^{\mathrm{i}\theta k}$ with $\theta = 2\pi j/(N/2) \neq 0$; their real and imaginary parts satisfy $\sum_i g_i\sigma_i = 0$. Per cell $\mathcal{Q}$ is the Hermitian form with diagonal $(m_\mathrm{e}, m_\mathrm{o})$ and off-diagonal entry of modulus $\frac{1}{2}|m_\mathrm{e}e^{\mathrm{i}\theta} + m_\mathrm{o}|$, whose determinant $-\frac{1}{4}(m_\mathrm{e} - m_\mathrm{o})^2 + \frac{1}{2}m_\mathrm{e}m_\mathrm{o}(1 - \cos\theta)$ is negative for $\theta = 4\pi/N$ once N is large. For gaps alternating between $0.9h$ and $1.1h$ ($\kappa = 1.5$) the smallest eigenvalue of $\mathcal{Q}$ on admissible slopes is positive for $N \le 64$ and negative from $N = 128$ (`ho_nn_admissible.py`): such grids are contractive in $\|\cdot\|_V$ up to $N = 64$. □

The telescoping term in (9.3) is the second obstruction. It has zero mean over the grid and is of divergence form, so it cancels for slowly varying σ to leading order, but not exactly. A scaling argument (not a proof) suggests why it eventually matters: for a smooth mode it is a random sum of N terms of size $O(\varepsilon h^2)\cdot O(h)$, i.e. $O(\varepsilon h^{5/2})$ in one dimension, against the fourth-order dissipation $\frac{1}{2}\sum m(\sigma_i - \sigma_{i-1})^2 = O(h^3)$ of the same mode, so it would dominate once $h \lesssim \varepsilon^2$. The matrix T of $\mathcal{Q}$ is a periodic Jacobi matrix $\bar{m}\,(I - \frac{1}{2}(E + E^{\mathsf{T}}))$ plus a random perturbation of relative size ε; the unperturbed matrix is positive semidefinite with spectrum reaching down to zero, so the perturbation produces negative eigenvalues of size $O(\varepsilon\bar{m})$ with eigenvectors localized near unfavourable configurations of three or four consecutive gaps. This explains how ω_V can grow like $1/h$ without any eigenvalue leaving the left half-plane: the growth lasts only for a time $O(h)$, after which the rough vector has been damped (measured in Observations 12.2 and 12.7).

9.3 Local symbols and the parity of the order

For a particle i let the *local symbol* be

$$\lambda_i(\theta) = \sum_j L_{ij}\, e^{\mathrm{i}\theta\cdot x_{ij}/h}, \qquad \text{so that} \qquad (L_h e_\theta)_i = e^{\mathrm{i}\theta\cdot x_i/h}\lambda_i(\theta), \quad e_\theta = (e^{\mathrm{i}\theta\cdot x_l/h})_l. \tag{9.4}$$

On a lattice $\lambda_i = \lambda$ is the von Neumann symbol. In general λ_i is not an eigenvalue of anything, but $\mathrm{Re}\langle e_\theta, L_h e_\theta\rangle_V = \sum_i V_i\,\mathrm{Re}\,\lambda_i(\theta)$ is the Rayleigh quotient of the Fourier mode, and λ_i carries the local consistency and dissipation properties.

Proposition 9.3 (Parity)**.** *Let the hypotheses of [9, Prop. 4.2] hold at particle i (unisolvent, unregularized fits at i and its neighbours, all orders K, no limiter), so that $\sum_j L_{ij}\,p(x_j) = -a\cdot\nabla p(x_i)$ for all $p \in \Pi_{K-1}$. Let $2k$ be the smallest even integer $\ge K$, i.e. $2k = K$ for even and $2k = K+1$ for odd K. Then*

$$\mathrm{Re}\,\lambda_i(\theta) = \frac{(-1)^k}{(2k)!}\sum_j L_{ij}\,(\theta\cdot x_{ij}/h)^{2k} + O\big(|\theta|^{2k+2}/h\big), \tag{9.5}$$

with a constant in $O(\cdot)$ depending on κ and the bound of [9, Lem. 3.5]. In particular:

(i) For odd K, $\operatorname{Re}\lambda_i(\theta) = O(|\theta|^{K+1}/h)$ on every stencil, the same order as the dissipation $-c|\theta|^{K+1}/h$ of the lattice symbol (Section 5): an irregular stencil changes the coefficient of the leading dissipative term but creates no lower-order term.

(ii) For even K, $\operatorname{Re}\lambda_i(\theta) = (-1)^{K/2}\sum_j L_{ij}(\theta\cdot x_{ij}/h)^K/K! + O(|\theta|^{K+2}/h)$, a term of order $|\theta|^K/h$ that vanishes on centrally symmetric stencils (where the lattice symbol has $\operatorname{Re}\lambda = O(|\theta|^{K+2}/h)$) but not on irregular ones, with a coefficient of either sign. For $K = 2$ in one dimension, $\operatorname{Re}\lambda_i(\theta) = -\theta^2 D_i/h + O(\theta^4/h)$ with the local diffusion coefficient $D_i = \frac{h}{2}\sum_j L_{ij}(x_{ij}/h)^2$.

Proof. $\operatorname{Re}\lambda_i(\theta) = \sum_j L_{ij}\cos(\theta\cdot x_{ij}/h)$ because L is real. The Taylor polynomial of $\cos(\theta\cdot y/h)$ of degree $K-1$ in y contains only even powers and is annihilated by $\sum_j L_{ij}(\cdot)(x_j)$ together with the constant: $\sum_j L_{ij}p(x_{ij}) = \sum_j L_{ij}\tilde{p}(x_j)$ with $\tilde{p}(x) = p(x-x_i) \in \Pi_{K-1}$ and $-a\cdot\nabla\tilde{p}(x_i) = -a\cdot\nabla p(0) = 0$ for an even polynomial p. The first surviving term has degree $2k$, and the remainder is bounded by $\sum_j |L_{ij}|\,(2\kappa|\theta|)^{2k+2}/(2k+2)!$ (the row of L reaches two stencil layers, $|x_{ij}| \le 2\kappa h$) with $\sum_j |L_{ij}| = O(1/h)$. On a centrally symmetric lattice the terms of degree $2k$ with $2k = K$ (even K) vanish by Lemma 5.1 and [9, Prop. 4.3], whose proof shows $\operatorname{Re}\lambda = O(|\theta|^{K+2}/h)$. □

Proposition 9.3 is exact, but turning it into a stability statement requires controlling how the fluctuating local coefficients couple Fourier modes. We have only a heuristic for this (Heuristic 12.5 in Part II), which is consistent with all measurements: odd orders are stable in every setting tested, and every instability found is at even order.

Part II
Numerical results (not proved)

Everything in this part is a computation with the independent implementation of Appendix A: periodic clouds, linear advection, Rusanov dissipation $s_d = |a_d|$, no limiter, no MOOD, dense linear algebra ($N \le 1600$ in 1-D, $N \le 576$ in 2-D in most studies). Each Observation names its script and data file. Measurements on finitely many clouds decide stability for those clouds and their dilations (Remark 2.1); they do not prove uniformity in h.

10 Lattices: orders three to five

Table 3 lists $\max_\theta h\operatorname{Re}\lambda(\theta)/\max_d|a_d|$ on a grid of 4001 (1-D) and 121^2 (2-D) wave numbers, and the largest CFL number, in the normalization $\Delta t = \mathrm{CFL}/(D\sum_d|a_d|\sum_j|c_{ij}^{(d)}|)$ of the implementation [9, Rem. 6.5(iii)], for which $\max_\theta|R(\Delta t\lambda(\theta))| \le 1$ with the stability polynomial R of RK3 and SSPRK(5,4).

Observation 10.1 (Lattices)**.** (i) All tested configurations are stable with the aligned orientation, and all are stable with the geometric orientation except $K = 4$ ($q = 3$) in two dimensions with $\kappa = 3.5$, for $a = (1, 0.5)$ ($h\operatorname{Re}\lambda = 7.7\cdot10^{-5}$) and $a = (1, 0)$ ($2.2\cdot10^{-3}$), and, marginally, with $\kappa = 5.5$ and $a = (1, 0)$ ($2.2\cdot10^{-8}$). The unstable modes vary only in the second coordinate, with wavelengths of about $9h$ ($a = (1, 0)$) and $15h$ ($a = (1, 0.5)$) for $\kappa = 3.5$. In these cases the dissipation of the outer neighbours, whose degree-three weights have the opposite sign of their offsets, is anti-dissipative, and Lemma 5.1 says that nothing else can compensate. The aligned orientation makes every dissipation coefficient non-negative and is stable in all cases. The instability persists under jitter $0.05h$ and is gone at $0.2h$ (Observation 12.1(iv)).

(ii) RK2 is not stable for $K = 4, 5$ at any time step: their dissipation is of order $|\theta|^6$ at small θ (in general of order $|\theta|^{K+1}$ for odd and $|\theta|^{K+2}$ for even K, since for even K the leading

Table 3: Von Neumann analysis on periodic lattices (`lattice_symbols.py`, `data/lattice_symbols_*.md`): maximal real part of the scaled symbol for the geometric and the aligned orientation at full order, and the largest stable CFL number of RK3 and SSPRK(5,4) ("–": unstable for every time step). Bold: positive real part, i.e. instability with growth rate proportional to $1/h$.

					$\max_\theta h\,\mathrm{Re}\,\lambda/\lvert a\rvert_\infty$		$\mathrm{CFL}_{\max}$, RK3		$\mathrm{CFL}_{\max}$, SSPRK(5,4)	
D	κ	a	K	q	geometric	aligned	geom.	aligned	geom.	aligned
1	2.5	(1)	2	1	$\le 10^{-15}$	$\le 10^{-15}$	1.361	1.361	1.962	1.962
			3	2	$\le 10^{-15}$	$\le 10^{-15}$	2.028	2.028	2.552	2.552
			4	3	$\le 10^{-15}$	$\le 10^{-15}$	1.387	1.287	2.010	1.895
			5	3	$\le 10^{-15}$	$\le 10^{-15}$	2.152	2.152	3.048	3.048
1	5.5	(1)	2	1	$\le 10^{-15}$	$\le 10^{-15}$	1.431	1.431	2.057	2.057
			3	3	$\le 10^{-15}$	$\le 10^{-15}$	2.178	2.087	3.719	3.414
			4	3	$\le 10^{-15}$	$\le 10^{-15}$	1.885	1.692	2.629	2.440
			5	5	$\le 10^{-15}$	$\le 10^{-15}$	2.470	2.222	4.235	4.087
2	2.5	(1, 0.5)	2	1	$\le 10^{-15}$	$\le 10^{-15}$	3.479	3.479	5.284	5.284
			3	3	$\le 10^{-15}$	$\le 10^{-15}$	4.471	3.855	> 8	7.191
			4	3	$\le 10^{-15}$	$\le 10^{-15}$	4.372	3.603	6.764	5.835
			5	4	$\le 10^{-15}$	$\le 10^{-15}$	6.986	6.619	> 8	> 8
2	3.5	(1, 0.5)	2	1	$\le 10^{-15}$	$\le 10^{-15}$	3.568	3.568	5.346	5.346
			3	3	$\le 10^{-15}$	$\le 10^{-15}$	5.733	4.807	> 8	> 8
			4	3	$\mathbf{7.7 \cdot 10^{-5}}$	$\le 10^{-15}$	–	4.204	–	6.600
			5	5	$\le 10^{-15}$	$\le 10^{-15}$	5.593	4.430	> 8	> 8
2	3.5	(1, 0)	2	1	$\le 10^{-15}$	$\le 10^{-15}$	2.899	2.899	4.152	4.152
			3	3	$\le 10^{-15}$	$\le 10^{-15}$	4.816	4.125	> 8	6.948
			4	3	$\mathbf{2.2 \cdot 10^{-3}}$	$\le 10^{-15}$	–	3.539	–	5.362
			5	5	$\le 10^{-15}$	$\le 10^{-15}$	4.650	3.739	> 8	7.934
2	5.5	(1, 0.5)	2	1	$\le 10^{-15}$	$\le 10^{-15}$	3.634	3.634	5.506	5.506
			3	3	$\le 10^{-15}$	$\le 10^{-15}$	5.928	5.018	> 8	> 8
			4	3	$\le 10^{-15}$	$\le 10^{-15}$	5.516	4.465	> 8	7.041
			5	5	$\le 10^{-15}$	$\le 10^{-15}$	7.077	5.558	> 8	> 8
2	5.5	(1, 0)	2	1	$\le 10^{-15}$	$\le 10^{-15}$	2.938	2.938	4.203	4.203
			3	3	$\le 10^{-15}$	$\le 10^{-15}$	5.054	4.313	> 8	7.201
			4	3	$\mathbf{2.2 \cdot 10^{-8}}$	$\le 10^{-15}$	–	3.662	–	5.525
			5	5	$\le 10^{-15}$	$\le 10^{-15}$	6.166	5.027	> 8	> 8

dissipative term in the proof of [9, Prop. 4.3] vanishes), while $|R(\mathrm{i}y)|^2 = 1 + y^4/4$ for RK2. The implementation uses RK3 or SSPRK(5,4) for $K \ge 3$, which are stable for CFL up to the values in Table 3; these lie well above the CFL = 0.2–0.8 of [10].

(iii) For $K = 2$ beyond the proved range (Remark 5.3): a random search over $2 \cdot 10^4$ positive weight vectors with $m \le 6$, and $5 \cdot 10^3$ each for $m = 9, 10, 12, 14$, found no violation of $\mathrm{Re}\,\lambda \le 0$ in 1-D (`k2_random_search.py`, `data/k2_random_search.txt`), and the 2-D rows of Table 3 with $\kappa \ge \sqrt{2}$ are stable.

11 The first-order scheme on perturbed clouds

Table 4 lists, for the first-order aligned scheme, the spread ρ_h of the stationary distribution of Theorem 4.1, the π-average $\bar{\beta}$ of the first moments (4.2), the corrector size of Theorem 4.4, and the effective speed fitted from computed solutions.

Table 4: First-order aligned scheme for $u_t + u_x = 0$ (and $a = (1, 0.5)$ in 2-D) on periodic perturbed lattices: spread ρ_h of the stationary distribution (range over two clouds), π-average $\bar{\beta}$ of the first moments of the dissipation weights, corrector size $\max_i |\delta_i|/h$ (Theorem 4.4), and the effective speed $a_{\text{eff}} - 1$ fitted from the solution at $T = 10$ (`first_order_markov.py`, `first_order_corrector.py`, `first_order_phase.py`; `data/first_order_markov.md`).

cloud	N	ρ_h	$\bar{\beta}$	$\max \lvert\delta\rvert/h$	$a_{\text{eff}} - 1$
1-D, jitter $0.2h$, $\kappa = 2.5$	400	1.31	$1.0 \cdot 10^{-3}$	0.19	$-1.23 \cdot 10^{-3}$
	1600	1.32–1.33	$1.0 \cdot 10^{-3}$	0.25	$-1.01 \cdot 10^{-3}$
	6400	–	$9.6 \cdot 10^{-4}$	0.27	$-9.6 \cdot 10^{-4}$
1-D, jitter $0.4h$, $\kappa = 2.5$	400	5.1–6.7	–	–	$-4.9 \cdot 10^{-3}$
	800–3200	6.7–16	4.5–$4.8 \cdot 10^{-3}$	0.59–0.90	–
	6400	–	–	–	$-4.5 \cdot 10^{-3}$
	25600	–	–	–	$-4.4 \cdot 10^{-3}$
1-D, jitter $0.4h$, $\kappa = 5.5$	400	1.7–2.0	$1.7 \cdot 10^{-3}$	0.56	–
	1600	1.9–2.1	$1.6 \cdot 10^{-3}$	0.57	$-1.62 \cdot 10^{-3}$
	6400	–	$1.6 \cdot 10^{-3}$	0.62–0.71	$-1.63 \cdot 10^{-3}$
2-D, jitter $0.2h$, $\kappa = 2.5$	16^2–128^2	1.6–1.9	–	–	–
lattice, $\kappa = 2.5$	1600	1	0	0	$-8 \cdot 10^{-6}$

Observation 11.1 (Spread of π)**.** For jitter $0.2h$, the perturbation of the computations of [10], ρ_h is 1.21–1.34 in one dimension, nearly constant in h, and 1.6–1.9 in two dimensions, where it grows slowly and monotonically (1.61, 1.62, 1.73, 1.78, 1.89 for $n = 16, 32, 48, 64, 128$, one cloud), as the extreme-value picture of Remark 4.2(ii) suggests; for jitter $0.4h$ with $\kappa = 5.5$ it is about two. For jitter $0.4h$ with $\kappa = 2.5$, where some particles have only three neighbours, it grows to 16 at $N = 3200$. So the L^2 constant $\sqrt{\rho_h}$ of Theorem 4.1(v) is about 1.1–1.4 in the settings of the paper.

Observation 11.2 (Drift)**.** On perturbed lattices the first moments β_i^d fluctuate (in 1-D, $\max_i |\beta_i| \approx 0.2$ for jitter $0.2h$ and 0.45 for $0.4h$). The corrector size of Theorem 4.4 satisfies $\max |\delta|/h < 1$ up to $N = 6400$, but it is not $O(h)$: for jitter $0.2h$ and $\kappa = 2.5$ (two clouds per N; `first_order_corrector_large.py`, `data/first_order_corrector_large.md`) $\max |\delta|/h =$ 0.19, 0.21–0.25, 0.25–0.27, 0.38–0.47, 0.57–0.64 and the π-r.m.s. value 0.08, 0.08, 0.09, 0.11–0.14, 0.20–0.26 for $N = 400$, 1600, 6400, 25600, 102400. The maxima grow by a factor 3.4 over a 256-fold refinement (a local exponent of about 0.2, not $\frac{1}{2}$) and fit $\max |\delta|/h \approx a + b\sqrt{N}$ with $a \approx 0.16$ and $b \approx 1.4 \cdot 10^{-3}$ at all five N, i.e. $\eta_h \approx a\,h + b\,h^{1/2}$: consistent with an $h^{1/2}$ term, but with a small constant, dominant only for $N \gtrsim (a/b)^2 \approx 1.3 \cdot 10^4$. $\bar{\beta}$ does not decrease with h ($0.9 \cdot 10^{-3}$ at $N = 102400$), and the effective speed fitted from the computed solutions agrees with $a_{\text{eff}} = 1 - \bar{\beta}$ to two digits. The first-order scheme therefore advects too slowly by 0.1% (jitter $0.2h$, $\kappa = 2.5$), 0.16% (jitter $0.4h$, $\kappa = 5.5$) and 0.45% (jitter $0.4h$, $\kappa = 2.5$), at every resolution. At $T = 10$ this phase error dominates: for jitter $0.4h$ and $\kappa = 2.5$ the r.m.s. error of $\sin(2\pi x)$ stalls at 0.19 between $N = 6400$ and $N = 25600$.

Observation 11.3 (Moment correction)**.** In the 1-D test of `first_order_moment_fix.py` (jitter $0.4h$, $\kappa = 2.5$, the correction ϵ of Proposition 4.5 placed on the farthest neighbour on the side opposite to β_i) the effective speed error drops from $-4.4 \cdot 10^{-3}$ to $-3 \cdot 10^{-8}$ and the error at $T = 10$ converges at first order (0.148, 0.040, 0.0103 for $N = 1600$, 6400, 25600, against 0.230, 0.196, 0.194 without the correction).

In the MOOD computations of [10] the first-order operator acts only on a few particles near discontinuities, where an $O(1)$ local consistency error is of the same size as the error that the fallback is meant to accept; the drift matters for the pure first-order runs (RK2MUSCL1), whose waves move about 0.1% too slowly on the clouds with jitter $0.2h$.

12 High order on jittered and periodic clouds

On irregular clouds neither Fourier analysis nor the Markov argument applies, and a perturbation argument from the lattice fails, because a jitter εh changes L_h by $O(\varepsilon/h)$. We therefore measure the spectral abscissa α, the numerical abscissa ω_V and the Kreiss constant $\mathcal{K}_V$ (Section 2).

12.1 Spectra and Kreiss constants on random clouds

Observation 12.1 (Spectra)**.** From Table 5:

(i) In the settings of the computations of [10] (2-D with jitter $0.2h$ and $\kappa = 2.5$ or 5.5; 1-D with jitter $0.2h$ and $\kappa = 2.5$, or jitter $0.4h$ and $\kappa = 5.5$) the spectral abscissa is zero up to round-off, except $\alpha = 3.1 \cdot 10^{-5}$ for $K = 2$, $\kappa = 5.5$, jitter $0.4h$ (see (ii)), and $\|e^L\|_2 \le 1.1$. For jitter $0.4h$ and $\kappa = 3.5$ the transient growth reaches 2.4 for $K = 5$.

(ii) For $K = 2$ and jitter $0.4h$, a band of smooth modes is weakly anti-diffused. For $\kappa = 2.5$ the unstable eigenvalues belong to the Fourier wave numbers 1–11 (for $n \le 1600$), and the largest rate grows slowly with the resolution: $\alpha = 4 \cdot 10^{-4}$, $1.1 \cdot 10^{-3}$, $1.6 \cdot 10^{-3}$, $1.9 \cdot 10^{-3}$ for $n = 200$, 400, 800, 1600 ($n = 1600$: three clouds, $\alpha = 1.2$–$1.9 \cdot 10^{-3}$, `data/spectra_1d_n1600_j04.jsonl`), with αh between $1 \cdot 10^{-6}$ and $3 \cdot 10^{-6}$. The same holds with smaller constants for $\kappa = 3.5$ ($\alpha h \approx 2 \cdot 10^{-7}$) and $\kappa = 5.5$ ($4 \cdot 10^{-8}$). Over the times of interest the growth is invisible, e.g. $\alpha h \approx 4 \cdot 10^{-8}$ extrapolated to $N = 3163$ and $\kappa = 5.5$ gives $e^{\alpha t} < 1.002$ for $t = 10$. The Bloch analysis (Observation 12.3) shows that this is a genuine $1/h$ instability, not a finite-size effect.

(iii) For $K = 4$, $\kappa = 2.5$ and jitter $0.4h$ in one dimension, the geometric orientation has eigenvalues with $\alpha h \approx 1$, localized on two neighbouring particles (inverse participation ratio 0.5). These particles have four or five neighbours, i.e. their degree-three fits are nearly interpolatory on one-sided stencils, and the instability grows like $1/h$. With the aligned orientation the localized blow-up disappears, but a weaker growth with $\alpha h \approx 8 \cdot 10^{-5}$ remains. The safeguards [9, Rem. 3.10] allow these stencils, since they only require $|\nu(i)| \ge B_q + 1$. In the settings of the paper this does not occur.

(iv) The 2-D lattice instability of Observation 10.1(i) ($K = 4$, $\kappa = 3.5$, geometric) is present for jitter $0.05h$ ($\alpha h = 7.0 \cdot 10^{-5}$ at $n = 30$, against $7.7 \cdot 10^{-5}$ on the lattice; `data/spectra_2d_j005.jsonl`) and absent for $0.2h$. The aligned orientation is stable in both.

Observation 12.2 (Kreiss constants)**.** From Table 6:

(i) At jitter $0.2h$ (constant topology), $\alpha = 0$ to round-off and $\mathcal{K}_V$ is constant in n: 1.002 ($K = 2$), 1.01–1.02 ($K = 3$), 1.23–1.33 ($K = 4$) in 1-D, and 1.003, 1.01–1.02 in 2-D. The numerical abscissa scales exactly like $1/h$, with $\omega_V h \approx 10^{-3}$, 10^{-2} and 0.5 for $K = 2, 3, 4$, as Proposition 9.1 predicts (its lower bound is about $2.6 \cdot 10^{-4}$, $7 \cdot 10^{-4}$, $2 \cdot 10^{-3}$, Table 9). The $1/h$ growth of ω_V is a transient of duration $O(h)$: the schemes behave stably on these clouds, in a norm that is not a volume norm.

(ii) At jitter $0.4h$ (three to five neighbours): $K = 2$ has $\alpha h \approx 2 \cdot 10^{-6}$ for both orientations, $K = 3$ has $\alpha = 0$ with $\mathcal{K}_V \le 1.7$ (geometric) and ≤ 1.2 (aligned), $K = 4$ blows up with the geometric orientation and has αh up to $8 \cdot 10^{-5}$ with the aligned one.

(iii) For the nearest-neighbour scheme of Proposition 9.2 (`ho_nn_identity.py`, which also checks the identity (9.3) to 10^{-16}), $\omega_V h$ is between $1.8 \cdot 10^{-4}$ and $5.4 \cdot 10^{-4}$ for jitter $0.2h$ and $N = 100$ to 1600, attained by localized rough vectors rather than smooth modes, so ω_V grows like $1/h$ for this scheme as well, while $\mathcal{K}_V$ stays close to one.

Table 5: Spectra of the linear operator on periodic perturbed lattices (`spectra_study.py`, `data/spectra_*.jsonl`, $a = 1$ in 1-D and $a = (1, 0.5)$ in 2-D, one to three clouds per resolution; geometric orientation at full order unless stated). α: largest real part over the clouds and resolutions; "$\leq 10^{-12}$": zero up to round-off (the constant mode); $\|e^L\|_2$: largest over all runs (∞: blow-up).

D	κ	jitter	K	orientation	n (per dim.)	α	αh	$\|e^L\|_2$
1	2.5	0.2	1	aligned	100–800	$\leq 10^{-12}$		1.00
			2	geometric	100–800	$\leq 10^{-12}$		1.00
			3	geometric	100–800	$\leq 10^{-12}$		1.03
			4	geometric	100–800	$\leq 10^{-12}$		1.04
1	2.5	0.4	1	aligned	100–800	$\leq 10^{-12}$		1.04
			2	aligned	200–800	$1.6 \cdot 10^{-3}$	$2.8 \cdot 10^{-6}$	1.07
			2	geometric	100–800	$1.6 \cdot 10^{-3}$	$2.8 \cdot 10^{-6}$	1.08
			3	aligned	200–800	$\leq 10^{-12}$		1.12
			3	geometric	100–800	$\leq 10^{-12}$		1.54
			4	aligned	200–800	$5.0 \cdot 10^{-2}$	$8.1 \cdot 10^{-5}$	2.12
			4	geometric	100–800	$1.2 \cdot 10^{3}$	$1.5 \cdot 10^{0}$	∞
1	3.5	0.4	1	aligned	100–800	$\leq 10^{-12}$		1.01
			2	geometric	100–800	$1.6 \cdot 10^{-4}$	$2.1 \cdot 10^{-7}$	1.02
			3	geometric	100–800	$\leq 10^{-12}$		1.18
			4	geometric	100–800	$\leq 10^{-12}$		1.61
			5	geometric	100–800	$\leq 10^{-12}$		2.42
1	5.5	0.4	1	aligned	100–800	$\leq 10^{-12}$		1.01
			2	geometric	100–800	$3.1 \cdot 10^{-5}$	$3.9 \cdot 10^{-8}$	1.01
			3	geometric	100–800	$\leq 10^{-12}$		1.02
			4	geometric	100–800	$\leq 10^{-12}$		1.02
			5	geometric	100–800	$\leq 10^{-12}$		1.10
2	2.5	0.2	1	aligned	12–32	$\leq 10^{-12}$		1.00
			2	aligned	12–32	$\leq 10^{-12}$		1.00
			2	geometric	12–32	$\leq 10^{-12}$		1.00
			3	aligned	12–32	$\leq 10^{-12}$		1.01
			3	geometric	12–32	$\leq 10^{-12}$		1.01
			4	aligned	12–32	$\leq 10^{-12}$		1.01
			4	geometric	12–32	$\leq 10^{-12}$		1.04
2	3.5	0.2	2	aligned	12–24	$\leq 10^{-12}$		1.00
			2	geometric	12–24	$\leq 10^{-12}$		1.00
			3	aligned	12–24	$\leq 10^{-12}$		1.00
			3	geometric	12–24	$\leq 10^{-12}$		1.00
			4	aligned	12–24	$\leq 10^{-12}$		1.00
			4	geometric	12–24	$\leq 10^{-12}$		1.01
			5	aligned	12–24	$\leq 10^{-12}$		1.01
			5	geometric	12–24	$\leq 10^{-12}$		1.02
2	5.5	0.2	2	aligned	12–20	$\leq 10^{-12}$		1.00
			2	geometric	12–20	$\leq 10^{-12}$		1.00
			3	aligned	12–20	$\leq 10^{-12}$		1.00
			3	geometric	12–20	$\leq 10^{-12}$		1.00
			4	aligned	12–20	$\leq 10^{-12}$		1.00
			4	geometric	12–20	$\leq 10^{-12}$		1.00
			5	aligned	12–20	$\leq 10^{-12}$		1.00
			5	geometric	12–20	$\leq 10^{-12}$		1.00

Table 6: Jittered clouds (`ho_random.py`, `data/ho_random_*.jsonl`, $\kappa = 2.5$, $a = 1$ in 1-D and $(1, 0.5)$ in 2-D, two clouds per n in 1-D): spectral abscissa, numerical abscissa in the volume norm times h, and Kreiss constant (sampled on 40 log-spaced times in $[h/100, 20]$; dual volumes in 1-D, uniform in 2-D), ranges over the clouds and resolutions.

D	κ	jitter	K	orientation	n	αh	$\omega_V h$	$\mathcal{K}_V$
1	2.5	0.2	2	geometric	100–400	$\leq 10^{-12}$	0.0010–0.0016	1.002–1.003
			3	geometric	100–400	$\leq 10^{-12}$	0.0053–0.0096	1.011–1.019
			4	geometric	100–400	$\leq 10^{-12}$	0.4203–0.5524	1.227–1.326
1	2.5	0.4	2	aligned	100–200	$2.2 \cdot 10^{-6}$	0.0342–0.0404	1.055–1.059
			2	geometric	100–400	$2.1 \cdot 10^{-6}$	0.0342–0.0463	1.052–1.059
			3	aligned	100–200	$\leq 10^{-12}$	0.0769–0.1051	1.096–1.151
			3	geometric	100–200	$\leq 10^{-12}$	0.0975–0.2063	1.165–1.687
			4	aligned	100–200	$1.7 \cdot 10^{-5}$ – $-7.8 \cdot 10^{-5}$	1.4580–2.7273	1.875–2.559
			4	geometric	100–200	$9.5 \cdot 10^{-1}$	2.0248–2.7899	2.636–∞
2	2.5	0.2	2	geometric	12–24	$\leq 10^{-12}$	0.0015–0.0021	1.003
			3	geometric	12–24	$\leq 10^{-12}$	0.0107–0.0227	1.011–1.019

Table 7: Bloch analysis of periodic jittered supercells (`ho_bloch.py`, `data/ho_bloch_*.jsonl`, $\kappa = 2.5$, $a = 1$ in 1-D and $(1, 0.5)$ in 2-D, two random patterns per m, 200 (1-D) or 8^2 (2-D) values of θ, $s/h \in [10^{-2}, 10^3]$). Per m: $\max_\theta \alpha(\hat{A}(\theta))\, h$ / Bloch–Kreiss constant, worst of the two patterns; ∞: blow-up (overflow); Kreiss constants are maxima over sampled times and θ, hence lower bounds; geom./alig.: geometric/aligned orientation.

D	κ	jitter	K	orient.	m	$\max_\theta \alpha h$ / $\mathcal{K}_Y$ for the listed m (per direction)		
1	2.5	0.2	2	geom.	8, 16, 32	$\leq 10^{-12}$ / 1.001	$8.6 \cdot 10^{-10}$ / 1.006	$1.0 \cdot 10^{-11}$ / 1.005
			3	geom.	8, 16, 32	$\leq 10^{-12}$ / 1.013	$\leq 10^{-12}$ / 1.049	$\leq 10^{-12}$ / 1.049
			4	geom.	8, 16, 32	$\leq 10^{-12}$ / 1.153	$\leq 10^{-12}$ / 1.146	$\leq 10^{-12}$ / 1.172
1	2.5	0.4	2	alig.	8, 16, 32	$1.1 \cdot 10^{-6}$ / 1.040	$2.1 \cdot 10^{-6}$ / 1.165	$1.0 \cdot 10^{-6}$ / 1.171
			2	geom.	8, 16, 32	$1.1 \cdot 10^{-6}$ / 1.040	$2.1 \cdot 10^{-6}$ / 1.165	$1.2 \cdot 10^{-6}$ / 1.172
			3	alig.	8, 16, 32	$\leq 10^{-12}$ / 1.137	$\leq 10^{-12}$ / 1.284	$\leq 10^{-12}$ / 1.287
			3	geom.	8, 16, 32	$\leq 10^{-12}$ / 1.144	$\leq 10^{-12}$ / 1.317	$\leq 10^{-12}$ / 1.319
			4	alig.	8, 16, 32	$2.0 \cdot 10^{-5}$ / 2.581	$\leq 10^{-12}$ / 2.582	$1.5 \cdot 10^{-4}$ / 2.619
			4	geom.	8, 16, 32	$1.4 \cdot 10^{-2}$ / $2.4 \cdot 10^{6}$	$\leq 10^{-12}$ / 2.605	$5.2 \cdot 10^{-4}$ / 2.609
2	2.5	0.2	2	geom.	4, 6	$\leq 10^{-12}$ / 1.004	$\leq 10^{-12}$ / 1.003	
			3	geom.	4, 6	$\leq 10^{-12}$ / 1.008	$\leq 10^{-12}$ / 1.008	

12.2 Bloch analysis of jittered supercells

Observation 12.3 (Bloch symbols)**.** Table 7 evaluates the quantities of Theorem 6.1 for supercells with m jittered particles per direction.

(i) *Jitter* $0.2h$ (constant stencil topology, four neighbours in 1-D, 17–21 in 2-D): $\alpha = 0$ to round-off for $K = 3, 4$ and all m, and for $K = 2$ except one of the two patterns, which has $\alpha h = 8.6 \cdot 10^{-10}$ ($m = 16$) and $1.0 \cdot 10^{-11}$ ($m = 32$). These are genuine: the eigenvector condition number is 1.001, a 40-digit recomputation agrees, and $\alpha > 0$ on a band of 84 of 2001 sampled θ near $\theta = -0.09$ (`ho_bloch_mpcheck.py`, `data/ho_bloch_mpcheck.md`); by Corollary 6.4 that refinement family grows like $e^{ct/h}$ with $c \approx 10^{-9}$, invisible in practice. The Bloch–Kreiss constants are 1.001–1.006 ($K = 2$), 1.01–1.05 ($K = 3$) and 1.15–1.17 ($K = 4$, worst pattern; 1.06 for the other at $m = 8$), with no visible growth from $m = 8$ to $m = 32$. In 2-D ($m = 4, 6$, $K = 2, 3$) the constants are below 1.01. Up to the sampling in θ and s, the $K = 3, 4$ patterns thus generate stable refinement families by Theorem 6.1.

(ii) *Jitter* $0.4h$ (three to five neighbours): for $K = 2$, αh between 10^{-7} and $2 \cdot 10^{-6}$ at interior θ for most patterns, which by Corollary 6.4 means an $e^{ct/h}$ instability of the refinement family

Table 8: Local symbols on jittered clouds (`ho_local_symbol.py`, `data/ho_local_symbol_*.md`, 1-D, geometric orientation): volume-weighted mean and standard deviation of the local diffusion coefficient D_i (the θ^2 term, Proposition 9.3) and of the fourth-order coefficient $D_i^{(4)} = -\frac{h}{24}\sum_j L_{ij}(x_{ij}/h)^4$, and the measured spectral abscissa.

κ	jitter	n	cloud	K	$\lvert\nu(i)\rvert$	$\overline{D}$	std D	$\overline{D^{(4)}}$	std $D^{(4)}$	αh
1.5	0.2	400	0	2	2	$\le 10^{-12}$	$1.5\cdot10^{-1}$	0.126	0.042	$\le 10^{-12}$
		400	0	3	2	$\le 10^{-12}$	$7.7\cdot10^{-2}$	0.068	0.015	$\le 10^{-12}$
		800	0	2	2	$\le 10^{-12}$	$1.4\cdot10^{-1}$	0.126	0.040	$\le 10^{-12}$
		800	0	3	2	$\le 10^{-12}$	$7.3\cdot10^{-2}$	0.068	0.015	$\le 10^{-12}$
2.5	0.2	400	0	2	4	$1.1\cdot10^{-3}$	$1.0\cdot10^{-1}$	0.700	0.091	$\le 10^{-12}$
		400	0	3	4	$\le 10^{-12}$	$\le 10^{-12}$	0.324	0.025	$\le 10^{-12}$
		400	1	2	4	$8.7\cdot10^{-4}$	$9.2\cdot10^{-2}$	0.701	0.093	$\le 10^{-12}$
		400	1	3	4	$\le 10^{-12}$	$\le 10^{-12}$	0.324	0.026	$\le 10^{-12}$
		800	0	2	4	$1.0\cdot10^{-3}$	$9.9\cdot10^{-2}$	0.700	0.090	$\le 10^{-12}$
		800	0	3	4	$\le 10^{-12}$	$\le 10^{-12}$	0.324	0.025	$\le 10^{-12}$
		800	1	2	4	$9.4\cdot10^{-4}$	$9.8\cdot10^{-2}$	0.700	0.090	$\le 10^{-12}$
		800	1	3	4	$\le 10^{-12}$	$\le 10^{-12}$	0.324	0.025	$\le 10^{-12}$
2.5	0.4	400	0	2	3, 4, 5	$3.0\cdot10^{-3}$	$1.9\cdot10^{-1}$	0.706	0.251	$1.5\cdot10^{-6}$
		400	0	3	3, 4, 5	$\le 10^{-12}$	$\le 10^{-12}$	0.325	0.074	$\le 10^{-12}$
		400	1	2	3, 4, 5	$3.7\cdot10^{-3}$	$1.8\cdot10^{-1}$	0.703	0.240	$2.8\cdot10^{-6}$
		400	1	3	3, 4, 5	$\le 10^{-12}$	$\le 10^{-12}$	0.328	0.073	$\le 10^{-12}$
		800	0	2	3, 4, 5	$3.4\cdot10^{-3}$	$1.8\cdot10^{-1}$	0.698	0.242	$6.4\cdot10^{-7}$
		800	0	3	3, 4, 5	$\le 10^{-12}$	$\le 10^{-12}$	0.324	0.071	$\le 10^{-12}$
		800	1	2	3, 4, 5	$5.2\cdot10^{-3}$	$1.8\cdot10^{-1}$	0.690	0.240	$2.0\cdot10^{-6}$
		800	1	3	3, 4, 5	$\le 10^{-12}$	$\le 10^{-12}$	0.323	0.073	$\le 10^{-12}$

with $c \approx 2\cdot10^{-6}$. For $K = 3$, $\alpha \le 0$ in all cases with constants 1.06–1.32. For $K = 4$ the geometric orientation fails catastrophically on one pattern ($\alpha h = 0.014$, constant $2\cdot10^6$, the near-interpolatory stencil of Observation 12.1(iii)) and weakly on another ($\alpha h = 5\cdot10^{-4}$); the aligned orientation removes the catastrophe but keeps weak instabilities (αh up to $1.5\cdot10^{-4}$).

12.3 Local symbols and the fluctuation heuristic

Observation 12.4 (Local symbols)**.** For $K = 3$ the local diffusion coefficient D_i is zero to round-off, as Proposition 9.3(i) predicts. For $K = 2$, $\bar{D} > 0$ on all clouds tested (10^{-3} at jitter $0.2h$, 3–$5\cdot10^{-3}$ at jitter $0.4h$) with $\overline{D^{(4)}} \approx 0.70$, and the fluctuations $\tilde{D}_i = D_i - \bar{D}$ have standard deviation 0.10 (jitter $0.2h$) and 0.18 (jitter $0.4h$). For $K = 3$ the fluctuating coefficient is that of θ^4, with relative standard deviation 8% at jitter $0.2h$ and 22% at $0.4h$.

Heuristic 12.5 (Smooth modes on jittered clouds). Consider $K = 2$ in one dimension and a smooth mode e_θ, $|\theta| \ll 1$. Its Rayleigh quotient is $-\theta^2\bar{D}/h - \theta^4\overline{D^{(4)}}/h + \dots$: on average the irregularity adds a small first-order dissipation, and the Fourier modes themselves are damped. The fluctuations $\tilde{D}_i$ couple e_θ to the other modes with matrix elements $\langle e_{\theta'}, L_h e_\theta\rangle_V \approx -(\theta^2/h)\,N^{-1}\sum_i \tilde{D}_i e^{\mathrm{i}(\theta-\theta')x_i/h}$ of size $\theta^2\,\mathrm{std}(D)/(h\sqrt{N})$. Second-order perturbation theory for the eigenvalue near λ_θ gives the real shift $\sum_{\theta'} |c_{\theta\theta'}|^2\,\mathrm{Re}\,(\lambda_\theta - \lambda_{\theta'})^{-1}$, and $\mathrm{Re}(\lambda_\theta - \lambda_{\theta'})^{-1} > 0$ when θ' is more damped than θ: coupling to damped modes pushes the smooth eigenvalue to the *right*. Summing over the N modes θ' with $|\,\mathrm{Im}(\lambda_\theta - \lambda_{\theta'})| \sim |\theta - \theta'|/h$ and $-\,\mathrm{Re}\,\lambda_{\theta'} \sim \theta'^4/h$ gives a shift of order $\theta^4\,\mathrm{std}(D)^2\,c'/h$, the same order as the lattice dissipation. The eigenvalue of the smooth mode therefore behaves like

$$\mathrm{Re}\,\lambda \approx -\frac{\bar{D}\,\theta^2}{h} + \frac{\big(c'\,\mathrm{std}(D)^2 - \overline{D^{(4)}}\big)\theta^4}{h}, \tag{12.1}$$

which is negative for all θ if the variance of the local diffusion coefficients is small enough, and otherwise has the positive maximum $\alpha h = \bar{D}^2/\big(4(c'\mathrm{std}(D)^2 - \overline{D^{(4)}})\big)$ at $\theta_*^2 = \bar{D}/(2(c'\mathrm{std}(D)^2 - \overline{D^{(4)}}))$. With $\bar{D} \approx 3 \cdot 10^{-3}$ at jitter $0.4h$, the observed $\alpha h \approx 2 \cdot 10^{-6}$ corresponds to $c'\mathrm{std}(D)^2 - \overline{D^{(4)}} \approx 1$ and $\theta_* \approx 0.04$, i.e. wavelengths of about $150h$, consistent with the unstable wave numbers of Observation 12.1(ii) ($\theta \le 0.043$). At jitter $0.2h$ the variance is three times smaller and the dissipation wins on most patterns, though not on all: one periodic pattern still shows the same mechanism with $\alpha h \approx 10^{-9}$ (Observation 12.3). For odd K there is no θ^K term; the fluctuating coefficient is that of θ^{K+1} itself, and its second-order effect is of order θ^{2K+2}/h, which cannot compete with the dissipation θ^{K+1}/h at small θ. This is why $K = 3$ is stable in every setting tested while $K = 2$ and $K = 4$ are not.

The heuristic is not a proof: the perturbation series is not controlled, and the mode coupling is a random-matrix problem. It identifies the quantity a proof would have to bound, the variance of the local even-order coefficient relative to the dissipation. A natural regularity assumption for a rigorous version is Kadec's 1/4 condition [11]: for $x_n = h(n + \delta_n)$ with $|\delta_n| \le \varepsilon < \frac{1}{4}$ the exponentials form a Riesz basis with constants depending on ε only, so the mode-coupling picture is meaningful for jitter $0.2h$ and not for $0.4h$. This is a coincidence of thresholds rather than an explanation, since the mechanism is the stencil topology.

12.4 The weight π and diagonal Lyapunov weights

Observation 12.6 (The weight π)**.** (i) At jitter $0.2h$, π is positive for $K = 1, 2, 3$ in 1-D and 2-D, with $\rho \le 1.4$ for $K \le 2$ and ≤ 2.7 for $K = 3$; for $K = 4$ in 1-D it is positive with $\rho \approx 2$, but it does not help: $\omega_\pi h = 0.65$ against $\omega_V h = 0.55$.

(ii) For $K = 2$, $\omega_\pi h$ is zero to round-off for $n \le 100$ (1-D) and $n \le 16$ (2-D, where also $K = 3$ gives zero), and $6 \cdot 10^{-7}$ to $1.1 \cdot 10^{-6}$ for $n = 200$ to 1600 in 1-D, i.e. ω_π between $2 \cdot 10^{-4}$ and $2 \cdot 10^{-3}$, growing under refinement. The eigenvector of the positive eigenvalue of the π-symmetric part is smooth (Fourier wave numbers 1–5, inverse participation ratio $\approx 1/N$), and the Rayleigh quotients of the pure Fourier modes are negative: this is the fluctuation mechanism of Proposition 9.2 and Heuristic 12.5. The weight π thus removes the first obstruction entirely and leaves a second one that is tiny but nonzero. For $K = 3$, $\omega_\pi h \approx 2 \cdot 10^{-3}$ is constant in h, a quarter of $\omega_V h$.

(iii) At jitter $0.4h$ in 1-D, π is not positive for $K = 3$ (geometric orientation, all clouds; aligned, most clouds), and for $K = 2$ $\omega_\pi h \approx 2$–$6 \cdot 10^{-3}$, consistent with the positive spectral abscissa.

Observation 12.7 (Diagonal Lyapunov weights)**.** A diagonal weight $H > 0$ with $M = HL_h + L_h^\mathsf{T} H \preceq 0$ is a multiple of the left null vector π: since $L_h \mathbf{1} = 0$, $\mathbf{1}^\mathsf{T} M \mathbf{1} = 0$, so $M \preceq 0$ forces $M\mathbf{1} = 0$, i.e. $L_h^\mathsf{T} H \mathbf{1} = 0$. A diagonal Lyapunov weight therefore exists if and only if $\pi > 0$ and $\omega_\pi \le 0$, and the minimal spread is then exactly $\max \pi / \min \pi$ (Table 10, `ho_lyapunov_exact.py`). For $K = 1$ this is Theorem 4.1 (spread 1.24). For $K = 2$ with $\kappa = 2.5$, $\omega_\pi h$ is zero to round-off at $n = 50, 100$ (spread 1.24), but $\omega_\pi h = 7.9 \cdot 10^{-7} > 0$ at $n = 200$, so no diagonal weight exists there; for $K = 3$ ($\omega_\pi h \approx 10^{-3}$) and for the nearest-neighbour scheme of Proposition 9.2 ($\kappa = 1.5$, $K = 2$, $N = 50, 100$) none exists. An earlier version of this observation solved the semidefinite programme "minimise ρ subject to $1 \le H_i \le \rho$ and $M \preceq 0$" numerically (`ho_lyapunov.py`, SCS [21]); its feasible set has empty interior, and the points the solver returned ($\rho \approx 1.17$–1.21, residual $\approx 2 \cdot 10^{-5}$) are not feasible. For $K \ge 3$, and for $K = 2$ on fine clouds, a Lyapunov weight, if it exists, is not diagonal: the energy method with a point-wise weight seems unavailable in principle, not only with the volumes.

Table 9: The left null vector π of L_h as a weight (`ho_pi.py`, `data/ho_pi_*.md`): positivity, spread ρ relative to the volumes, numerical abscissa in the volume and the π norm (times h), and the lower bound (9.1) for $\omega_V h$. "not positive": π has entries of both signs. One cloud (seed 0) per resolution; the second cloud behaves alike.

D	κ	jitter	n	cloud	K	orientation	$\min\pi/\max\pi$	ρ	$\omega_V h$	$\omega_\pi h$	bound
1	2.5	0.2	100	0	1	aligned	0.806	1.33	$7.2\cdot10^{-4}$	$\le 10^{-12}$	$2.6\cdot10^{-4}$
			100	0	2	geometric	0.805	1.33	$1.1\cdot10^{-3}$	$\le 10^{-12}$	$2.6\cdot10^{-4}$
			100	0	3	geometric	0.305	2.45	$5.3\cdot10^{-3}$	$2.4\cdot10^{-3}$	$8.0\cdot10^{-4}$
			100	0	4	geometric	0.389	2.15	$5.5\cdot10^{-1}$	$6.5\cdot10^{-1}$	$2.5\cdot10^{-3}$
			200	0	1	aligned	0.797	1.36	$8.5\cdot10^{-4}$	$\le 10^{-12}$	$3.1\cdot10^{-4}$
			200	0	2	geometric	0.794	1.36	$1.3\cdot10^{-3}$	$7.9\cdot10^{-7}$	$3.0\cdot10^{-4}$
			200	0	3	geometric	0.298	2.50	$9.3\cdot10^{-3}$	$2.2\cdot10^{-3}$	$8.3\cdot10^{-4}$
			200	0	4	geometric	0.389	2.15	$5.5\cdot10^{-1}$	$6.5\cdot10^{-1}$	$2.5\cdot10^{-3}$
			400	0	1	aligned	0.765	1.36	$8.1\cdot10^{-4}$	$\le 10^{-12}$	$2.8\cdot10^{-4}$
			400	0	2	geometric	0.765	1.36	$1.4\cdot10^{-3}$	$6.1\cdot10^{-7}$	$2.8\cdot10^{-4}$
			400	0	3	geometric	0.297	2.50	$9.6\cdot10^{-3}$	$2.2\cdot10^{-3}$	$7.1\cdot10^{-4}$
			400	0	4	geometric	0.389	2.15	$5.5\cdot10^{-1}$	$6.5\cdot10^{-1}$	$2.2\cdot10^{-3}$
			800	0	2	geometric	0.748	1.37	$1.4\cdot10^{-3}$	$8.7\cdot10^{-7}$	$2.7\cdot10^{-4}$
			800	0	3	geometric	0.286	2.68	$9.6\cdot10^{-3}$	$2.2\cdot10^{-3}$	$6.7\cdot10^{-4}$
			1600	0	2	geometric	0.748	1.38	$1.5\cdot10^{-3}$	$1.1\cdot10^{-6}$	$2.7\cdot10^{-4}$
			1600	0	3	geometric	0.273	2.72	$1.1\cdot10^{-2}$	$2.2\cdot10^{-3}$	$6.6\cdot10^{-4}$
1	2.5	0.4	100	0	1	aligned	0.271	4.07	$1.9\cdot10^{-2}$	$\le 10^{-12}$	$3.4\cdot10^{-3}$
			100	0	2	geometric	0.260	4.09	$3.9\cdot10^{-2}$	$6.5\cdot10^{-3}$	$3.0\cdot10^{-3}$
			100	0	2	aligned	0.258	4.12	$4.0\cdot10^{-2}$	$5.7\cdot10^{-3}$	$3.0\cdot10^{-3}$
			100	0	3	geometric	-0.325	–	$9.8\cdot10^{-2}$	–	$5.4\cdot10^{-3}$
			100	0	3	aligned	0.067	12.97	$7.7\cdot10^{-2}$	$1.2\cdot10^{-1}$	$5.1\cdot10^{-3}$
			200	0	1	aligned	0.156	5.79	$1.8\cdot10^{-2}$	$\le 10^{-12}$	$3.2\cdot10^{-3}$
			200	0	2	geometric	0.161	5.46	$3.9\cdot10^{-2}$	$6.5\cdot10^{-3}$	$3.1\cdot10^{-3}$
			200	0	2	aligned	0.161	5.46	$4.0\cdot10^{-2}$	$5.7\cdot10^{-3}$	$3.1\cdot10^{-3}$
			200	0	3	geometric	-0.564	–	$2.1\cdot10^{-1}$	–	$5.4\cdot10^{-3}$
			200	0	3	aligned	-0.069	–	$1.0\cdot10^{-1}$	–	$4.5\cdot10^{-3}$
			400	0	1	aligned	0.150	5.79	$1.8\cdot10^{-2}$	$\le 10^{-12}$	$2.6\cdot10^{-3}$
			400	0	2	geometric	0.158	5.46	$4.6\cdot10^{-2}$	$6.5\cdot10^{-3}$	$2.5\cdot10^{-3}$
			400	0	2	aligned	0.158	5.46	$4.6\cdot10^{-2}$	$5.7\cdot10^{-3}$	$2.5\cdot10^{-3}$
			400	0	3	geometric	-0.564	–	$2.1\cdot10^{-1}$	–	$4.4\cdot10^{-3}$
			400	0	3	aligned	-0.065	–	$1.0\cdot10^{-1}$	–	$3.7\cdot10^{-3}$
2	2.5	0.2	12	0	1	aligned	0.706	1.42	$1.5\cdot10^{-3}$	$\le 10^{-12}$	$5.1\cdot10^{-4}$
			12	0	2	geometric	0.712	1.41	$1.8\cdot10^{-3}$	$\le 10^{-12}$	$5.1\cdot10^{-4}$
			12	0	3	geometric	0.479	2.09	$1.1\cdot10^{-2}$	$\le 10^{-12}$	$2.0\cdot10^{-3}$
			16	0	1	aligned	0.622	1.61	$1.3\cdot10^{-3}$	$\le 10^{-12}$	$4.5\cdot10^{-4}$
			16	0	2	geometric	0.614	1.63	$1.5\cdot10^{-3}$	$\le 10^{-12}$	$4.6\cdot10^{-4}$
			16	0	3	geometric	0.457	2.19	$1.2\cdot10^{-2}$	$\le 10^{-12}$	$1.9\cdot10^{-3}$

Table 10: Diagonal Lyapunov weights (`ho_lyapunov_exact.py`, `data/ho_lyapunov_exact_1d.jsonl`, 1-D, jitter $0.2h$): a diagonal H with $HL_h + L_h^{\mathsf{T}}H \preceq 0$ exists if and only if $\pi > 0$ and $\omega_\pi \le 0$ (Observation 12.7); numerical abscissa $\omega_\pi h$ (round-off threshold 10^{-12}), feasibility, and the minimal spread $\rho = \max \pi_i / \min \pi_i$.

D	κ	jitter	n	K	orientation	$\omega_\pi h$	status	ρ
1	2.5	0.2	50	1	aligned	$\le 10^{-12}$	feasible	1.236
1	2.5	0.2	50	2	geometric	$\le 10^{-12}$	feasible	1.243
1	2.5	0.2	50	3	geometric	$9.5 \cdot 10^{-4}$	infeasible	–
1	2.5	0.2	100	1	aligned	$\le 10^{-12}$	feasible	1.240
1	2.5	0.2	100	2	geometric	$\le 10^{-12}$	feasible	1.243
1	2.5	0.2	100	3	geometric	$2.4 \cdot 10^{-3}$	infeasible	–
1	1.5	0.2	50	2	geometric	$1.1 \cdot 10^{-4}$	infeasible	–
1	1.5	0.2	50	2	geometric	$7.8 \cdot 10^{-5}$	infeasible	–
1	1.5	0.2	100	2	geometric	$1.8 \cdot 10^{-4}$	infeasible	–
1	1.5	0.2	100	2	geometric	$1.5 \cdot 10^{-4}$	infeasible	–
1	2.5	0.2	200	2	geometric	$7.9 \cdot 10^{-7}$	infeasible	–

Table 11: Smoothly deformed lattices $x = \varphi(s)$, $\varphi(s) = s + \frac{A}{2\pi}\sin(2\pi s)$ per coordinate (`ho_deformed.py`, `data/ho_deformed_*.jsonl`, $\kappa = 2.5$, geometric orientation): spectral abscissa, numerical abscissa ω_V in the Jacobian-volume norm (which Theorem 7.3 bounds independently of h; compare $\omega_V \propto 1/h$ in Table 6), and the Kreiss constant minus one, for the listed n.

D	κ	A	K	n	αh	ω_V	$\mathcal{K}_V - 1$
1	2.5	0.15	2	100, 200, 400, 800	$\le 10^{-12}$	4.3e-4, 2.7e-4, 1.7e-4, 1.1e-4	9e-5, 5e-5, 3e-5, 2e-5
			3	100, 200, 400, 800	$\le 10^{-12}$	2.6e-4, 1.6e-4, 9.7e-5, 6.1e-5	5e-5, 3e-5, 2e-5, 1e-5
			4	100, 200, 400, 800	$\le 10^{-12}$	4.5e-4, 1.9e-4, 7.1e-5, 2.3e-5	8e-5, 3e-5, 1e-5, 4e-6
2	2.5	0.1	2	12, 16, 24	$\le 10^{-12}$	1.9e-3, 1.2e-3, 7.5e-4	5e-4, 3e-4, 2e-4
			3	12, 16, 24	$\le 10^{-12}$	3.2e-4, 1.3e-4, 3.5e-5	8e-5, 3e-5, 8e-6

13 High order on deformed lattices and lattices with defects

13.1 Smoothly deformed lattices

Observation 13.1 (Deformed lattices)**.** (i) *Hypothesis (D4).* The frozen symbols are ≤ 0 to round-off for $K = 3, 4$ in 1-D with $\varphi' \in [0.85, 1.15]$, and in 2-D for $K = 2, 3$ on rectangular lattices with aspect ratios in $[0.9, 1.1]^2$ (`ho_frozen.py`, `data/ho_frozen_1d.md` for $\varphi' \in [0.85, 1.20]$ and `data/ho_frozen_2d.md`). Rectangular frozen lattices are those of separable deformations $\varphi(s) = (\varphi_1(s_1), \varphi_2(s_2))$, as in Table 11; general deformations need (D4) on sheared lattices $A(s)\mathbb{Z}^2$, which is not checked. Together with Theorem 7.3 this makes stability on these deformed lattices a theorem up to this numerical check.

(ii) *Size of ω.* With $A = 0.15$ in 1-D ($\varphi' \in [0.85, 1.15]$, four neighbours) and $A = 0.1$ in 2-D (20 neighbours), ω_V is not of order $1/h$: it is between $2.3 \cdot 10^{-5}$ and $4.5 \cdot 10^{-4}$ in 1-D and decreasing under refinement, and $\mathcal{K}_V - 1 < 10^{-4}$ in 1-D and $\le 5 \cdot 10^{-4}$ in 2-D (Table 11). The constant ω of Theorem 7.3 is very small in practice.

(iii) *Local squares.* The remainder matrix R of Lemma 8.2 satisfies $\lambda_{\max}(R/h) \approx 2.3$–$2.5$ for $K = 2$ and stays bounded for $K = 3, 4$, for $n = 50$ to 400 (`ho_deformed_squares.py`, $A = 0.15$). The Fejér–Riesz factors of Lemma 8.1 reproduce (8.2) to 10^{-15} for $K = 2, 3, 4$ and $g \in [0.85, 1.2]$, and (D5) holds for $\kappa = 2.5$ with $m = 2$ ($K = 2, 3$) and $m = 3$ ($K = 4$) on $g \in [0.85, 1.2]$ (`ho_fr.py`, driver `d5_check.py`, `data/d5_check.txt`: 141 values of $g \in [0.85, 1.2]$, $\min q_g = 0.26$, 0.25, 0.027 and 0.63 for $K = 2$, 3, 4 geometric and 4 aligned).

(iv) *Varying topology.* With $A = 0.3$ (2 to 6 neighbours, (D1) violated), ω_V is 1.1–1.8 for

Table 12: Local certificates in 1-D (`ho_defect_cert.py`, `data/ho_defect_cert_1d.jsonl`, cvxpy/Clarabel, $\kappa = 2.5$, $c_W = 0.05$, $C_W = 20$): a posteriori residual res of the optimal block W_Z, its eigenvalue range in units of g, and the resulting numerical abscissa $\omega_W h$ of the single-defect cloud of 64 sites, for zone radii $\varrho = 3, 5, 8, 12, 16$ (listed in this order in each cell).

type	K	orient.	g	res(ϱ), $\varrho = 3,5,8,12,16$	$\lambda(W_Z)/g$ (range)	$\omega_W h$, $\varrho = 3,\dots,16$
insert	2	geom.	0.85	6e-4, 1e-4, 5e-6, 7e-7, 4e-7	0.62–7.4	7e-5, 2e-5, 9e-7, 2e-7, 9e-8
			1	4e-4, 8e-5, 2e-6, 6e-7, 4e-7	0.57–7.4	3e-5, 9e-6, 4e-7, 1e-7, 7e-8
			1.15	2e-4, 2e-5, 5e-7, 1e-8, 6e-9	0.54–7.7	8e-6, 2e-6, 6e-8, 2e-9, 9e-10
insert	3	geom.	0.85	6e-3, 5e-4, 2e-5, 2e-7, 7e-10	0.61–8.1	3e-3, 1e-4, 3e-6, 4e-8, 2e-10
			1	6e-3, 4e-4, 1e-5, 6e-8, 2e-10	0.60–7.3	3e-3, 9e-5, 1e-6, 1e-8, 5e-11
			1.15	3e-4, 2e-5, 7e-7, 1e-9, 2e-10	0.63–7.9	6e-5, 1e-6, 9e-8, 2e-10, 3e-11
insert	4	alig.	0.85	3e-3, 4e-5, 3e-7, 1e-7, 5e-8	0.57–5.9	1e-3, 1e-5, 5e-8, 1e-8, 5e-9
			1	3e-3, 3e-5, 1e-7, 5e-8, 2e-8	0.55–5.9	9e-4, 8e-6, 2e-8, 3e-9, 1e-9
			1.15	6e-4, 2e-5, 2e-7, 3e-7, 4e-9	0.35–6.1	2e-4, 7e-6, 5e-8, 6e-8, 3e-10
insert	4	geom.	0.85	2e-3, 4e-5, 2e-8, 8e-11, 7e-12	0.43–5.0	1e-3, 2e-5, 3e-9, 4e-12, 2e-12
			1	1e-3, 4e-5, 2e-8, 7e-11, 2e-9	0.44–5.0	7e-4, 1e-5, 2e-9, 1e-11, 5e-10
			1.15	7e-5, 3e-6, 1e-9, 5e-11, 2e-9	0.30–5.2	2e-5, 6e-7, 2e-10, 1e-11, 5e-10
merge	2	geom.	0.85	1e-2, 1e-3, 6e-5, 6e-7, 6e-8	0.59–8.6	3e-3, 2e-4, 8e-6, 1e-7, 1e-8
			1	8e-3, 8e-4, 4e-5, 2e-7, 3e-8	0.54–8.5	2e-3, 1e-4, 4e-6, 3e-8, 5e-9
			1.15	7e-3, 6e-4, 2e-5, 5e-8, 1e-8	0.55–8.5	2e-3, 5e-5, 2e-6, 7e-9, 2e-9
merge	3	geom.	0.85	3e-2, 3e-3, 7e-5, 1e-6, 5e-9	0.59–7.6	2e-2, 1e-3, 9e-6, 3e-7, 2e-9
			1	3e-2, 2e-3, 4e-5, 4e-7, 1e-9	0.57–9.3	1e-2, 6e-4, 4e-6, 9e-8, 3e-10
			1.15	3e-2, 2e-3, 2e-5, 1e-7, 2e-10	0.55–9.1	1e-2, 4e-4, 1e-6, 2e-8, 5e-11
merge	4	alig.	0.85	3e-2, 5e-4, 3e-6, 1e-8, 1e-10	0.52–11.4	1e-2, 2e-4, 1e-6, 3e-9, 4e-11
			1	3e-2, 5e-4, 3e-6, 1e-8, 3e-10	0.53–11.5	1e-2, 2e-4, 1e-6, 3e-9, 7e-11
			1.15	3e-2, 5e-4, 3e-6, 1e-8, 2e-9	0.52–13.2	1e-2, 2e-4, 9e-7, 2e-9, 2e-11
merge	4	geom.	0.85	9e-2, 3e-4, 2e-7, 3e-10, 1e-9	0.44–11.0	4e-2, 1e-4, 7e-8, 9e-11, 3e-10
			1	9e-2, 3e-4, 2e-7, 2e-10, 4e-9	0.43–11.0	3e-2, 1e-4, 6e-8, 6e-11, 1e-9
			1.15	9e-2, 3e-4, 2e-7, 1e-9, 9e-10	0.39–11.3	3e-2, 1e-4, 5e-8, 1e-10, 2e-10
shift	2	geom.	0.85	3e-5, 2e-6, 2e-7, 5e-8, 3e-8	0.54–7.7	4e-6, 2e-7, 3e-8, 1e-8, 6e-9
			1	2e-4, 2e-5, 3e-7, 4e-10, 1e-11	0.54–7.7	9e-6, 3e-6, 4e-8, 6e-11, 2e-12
			1.15	2e-3, 2e-4, 6e-6, 5e-8, 3e-8	0.56–8.0	3e-4, 1e-5, 6e-7, 7e-9, 4e-9
shift	3	geom.	0.85	7e-4, 9e-5, 6e-6, 3e-8, 2e-10	0.62–8.1	2e-4, 1e-5, 1e-6, 8e-9, 7e-11
			1	2e-4, 2e-5, 4e-7, 1e-9, 8e-11	0.60–8.1	3e-5, 2e-6, 7e-8, 3e-10, 1e-11
			1.15	2e-2, 1e-3, 2e-5, 1e-7, 4e-10	0.62–8.4	9e-3, 3e-4, 1e-6, 2e-8, 7e-11
shift	4	alig.	0.85	8e-3, 5e-5, 8e-8, 2e-10, 2e-10	0.50–6.4	4e-3, 1e-5, 2e-8, 4e-11, 3e-11
			1	3e-3, 6e-5, 8e-7, 4e-8, 2e-8	0.48–6.5	1e-3, 3e-5, 3e-7, 4e-9, 1e-9
			1.15	1e-3, 1e-5, 6e-8, 8e-10, 4e-9	0.56–10.3	3e-4, 4e-6, 1e-8, 1e-10, 1e-10
shift	4	geom.	0.85	2e-3, 3e-6, 1e-9, 1e-10, 9e-10	0.37–5.9	1e-3, 9e-7, 3e-10, 3e-11, 4e-10
			1	1e-4, 3e-6, 9e-10, 6e-11, 9e-10	0.38–5.7	4e-5, 7e-7, 2e-10, 1e-11, 3e-10
			1.15	2e-3, 7e-5, 3e-8, 1e-9, 3e-9	0.44–8.1	1e-3, 2e-5, 5e-9, 2e-10, 9e-10

$N = 100$ instead of 10^{-4}, while the Kreiss constants stay at 1.02.

13.2 Local certificates for single defects

Observation 13.2 (Certificates)**.** (i) For $K = 3$ the residual decreases roughly exponentially in ϱ for all three types, from about 10^{-2} at $\varrho = 3$ to between $8 \cdot 10^{-11}$ and $5 \cdot 10^{-9}$ at $\varrho = 16$, and it is below $1.1 \cdot 10^{-6}$ at $\varrho = 12$. For $K = 4$ with the geometric orientation it reaches about 10^{-10}, the solver accuracy, at $\varrho = 8$–12 and is then not monotone (insert at $g = 1$: $6.8 \cdot 10^{-11}$ at $\varrho = 12$, $1.5 \cdot 10^{-9}$ at $\varrho = 16$). The optimal residual is non-increasing in ϱ (extend the weight by gI on the added sites; the far squares that are removed are positive semidefinite), so such values are solver-limited upper bounds, still valid but not evidence of decay. For $K = 4$ with the aligned orientation it stalls at shift defects ($g = 1$: $7.7 \cdot 10^{-7}$, $4.4 \cdot 10^{-8}$, $2.2 \cdot 10^{-8}$ at $\varrho = 8, 12, 16$), as the weak instability of the periodic shift array forces (Observation 13.3(iii)), and near 2–$5 \cdot 10^{-8}$ at insert defects for $g \le 1$, whose periodic array is weakly unstable as well. For $K = 2$ it decays much more slowly at

Table 13: Periodic defect arrays, one defect per m sites (`ho_defect_bloch.py`, `data/ho_defect_bloch_*.jsonl`, Bloch analysis of Theorem 6.1, $\kappa = 2.5$, 64 values of θ): spectral abscissa times h and Bloch–Kreiss constant for merge defects with $m = 8, 16, 32, 64, 128$ (1-D; $m \le 64$ for $K = 4$; the insert and shift arrays of Observation 13.3 are not in the table; their data are in `data/ho_defect_bloch_1d_insert_shift.jsonl`) and $m = 4, 6, 8$ per direction (2-D, $a = (1, 0.5)$, 8^2 values of θ).

D	type	K	orient.	m	αh	Bloch–Kreiss constant
1	merge	2	alig.	8, 16, 32, 64, 128	1.6e-6, 7.5e-8, 1.8e-8, 4.5e-9, 1.1e-9	1.199, 1.170, 1.169, 1.169, 1.169
		2	geom.	8, 16, 32, 64, 128	1.6e-6, 7.5e-8, 1.8e-8, 4.5e-9, 1.1e-9	1.199, 1.170, 1.169, 1.169, 1.169
		3	alig.	8, 16, 32, 64, 128	0, 0, 0, 0, 0	1.319, 1.301, 1.301, 1.301, 1.301
		3	geom.	8, 16, 32, 64, 128	0, 0, 0, 0, 0	1.319, 1.301, 1.301, 1.301, 1.301
		4	alig.	8, 16, 32, 64	0, 0, 0, 0	1.607, 1.597, 1.597, 1.597
		4	geom.	8, 16, 32, 64	0, 0, 0, 0	1.646, 1.644, 1.644, 1.644
2	insert	2	geom.	4, 6, 8	0, 0, 0	1.007, 1.004, 1.003
		3	geom.	4, 6, 8	0, 0, 0	1.012, 1.007, 1.006
2	merge	2	geom.	4, 6, 8	0, 0, 0	1.005, 1.004, 1.003
		3	geom.	4, 6, 8	0, 0, 0	1.022, 1.013, 1.010

insert defects ($3.8 \cdot 10^{-7}$ at $\varrho = 16$) and, beyond $\varrho = 12$, at merge defects, consistent with Observation 13.3; at shift defects it decays exponentially at $g = 1$ ($1.2 \cdot 10^{-11}$ at $\varrho = 16$) but stalls near $3 \cdot 10^{-8}$ at $g = 0.85$ and 1.15.

(ii) The left null vector of a single-defect cloud is positive for all three types and $K = 2, 3, 4$, and it is not constant outside any zone, as Proposition 8.6 requires for a positive residual: $|\pi_i - 1|$ at distance i from a merge defect is 0.5, 0.1, 0.025, 0.006, 0.0013 for $i = 2, 4, 6, 8, 10$ at $K = 2$, similar at $K = 3$ and faster at $K = 4$ (`ho_defect_pi.py`, `data/ho_defect_pi.md`).

The certificates are floating-point solutions verified by floating-point eigenvalues of $E_\tau(W)|_{Z' \times Z'}$; they are the inputs of Theorem 8.7.

13.3 Periodic defect arrays

Observation 13.3 (Defect arrays and even order)**.** The periodic array of one defect every m sites is a periodic cloud, so Theorem 6.1 decides its stability.

(i) For the merge and insert defects with $K = 2$ the Bloch spectral abscissa is positive and decays algebraically in the spacing, $\alpha h = 1.6 \cdot 10^{-6}, 7.5 \cdot 10^{-8}, 1.8 \cdot 10^{-8}, 4.5 \cdot 10^{-9}$ for $m = 8, 16, 32, 64$ (merge) and $2.6 \cdot 10^{-6}, 8.6 \cdot 10^{-7}, 2.2 \cdot 10^{-7}, 5.7 \cdot 10^{-8}$ (insert), roughly m^{-2} for $m \ge 16$: the periodic defect array is weakly unstable at every spacing, with a rate that vanishes as the spacing grows. Since Theorem 8.7 applies to this array with $\varrho \approx m/2 - (d + 6\kappa + 1)$ and gives $\alpha h \le \mathrm{res}^*(\varrho)/(2c_W)$, the residual of any certificate must satisfy $\mathrm{res}_\tau(\varrho) \gtrsim c\,\varrho^{-2}$ for $K = 2$ (the separation condition gives $\varrho \ge 0$ only for $m \ge 40$, so this rests on $m = 64$ and 128): it cannot decay exponentially, and Corollary 8.8(i) is not available. The instability is harmless at fixed physical defect spacing $\ell = mh$: $\alpha = \alpha h/h \approx c\,(h/\ell)^2/h = c\,h/\ell^2 \to 0$.

(ii) For $K = 3$ (all types) and $K = 4$ (merge with both orientations, insert with the geometric one) $\alpha = 0$ to round-off and the Bloch–Kreiss constants converge as $m \to \infty$ (1.30, 1.10, 1.03 for merge, insert, shift at $K = 3$), consistent with the exponential decay of the residual.

(iii) The shift and insert defects with $K = 4$ and the aligned orientation are weakly unstable: $\alpha h \approx 3 \cdot 10^{-4} m^{-3}$ for shift ($6.6 \cdot 10^{-7}$ at $m = 8$, $1.3 \cdot 10^{-9}$ at $m = 64$) and $3.7 \cdot 10^{-7}, 7.0 \cdot 10^{-8}, 1.1 \cdot 10^{-8}, 1.6 \cdot 10^{-9}$ for insert with $m = 8, 16, 32, 64$ (roughly $m^{-2.5}$ to m^{-3}): the even orders are again the fragile ones, and the parity mechanism of Proposition 9.3 is at work at a single defect. By the argument of (i), $\mathrm{res}_\tau(\varrho) \gtrsim c\,\varrho^{-3}$ for these types, so their residuals cannot decay exponentially either; for insert defects this is consistent with the

Table 14: Superposition on clouds with $n/50$ random defects (`ho_defect_cert.py super`, `data/ho_defect_super_1d.jsonl`, $\kappa = 2.5$, geometric orientation): zone radius ϱ, largest residual res* of the certificates used, numerical abscissa in the assembled weight W_h and in the volume norm (times h), spectral abscissa, eigenvalue range of W_h/h and Kreiss constant in the volume norm.

K	A	ϱ	n	defects	m/i/s	res*	$\omega_W h$	$\omega_V h$	αh	$\lambda(W_h/h)$, $\mathcal{K}_V$
2	0	8	200	4	1/3/0	$3.8 \cdot 10^{-5}$	$4.3 \cdot 10^{-6}$	$9.0 \cdot 10^{-2}$	$\leq 10^{-12}$	0.64–4.1, 1.169
		8	400	8	4/3/1	$3.8 \cdot 10^{-5}$	$6.2 \cdot 10^{-6}$	$9.0 \cdot 10^{-2}$	$2.2 \cdot 10^{-8}$	0.60–4.1, 1.169
		12	200	4	1/1/2	$6.4 \cdot 10^{-7}$	$8.8 \cdot 10^{-8}$	$9.0 \cdot 10^{-2}$	$\leq 10^{-12}$	0.54–6.5, 1.169
		12	400	8	4/3/1	$6.4 \cdot 10^{-7}$	$1.4 \cdot 10^{-7}$	$9.0 \cdot 10^{-2}$	$2.2 \cdot 10^{-8}$	0.54–6.5, 1.169
2	0.15	8	200	4	1/3/0	$5.6 \cdot 10^{-5}$	$1.1 \cdot 10^{-3}$	$1.0 \cdot 10^{-1}$	$\leq 10^{-12}$	0.57–3.7, 1.168
		8	400	8	4/3/1	$5.3 \cdot 10^{-5}$	$4.4 \cdot 10^{-4}$	$1.0 \cdot 10^{-1}$	$\leq 10^{-12}$	0.48–4.7, 1.169
		12	200	4	1/1/2	$7.1 \cdot 10^{-7}$	$1.7 \cdot 10^{-3}$	$8.0 \cdot 10^{-2}$	$\leq 10^{-12}$	0.58–7.2, 1.168
		12	400	8	4/3/1	$7.3 \cdot 10^{-7}$	$9.0 \cdot 10^{-4}$	$1.0 \cdot 10^{-1}$	$5.6 \cdot 10^{-9}$	0.57–7.2, 1.169
3	0	8	200	4	1/3/0	$3.8 \cdot 10^{-5}$	$3.9 \cdot 10^{-6}$	$1.7 \cdot 10^{-1}$	$\leq 10^{-12}$	0.69–5.0, 1.301
		8	400	8	4/3/1	$3.8 \cdot 10^{-5}$	$5.1 \cdot 10^{-6}$	$1.7 \cdot 10^{-1}$	$\leq 10^{-12}$	0.68–5.0, 1.301
		8	800	16	3/6/7	$3.8 \cdot 10^{-5}$	$3.8 \cdot 10^{-6}$	$1.7 \cdot 10^{-1}$	$\leq 10^{-12}$	0.68–5.0, 1.301
		12	200	4	1/1/2	$4.0 \cdot 10^{-7}$	$3.6 \cdot 10^{-8}$	$1.7 \cdot 10^{-1}$	$\leq 10^{-12}$	0.60–7.3, 1.301
		12	400	8	4/3/1	$4.0 \cdot 10^{-7}$	$8.7 \cdot 10^{-8}$	$1.7 \cdot 10^{-1}$	$\leq 10^{-12}$	0.60–7.3, 1.301
3	0.15	8	200	4	1/3/0	$5.8 \cdot 10^{-5}$	$1.4 \cdot 10^{-3}$	$1.9 \cdot 10^{-1}$	$\leq 10^{-12}$	0.63–4.6, 1.301
		8	400	8	4/3/1	$5.5 \cdot 10^{-5}$	$6.1 \cdot 10^{-4}$	$1.8 \cdot 10^{-1}$	$\leq 10^{-12}$	0.62–5.5, 1.301
		12	200	4	1/1/2	$1.6 \cdot 10^{-7}$	$2.2 \cdot 10^{-3}$	$1.5 \cdot 10^{-1}$	$\leq 10^{-12}$	0.54–8.1, 1.300
		12	400	8	4/3/1	$8.2 \cdot 10^{-7}$	$1.0 \cdot 10^{-3}$	$1.9 \cdot 10^{-1}$	$\leq 10^{-12}$	0.53–8.6, 1.301
4	0	8	200	4	1/3/0	$2.2 \cdot 10^{-7}$	$5.7 \cdot 10^{-8}$	$2.7 \cdot 10^{-1}$	$\leq 10^{-12}$	0.45–6.8, 1.796
		8	400	8	4/3/1	$2.2 \cdot 10^{-7}$	$6.5 \cdot 10^{-8}$	$2.7 \cdot 10^{-1}$	$\leq 10^{-12}$	0.45–6.8, 1.644
		12	200	4	1/1/2	$2.4 \cdot 10^{-10}$	$3.1 \cdot 10^{-11}$	$2.7 \cdot 10^{-1}$	$\leq 10^{-12}$	0.46–9.1, 1.670
		12	400	8	4/3/1	$2.4 \cdot 10^{-10}$	$5.0 \cdot 10^{-11}$	$2.7 \cdot 10^{-1}$	$\leq 10^{-12}$	0.46–9.1, 1.660
4	0.15	8	200	4	1/3/0	$2.2 \cdot 10^{-7}$	$3.5 \cdot 10^{-3}$	$3.1 \cdot 10^{-1}$	$\leq 10^{-12}$	0.35–5.5, 1.653
		8	400	8	4/3/1	$2.4 \cdot 10^{-7}$	$2.1 \cdot 10^{-3}$	$3.2 \cdot 10^{-1}$	$\leq 10^{-12}$	0.35–10.4, 1.646
		12	200	4	1/1/2	$5.1 \cdot 10^{-10}$	$9.7 \cdot 10^{-3}$	$2.4 \cdot 10^{-1}$	$\leq 10^{-12}$	0.46–10.5, 1.636
		12	400	8	4/3/1	$3.4 \cdot 10^{-9}$	$4.2 \cdot 10^{-3}$	$3.1 \cdot 10^{-1}$	$\leq 10^{-12}$	0.37–13.5, 1.661

stalling residual of Observation 13.2.

13.4 Superposition and random defects

Observation 13.4 (Superposition)**.** Table 14 assembles the weight W_h of Theorem 8.7 on clouds of $n = 200$ to 800 sites with $n/50$ defects of random types (m/i/s = numbers of merge/insert/shift defects) and positions (separation $\geq 2\varrho + 20$, somewhat less than the $\sigma = 2\varrho + 40$ of (8.7) for $\kappa = 2.5$, $d = 4$, which the support of the actual blocks permits), on the lattice ($A = 0$) and on the deformed lattice ($A = 0.15$, certificates computed at the local spacing $g(z_A)$). On the lattice the measured $\omega_W h$ is about a tenth of the residual res* (at $K = 3$, res$^* = 3.8 \cdot 10^{-5}$ and $\omega_W h = 4$–$5 \cdot 10^{-6}$ for $\varrho = 8$; res$^* = 4 \cdot 10^{-7}$ and $\omega_W h = 4$–$9 \cdot 10^{-8}$ for $\varrho = 12$), within the bound of (8.8), while the volume norm gives $\omega_V h = 0.17$ at $K = 3$ (0.09 for $K = 2$, 0.24–0.32 for $K = 4$; i.e. $\omega_V \propto 1/h$, the obstruction of Proposition 9.1). On the deformed lattice $\omega_W h$ is dominated by the $O(h)$ deformation term: ω_W itself is $O(1)$, as (8.8) says. The Kreiss constants in the volume norm are those of the worst single defect (1.30 for a merge at $K = 3$), independent of n.

Observation 13.5 (Random defects)**.** For lattices and deformed lattices in which a fraction $p = 0.02$ to 0.1 of the sites carries a random, non-overlapping defect (the model of a cloud after particle management; these clouds do not satisfy the separation (8.7)): $K = 3$ has $\alpha = 0$ and Kreiss constants ≤ 1.44 at every density and resolution, on the lattice and on the deformed lattice; $K = 2$ shows the weak instability of Observation 13.3 (αh from 10^{-8} at $p = 0.02$ to $7 \cdot 10^{-7}$ at $p = 0.1$); $K = 4$ is stable up to $p = 0.05$ but blows up at $p = 0.1$ on the lattice ($\alpha h = 0.5$: two merge defects three sites apart leave a particle with two neighbours on one side, which the degree-three fit does not tolerate) and has Kreiss constants up to 8 on the deformed

Table 15: Random defects of density p on lattices ($A = 0$) and deformed lattices (`ho_defect_random.py`, `data/ho_defect_random_*.jsonl`, $\kappa = 2.5$, geometric orientation, two seeds): ranges over n and seeds of αh, $\omega_V h$ and $\mathcal{K}_V$.

D	A	p	K	n	αh	$\omega_V h$	$\mathcal{K}_V$
1	0	0.02	2	100–400	$7.7 \cdot 10^{-9}$	0.0901–0.0905	1.169–1.170
			3	100–400	$\leq 10^{-12}$	0.1676–0.1679	1.300–1.301
			4	100–400	$\leq 10^{-12}$	0.2722–0.2723	1.634–1.782
1	0	0.05	2	100–400	$1.3 \cdot 10^{-7}$	0.0901–0.0910	1.169–1.238
			3	100–400	$\leq 10^{-12}$	0.1679–0.1681	1.301–1.395
			4	100–400	$\leq 10^{-12}$	0.2723	1.676–1.927
1	0	0.1	2	100–400	$7.2 \cdot 10^{-7}$	0.0900–0.0942	1.183–1.285
			3	100–400	$\leq 10^{-12}$	0.1679–0.1724	1.301–1.436
			4	100–400	$5.1 \cdot 10^{-1}$	0.2723–5.0362	1.663–∞
1	0.15	0.02	2	100–400	$2.0 \cdot 10^{-9}$	0.0831–0.1069	1.168–1.173
			3	100–400	$\leq 10^{-12}$	0.1564–0.1975	1.295–1.301
			4	100–400	$\leq 10^{-12}$	0.2698–0.3559	1.643–1.750
1	0.15	0.05	2	100–400	$6.6 \cdot 10^{-8}$	0.0831–0.1069	1.167–1.244
			3	100–400	$\leq 10^{-12}$	0.1564–0.1975	1.300–1.399
			4	100–400	$\leq 10^{-12}$	0.2948–0.3559	1.660–1.888
1	0.15	0.1	2	100–400	$3.9 \cdot 10^{-7}$	0.0831–0.1062	1.181–1.281
			3	100–400	$\leq 10^{-12}$	0.1564–0.2010	1.302–1.430
			4	100–400	$\leq 10^{-12}$	0.3429–3.0052	1.684–8.237
2	0	0.02	2	12–24	$\leq 10^{-12}$	0.0026–0.0042	1.005–1.006
			3	12–24	$\leq 10^{-12}$	0.0115–0.0216	1.013–1.023
2	0	0.05	2	12–24	$\leq 10^{-12}$	0.0053–0.0106	1.008–1.016
			3	12–24	$\leq 10^{-12}$	0.0265–0.0335	1.024–1.048
2	0.1	0.02	2	12–24	$\leq 10^{-12}$	0.0019–0.0037	1.003–1.007
			3	12–24	$\leq 10^{-12}$	0.0093–0.0198	1.012–1.024
2	0.1	0.05	2	12–24	$\leq 10^{-12}$	0.0044–0.0096	1.007–1.017
			3	12–24	$\leq 10^{-12}$	0.0212–0.0301	1.024–1.049

lattice. In two dimensions (merge and insert defects, $K = 2, 3$, $p \leq 0.05$) $\alpha = 0$ and the Kreiss constants are 1.00–1.05.

Observation 13.6 (Two dimensions)**.** For the 2-D lattice ($\kappa = 2.5$, $a = (1, 0.5)$) the semidefinite programme `ho_sos2d.py` (`data/ho_sos2d.md`) finds a Gram matrix G as in Remark 8.9 with $r = 4$ for $K = 2$ and $K = 3$: the coefficients match to 10^{-15} and the smallest eigenvalue of G is $-6 \cdot 10^{-12}$ (zero, as it must be, since $f(0) = 0$). The certificate programme for a 2-D defect (`ho_defect_cert2d.py`) has a zone block of $|Z| \approx (2\varrho + 1)^2$ unknowns and a matrix inequality of size $|Z'| \approx (2\varrho + 21)^2$; with the generic dense formulation it exceeded the memory of the machine used already at $\varrho = 2$, and no 2-D certificates have been computed. A sparse formulation (the inequality is affine in W_Z through a block of size $(2\varrho + 11)^2$ only) is the next step.

Part III
Consequences and open questions

14 Consequences for the scheme and the paper

(1) *What can be claimed as proved.* On arbitrary clouds: admissibility [9, Thm. 6.3]; a maximum principle for scalar laws under stage-wise MOOD with the strict DMP (Theorem 3.1);

ℓ^1 bounds for the Euler equations up to the conservation defect (Proposition 3.3); for linear advection, $\ell^p(\pi)$ contractivity of the first-order scheme (Theorem 4.1). For the linear high-order schemes: von Neumann stability of order two on lattices (Theorem 5.2); stability of periodic refinements of any pattern whose Bloch semigroups are uniformly bounded (Theorem 6.1); stability on smoothly deformed lattices with constant stencil topology, assuming the torus form of the Lax–Nirenberg inequality, unconditionally for $K = 2$ in 1-D and, for other cases, given the frozen symbol check (Theorem 7.3); in 1-D, stability on lattices with separated insertion/merging defects up to a residual, with the computed certificates (Theorem 8.7 for $\varphi = \mathrm{id}$), and on deformed lattices given certificates for every local spacing, which are computed here only at three spacings. Conditionally on an exponential decay of the residuals, which no computation can verify and which fails for $K = 2$ (merge, insert) and $K = 4$ aligned (shift), the residual can be removed on lattices with defects separated by $O(|\log h|)$ spacings (Corollary 8.8(i)).

(2) *What can be claimed as measured.* On jittered clouds with jitter $0.2h$ the high-order schemes show no visible growth, with Kreiss constants 1.002–1.33 (Observation 12.2), and the only instability found is a Bloch instability of one $K = 2$ pattern with $\alpha h \le 10^{-9}$ (Observation 12.3); orders three to five are von Neumann stable on lattices except order four with the geometric orientation in 2-D (Observation 10.1). A stability proof for the high-order schemes on random clouds is open, and so is a convergence proof for any order on irregular clouds.

(3) *Odd orders are safer.* Proposition 9.3 and all measurements single out even K as the orders that can lose stability on irregular stencils. For the paper this suggests $K = 3$ with RK3 as the robust high-order configuration, and a remark that the order-reduction mechanism of [9, Prop. 4.3] has a stability counterpart only for even K.

(4) *Topology, not jitter.* Every instability found arises when the number of neighbours varies from particle to particle (jitter $0.4h$ with $\kappa = 2.5$, or the $A = 0.3$ deformation). A neighbour selection that fixes the count (e.g. k nearest neighbours per half-space), or a stricter safeguard such as $|\nu(i)| \ge B_q + 2$ or a bound on the condition number of the scaled moment matrix, would exclude the near-interpolatory stencils of Observation 12.1(iii) and is the natural hypothesis for a theorem on random clouds. These stencils pass the present safeguard and are not regularized (their Cholesky factorization succeeds), so the analysis applies to them as implemented; the change would be local to the safeguard of [9, Rem. 3.10].

(5) *Aligned orientation at all orders.* With the aligned orientation on every edge, the lattice instability of the geometric orientation disappears (Observation 10.1), and so does the local blow-up of Observation 12.1(iii); consistency on lattices is unaffected, since the proof of [9, Prop. 4.3] uses only that the orientation is odd. The price is the symmetry of [9, Lem. 4.1] on pairs with $c_{ij}c_{ji} \ge 0$, which affects conservation at the level of [9, Prop. 4.4(ii)]. This seems worth a numerical test in HyperCloudCore.jl. In the settings of the paper the geometric orientation matters only for $K = 4$ in 2-D with $\kappa = 5.5$ and a along an axis ($h \operatorname{Re}\lambda = 2.2 \cdot 10^{-8}$ on the lattice, absent at jitter $0.2h$), which should be mentioned if $K = 4$ runs are reported.

(6) *Moment-corrected first-order dissipation* (Proposition 4.5) restores the consistency of the first-order scheme on irregular clouds and keeps all positivity results (Observation 11.3).

(7) *Clouds from particle management.* Keep the stencil topology constant except at insertion/merging sites, keep those sites apart ($\sigma = 2\varrho + 40$ spacings at $\kappa = 2.5$, i.e. 56–64 spacings for $\varrho = 8$–12, for a negligible residual), and prefer odd K; then Theorem 8.7 applies, given certificates at the local spacings (computed here at $g = 0.85, 1, 1.15$; a bound

between them is open), with a residual that is invisible over any practical number of time steps.

(8) *Antisymmetric weights* would make Proposition 3.3 exact and Corollary 3.7 a global entropy (and, for linear symmetric systems, L^2) stability result on arbitrary clouds, in addition to local conservation [9, Prop. 4.4].

15 Open questions and other routes

Open questions.

(a) *High order on random clouds.* A proof of $\sup_h \mathcal{K}_V(L_h) < \infty$ for jittered clouds with constant topology and odd K. The missing tool is a bound on the second-order mode coupling of Heuristic 12.5 in terms of the variance of the leading local coefficient, a random-matrix question. A non-diagonal Lyapunov weight would be needed for $K = 3$ (Observation 12.7).

(b) *The weight π for $K = 2$.* Whether $\omega_\pi(L_h)$ stays bounded at jitter $0.2h$ (it grows from $2 \cdot 10^{-4}$ to $2 \cdot 10^{-3}$ between $n = 400$ and 1600, Observation 12.6); if it does, π gives a Lax–Richtmyer proof for $K = 2$.

(c) *Uniformity of ρ_h and the size of the corrector* for the first-order scheme (Observations 11.1 and 11.2): a proof of $\eta_h = O(h^{1/2} \log^{1/2}(1/h))$ for independent jitter; for random clouds these are questions about Markov chains in random environments.

(d) *Bloch constants in the pattern size.* Uniformity of the Bloch–Kreiss constants in m for random patterns (bounded for $m \le 32$ in Observation 12.3); a statement in probability over the pattern would bridge the periodic and the random case.

(e) *Defects.* A residual-free statement: the single-defect problem on the infinite lattice is a Godunov–Ryabenkii/Kreiss normal-mode problem for an interface, and the semigroup results of Kreiss and Wu [15] for dissipative schemes may give it for finitely many defects. Also: rigorous (interval) verification of the certificates; 2-D certificates via a sparse formulation; a sum-of-squares decomposition of the frozen symbols of affine lattices in 2-D with Lipschitz dependence on the lattice, which would extend Theorem 8.7 to deformed lattices in 2-D.

(f) *Supraconvergence.* The high-order schemes converge at the design order on perturbed clouds in [10], although their truncation error loses one order [9, Prop. 4.3]. A corrector argument as in Theorem 4.4, in the spirit of supraconvergence on irregular grids [13], would explain this if a stable norm for the high-order operator were available. Entropy-stable high-order constructions [3, 23] combine antisymmetric weights with an analysis of the reconstruction jumps, on meshes.

(g) *MOOD and linear stability.* Theorem 3.1 bounds the solution but not perturbations; an L^1 contraction for the MOOD scheme is open even on lattices.

Other routes considered.

(1) *MOOD as a switched linear system.* The MOOD scheme at a fixed order pattern is a row-wise mixture of the high-order and the first-order operator, not a convex combination of operators. A common Lyapunov weight for all patterns would give stability of the switched system [19], but row-wise mixtures do not inherit $HL + L^{\mathsf{T}}H \preceq 0$ from the two extremes unless the dissipation inequality holds locally, as a sum of nonpositive pair terms. It does for $K = 1$ ((4.1)) but not for $K \ge 2$ (the pair terms in (9.3) have both signs). For scalar laws with the strict DMP, Theorem 3.1 is the nonlinear stability statement, and the linear analysis describes the scheme where MOOD is inactive.

(2) *GKS theory* concerns boundaries; the periodic problem has none. A GKS-type analysis of boundary stencils on a lattice interior would be a natural continuation of Section 5, but is outside the question of irregular interiors.

(3) *Perturbation of the lattice operator.* A jitter εh changes L_h by $O(\varepsilon/h)$ in norm, and no perturbation theorem absorbs this unless the dissipation is strict on all modes; it is not on smooth modes. Theorem 7.3 is the version of the perturbation argument that works: the perturbation is $O(\varepsilon/h)$ pointwise there as well, but its symmetric part is $O(1)$ because the coefficients vary on the scale 1, not h.

A Numerical method and scripts

All numbers come from an independent NumPy/SciPy implementation in `stability/scripts/` (not HyperCloud code). `linop.py` assembles the matrix of the linear scheme for $u_t + a \cdot \nabla u = 0$ on periodic jittered lattices of the unit torus (jitter uniform in $[-\varepsilon h, \varepsilon h]$ per coordinate, seeds $0, 1$; Bloch supercells use seeds $1000, 1001$), with the weight $w(r) = 1/(1 + y + y^2/2 + y^3/6 + y^4/24)$, $y = r^2/R^2$, of `scripts/stencil_diagnostics.py`, reconstruction fits of degree $K - 1$ through U_i, derivative fits of degree q with constant term and the safeguards of [9, Rem. 3.10], the Rusanov flux with $s_d = |a_d|$, no limiter and no MOOD. Fits are unregularized (plain solves). The first-order operator is assembled separately in sparse form. `ho_common.py` provides dual volumes, the numerical abscissa ω_H (generalized symmetric eigenvalue problem), Kreiss constants (eigen-decomposition or `expm` on 40 log-spaced times in $[h/100, 20]$), Bloch symbols (the operator on three copies of the supercell per direction, read off the middle block with the phases $e^{\mathrm{i}\theta\cdot(k-k_{\mathrm{mid}})}$) and deformed lattices. Dense linear algebra limits the sizes to $N \le 1600$ in 1-D and $N \le 576$ in 2-D for the high-order studies.

Script	Used for
`certify_K2_1d.py`	exact certificate in the proof of Theorem 5.2
`lattice_symbols.py`, `k2_random_search.py`	Table 3, Observation 10.1
`first_order_markov.py`, `_corrector.py`, `_corrector_large.py`, `_phase.py`, `_moment_fix.py`	Table 4, Observations 11.1 to 11.3
`spectra_study.py`, `summarize_spectra.py`	Table 5, Observation 12.1
`ho_random.py`, `ho_nn_identity.py`, `ho_nn_admissible.py`	Table 6, Observation 12.2, Proposition 9.2(iii)
`ho_bloch.py`, `ho_bloch_mpcheck.py`	Table 7, Observation 12.3
`ho_local_symbol.py`	Table 8, Observation 12.4
`ho_pi.py`, `ho_lyapunov_exact.py` (`ho_lyapunov.py`: superseded SDP, cvxpy, SCS)	Tables 9 and 10, Observations 12.6 and 12.7
`ho_deformed.py`, `ho_frozen.py`, `ho_fr.py`, `d5_check.py`, `ho_deformed_squares.py`	Table 11, Observation 13.1
`ho_defect_cert.py` (cvxpy, Clarabel), `exact_cert.py`, `ho_defect_pi.py`	Tables 12 and 14, Observations 13.2 and 13.4
`ho_defect_bloch.py`, `ho_defect_random.py`	Tables 13 and 15, Observations 13.3 and 13.5
`ho_sos2d.py`, `ho_defect_cert2d.py`	Observation 13.6
`make_tables.py`, `ho_make_tables.py`	the LaTeX tables in `tables/` from `data/`

The checks differ from HyperCloudCore.jl 0.7.8 in the boundary treatment (periodic only), in the absence of MOOD and limiters, in the dissipation speed ($|a_d|$ instead of the Euler wave

speeds), and possibly in details of the basis ordering and scaling of the fits; they test the scheme as defined in [A], not the implementation.

Declaration of generative AI use

This note, its proofs, the scripts and data of Appendix A and a partial Lean 4 formalization of Part I (`lean/STABILITY.md`, in which computer-assisted, cited and several analytic steps enter as hypotheses) were generated with Claude Fable 5.1 (Anthropic) in Claude Code. The note was reviewed by an independent Claude Opus 5.5 session without access to the generating conversation, which re-derived the proofs of Part I, reran all scripts and checked six certificates in exact rational arithmetic; the corrections of that review are included. P. Jung takes responsibility for the content.

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